

Title

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Abstract

Sparse high-dimensional signal recovery is only possible under certain conditions on the number of parameters, sample size, signal strength and underlying sparsity. We study how these mathematical limits can be pushed by leveraging external information on the likelihood of variables to be associated with the outcome. As an important particular case, we consider a flexible transfer learning framework robust to changes in domains and negative transfer where the set of variables selected in the source is transferred to ease variable selection in the target. We introduce a family of penalized linear regression methods motivated by connections to Bayesian variable selection where the penalties depend on the external information or the transferred set of variables. We show they attain variable selection consistency where otherwise it would not be possible, or attain it at a faster rate. We first precisely quantify the gains that are achievable for ideal penalties set by an oracle. Subsequently, we propose computationally fast algorithms for the incorporation of external information and transfer learning that do not require an oracle and are inspired by empirical Bayes techniques. We prove these algorithms recover most of the gains achievable by oracle penalties and are robust to negative transfer. We show that the proposed algorithm improves on standard variable selection methods on simulated and empirical data.♠

Paul: Not saying it's ℓ_0 penalty based to not scare frequentists

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1 Introduction

In a data-rich world, basing statistical inference on only a single source of information increasingly seems like a missed opportunity. As ever more data is being generated and stored, many have answered statistical questions by leveraging multiple datasets. Integrating multiple sources of data implies handling and possibly exploiting some form of heterogeneity. There may be heterogeneity in samples where, for example, samples are not identically distributed but rather feature a group structure XXX cite some Annie Qu’s work XXX. There may also be heterogeneity in variables in which case there is often knowledge, external to the dataset of interest, of varying likelihood of variables to be truly associated with the outcome of interest. This may occur when working with *multimodal* data, i.e. featuring different types of variables , where one block of variables may be expected to be less sparse than another, e.g. clinical vs genomic markers. Another example is when meta-data is available on the variables themselves, e.g functional annotations on genes. A setting of particular interest is transfer learning which can broadly be defined as the problem of improving learning for given *target* task and dataset by reusing knowledge acquired from similar *source* tasks and datasets. In that case, the set of variable selected in the source data and task may be seen as external knowledge on the likelihood of variables to be truly associated with the target outcome.

In high-dimensional settings, where the number of samples is smaller than the number of parameters, leveraging such external information is a particularly attractive idea. The inherent scarcity of information of these settings implies that strong assumptions are required to carry out meaningful inference. In structural learning problems such as variable selection, those assumptions regard sparsity and signal strength and, although mathematically necessary, they can be too strong in practice (see (?) for a discussion in the context of social sciences). Numerous applied and methodological works have employed such external information to

guide structural learning and consistently observed improvements when suitable external information is integrated. bousteleix..., codata,... As regards transfer learning, statistical literature has focused on the transfer from the source to the target of parameter estimates. Some proposed methods perform variable selection but do not consider the transfer of the set of selected variables as we consider here. perform variable selection example telesca.

Despite empirical success, a theoretical framework explaining precisely why and how leveraging external information on the likelihood of variables to be truly associated with the outcome as possible with data integration and transfer learning is not currently available. In this work, we investigate how, in linear regression, said information allows pushing the mathematical conditions under which consistently recovering the true subset of variables associated with the outcome is possible, and improving the corresponding rates. Consistency is understood here as recovering the true subset with probability going to 1 as the number of samples n and possibly the dimension p go to infinity.

To make ideas concrete, consider variable selection in the Gaussian linear regression

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta}^* + \epsilon, \quad \epsilon \sim N(0, \sigma^2 I_n), \quad (1)$$

where $\mathbf{X} \in \mathbb{R}^{n \times p}$, $\mathbf{y} \in \mathbb{R}^n$, $\sigma > 0$, and $\boldsymbol{\beta} \in \mathbb{R}^p$ are the data-generating parameters. A large class of methods operate by penalizing the size of an estimated $\hat{\boldsymbol{\beta}}$. For instance, penalized likelihood methods optimize the log-likelihood plus a penalty term driven by the ℓ_q "norm" of $\hat{\boldsymbol{\beta}}$ for $q \in [0, 1]$ (??). In particular, ℓ_0 penalties have been shown to have theoretically superior or even optimal selection properties ?? and to outperform other types of penalties in practice ?. Advances in optimization and MCMC methods have also made ℓ_0 penalized regression much more computationally tractable: it can be solved exactly for p in the thousands (?) and with probability going to 1 with linear cost in p using Markov Chain Monte Carlo (MCMC) under mild conditions (??). There is also a direct connection between ℓ_0 penalties and Bayesian variable selection based on posterior model probabilities

(???). Under mild regularity conditions, a choice of ℓ_0 penalty corresponds to a choice of prior probability for the inclusion of variables in the support of β^* . This connection is more formally discussed in Section 2. Motivated by the superior properties and the convenient Bayesian interpretation of ℓ_0 penalties, we study a class of ℓ_0 penalties that depend on external information.

Indeed, the external information may be seen as providing prior knowledge in a Bayesian setting. The Bayesian framework has indeed been frequently observed to be naturally suited for transfer learning ???. In such Bayesian setting, prior knowledge on the likelihood of variables to be associated with the outcome would be reflected in the choice of prior probability for their inclusion in the support of β^* . Given the connection between those probabilities and ℓ_0 penalties, in a frequentist setting, it is natural to let ℓ_0 penalties depend on external information. Consider, for example, the transfer learning problem for the task of variable selection. Among the candidates variables in the target dataset, one can form two blocks: a first one with the variables that were also selected the source datasets and a second one with the rest of the variables. The external information induces here a partition in two blocks where, if the transfer is informative for selection in the target, there is prior knowledge that the first block is less sparse than the second.

More generally, assume that we have a vector of external information $\mathbf{Z} = (z_1, \dots, z_p) \in \mathbb{R}^q$, where z_i is the external information for variable $i = 1, \dots, p$, that partitions the p variables into b blocks $B_j \subset \{1, \dots, p\}$, $j = 1, \dots, b$ so that key characteristics are thought to potentially vary across the B_j 's. That is, external information that induces a partition function

$$\Gamma : z_i \rightarrow B_j, \mathcal{J} \in \{1, \dots, b\}. \quad (2)$$

This framework encompasses, for example, the earlier transfer learning setting, the multi-modal data setting with blocks given by the different types of variables and setting of the

meta-data on variables with a single discrete meta-variable. We study here ℓ_0 penalties that depend on such external information $\mathbf{Z}$ through the induced partition. More precisely, we consider ℓ_0 penalties that allow modulating the strength of the penalty in each block. Unlike standard ℓ_0 penalties, such as BIC (?) or EBIC (?), informed penalties are *non-exchangeable* in the sense that the penalty for adding a variable i can depend on its block $B_j = B_j(z_i)$. We derive the variable selection properties of those informed ℓ_0 penalties and compare them to those of standard ℓ_0 penalties in relation to the relevance of the external information or transfer.

1.1 Related literature

External information has been used to guide variable selection in many applications. For instance, in Bayesian literature, ? and ? proposed Bayesian variable selection methods for gene expression, where prior inclusion probabilities depend on biological knowledge and meta-variables. ? similarly considers a regression of prior inclusion probabilities on external data and suggest an empirical Bayes approach to learn the regression parameter. ? proposed a variational Bayes approximation for the same setting. Recently ? considered a horseshoe type of prior that depends on external information. Beyond regression, node and edge variables have been incorporated in ? and ? to drive edge inclusion in Gaussian graphical models, while ? used meta-variables to determine non-zero loadings in factor models. In the frequentist literature, ? proposed an ℓ_1 penalty for linear regression on gene expression where the penalty varies depending on correlation between the gene expression and its copy number. In the context of multi-omics data, ? let the ℓ_1 penalty vary depending on the modality the variable belongs to. ? predicted disease outcomes by allowing LASSO penalties to depend on functional annotation categories. ? and ? proposed empirical Bayes approaches to setting for externally-informed ℓ_1 and elastic-net penalties respectively. See ? for a complete review. Also related is the causal analysis literature where the inclusion of

control variables may be driven by their degree of association with the variables of interest (referred to as treatments) (??).

The statistical properties of the transfer of parameter estimates in linear regression have previously been studied in different works. In the frequentist literature ? proposed a variation of LASSO, the prior LASSO, where the ℓ_1 penalization is supplemented by a penalization on the distance to the predicted values in the source datasets. They show improvements over LASSO in the efficiency in estimation and prediction. ? proposed the Trans-LASSO to transfer parameter estimates in linear regression, where deviations from parameter estimates in source datasets are penalized. They show improvements over LASSO in optimal rates of convergence for prediction and estimation. In concurrent works, ? and ? both extended Trans-LASSO to generalized linear models. In the Bayesian literature, XXXX . We refer to ? for a more complete review of Bayesian literature on transfer learning.

1.2 Organization and contribution

The rest of the paper is organized as follows. In Section 2 we formally introduce externally-informed ℓ_0 penalties and further detail their connection to Bayesian variable selection. We also introduce our transfer learning algorithm for the transfer of sets of selected variables which builds upon our externally-informed ℓ_0 penalties. Our algorithm is highly flexible, it does not require full access to the source data ensuring higher levels of privacy and communication-efficiency. It does not require that the source task to be linear regression either, and is robust to variations in *domains*, the sets of variables and their distributions, between the source and the task datasets. In this section, we also describe the general assumptions used across results.

Section 3.1 studies the properties of external information-dependent ℓ_0 penalties and how they ease consistent model recovery. We first show that an oracle, knowing the true pattern

of sparsity and signal strength, may take advantage of the external information so that variable selection consistency is either attained where otherwise it would not be possible, or is attained at a faster rate. These oracle results give an understanding of the maximum gains achievable by leveraging external information. As an independent contribution, our oracle results applied to standard ℓ_0 penalties improve on prior results on the fundamental limits on variable selection without external information [10], with new tighter sufficient and necessary conditions and nearly-optimal rate for variable selection consistency in linear regression. Subsequently, we propose two-step data-based procedure grounded in empirical Bayes methods to set externally-informed ℓ_0 penalties. We show the proposed procedure improves variable selection properties without requiring an oracle and compare its properties to those of oracle-set penalties.

Section 4 harnesses the results of Section 3.1 to derive the properties of our transfer learning algorithm for the transfer of selected variables. We first derive its properties for penalties set by an oracle knowing the true pattern of sparsity and signal strength. We then discuss its properties for a concrete, data-based, approach to set penalties. In particular, we show our data-based approach is robust to so-called *negative transfer*, when transfer learning worsens the performance in the target. To the best of our knowledge, this is the first theoretical analysis of a transfer learning algorithm centered on the transfer of the binary information on selection vs. no selection in the source data, rather than parameter estimates. This is relevant firstly because the transfer of selection information might be preferred when the similarity between source and target data is thought to lie in the sets of variables associated with the respective outcomes, rather than in the similarity between respective parameter vectors. Secondly, our work highlights the value of transferring the set of selected variables independently of parameter estimates. Thirdly, transferring binary information on selection can be seen as a robust alternative to transferring parameter estimates when the *domains*,

the variables and their distributions, differ greatly between the source and the target.

1.3 Notation

We start by describing the notations for the linear model of interest, that is, when we discuss a transfer learning setting, the target linear regression. The parameters of interest are denoted by $\beta \in \mathbb{R}^p$ and β^* denotes their true values. The set of variable indices is $V = \{1, \dots, p\}$. For any $A \subseteq V$ and any vector $\mathbf{x} \in \mathbb{R}^p$, $\mathbf{x}_A$ denotes the subvector of $\mathbf{x}$ with entries corresponding to indices in A . For any matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$, $\mathbf{X}_A$ denotes the submatrix of $\mathbf{X}$ obtained by selecting the *columns* with indices in A . $S = \{i \in V : \beta_i^* \neq 0\}$ denotes the true support of β^* , its size is $s = |S|$, and hence $p - s$ is the number of truly inactive parameters. The set of candidate models is denoted by $\mathcal{M}$ and is given by subsets $M \subseteq V$. Denote by $\beta_{\min}^* = \min_{i \in S} |\beta_i^*|$ the smallest true signal. The set V is partitioned into blocks, their number is fixed and noted b . The blocks are disjoint and noted $B_j \subseteq V$ for $j = 1, \dots, b$. $S_j = \{i \in B_j : \beta_i^* \neq 0\}$ denotes the set of active parameters in block B_j , $s_j = |S_j|$ its size, $p_j - s_j$ the number of inactive parameters in the block, and $\beta_{\min,j}^* = \min_{i \in S_j} |\beta_i^*|$.

When we discuss a transfer learning setting, $\mathcal{D} = \{\mathbf{y}, \mathbf{X}\}$ denotes the target dataset, V the set of variables in $\mathcal{D}$, and S the subset of V of variables truly associated to target outcome $\mathbf{y}$. The source datasets are denoted $\mathcal{D}^{(k)} = \{\mathbf{y}^{(k)}, \mathbf{X}^{(k)}\}$ for $k = 1, \dots, K$ where K is the number of source datasets. For $k = 1, \dots, K$, $S^{(k)}$ denotes the set of variables truly associated to source outcome $\mathbf{y}^{(k)}$.

Given sequences $f(n) > 0$ and $g(n) > 0$, $f(n) = O(g(n))$ means that there exists a constant $c < \infty$ such that $f(n) \leq cg(n)$ for all $n \geq n_0$ and some fixed n_0 , $f(n) = o(g(n))$ means that $\lim_{n \rightarrow \infty} f(n)/g(n) = 0$, $f(n) = \Theta(g(n))$ means that $f(n) = O(g(n))$ and $g(n) = O(f(n))$ and $f(n) \sim g(n)$ means that $\lim_{n \rightarrow \infty} f(n)/g(n) = 1$. For any set A , A^C denotes the complement of A .

2 Variable selection via externally-informed penalization and transfer learning

This section describes the main objects of study in this work: externally-informed ℓ_0 penalization (Section XXX) and a transfer learning algorithm for the transfer of selected variables (Section XXX). It also reviews the connection between externally-informed ℓ_0 penalization and Bayesian variable selection. Finally, it discusses general assumptions used across results (Section XXX).

2.1 Externally-informed penalization

Consider a set of candidate models $\mathcal{M}$ given by subsets $M \subseteq V$ and their corresponding coordinate subspaces

$$\mathcal{L}_M := \{\beta \in \mathbb{R}^p : \beta_i = 0 \text{ if } i \notin M\}.$$

A standard ℓ_0 selection procedure with penalty $\eta(M)$ that depends on M linearly through its cardinality, $\eta(M) = \kappa|M|$ for some $\kappa > 0$ selects the model

$$\hat{S}^{sd} = \arg \max_{M \in \mathcal{M}} \left\{ \max_{\beta \in \mathcal{L}_M} \ell(\mathbf{y}; \beta) - \kappa|M| \right\}, \quad (3)$$

where $\ell(\mathbf{y}; \beta)$ is the log-likelihood function.

A popular Bayesian framework for variable selection are discrete spike-and-slab priors that feature, for each variable i , a random binary indicator m_i for the inclusion of variable i in the model, or equivalently an indicator for the corresponding parameter to be nonzero, $m_i = I(\beta_i \neq 0)$. A model $M \in \mathcal{M}$ corresponds then to a particular value of the vector of m_i . The joint prior on parameters and models is

$$p(\beta, M \mid \theta) = p(\beta \mid M)p(M \mid \theta) \quad (4)$$

where $p(\beta \mid M)$ is a prior on regression coefficients given the model, and $p(M \mid \theta)$ the prior

probability of model M . Specifically, assume that variable inclusions are independent a priori and Bernoulli distributed with inclusion probability θ ,

$$p(M \mid \theta) \propto \prod_{i=1}^p \text{Bern}(m_i; \theta) I(M \in \mathcal{M}). \quad (5)$$

Posterior model probabilities are $p(M \mid \mathbf{y}, \theta) \propto p(\mathbf{y} \mid M)p(M \mid \theta)$, where $p(\mathbf{y} \mid M)$ is the so-called marginal likelihood of model M . The BIC approximation (?) to $p(M \mid \mathbf{y}, \theta)$ gives for a wide family of priors $p(\boldsymbol{\beta} \mid M)$, up to rescaling,

$$\ln p(M \mid \mathbf{y}, \theta) \approx \ln p(\mathbf{y} \mid \tilde{\boldsymbol{\beta}}^{(M)}) - \left(\frac{1}{2} \ln(n) + \ln(1/\theta - 1) \right) |M| + c_M, \quad (6)$$

where $\tilde{\boldsymbol{\beta}}^{(M)}$ is the maximum likelihood estimator (MLE) under model M , and c_M a constant that may depend on M . In particular, when $p(\boldsymbol{\beta} \mid M)$ is the Zellner's unit information prior (?), then c_M does not depend on M and the approximation is exact by replacing $\tilde{\boldsymbol{\beta}}^{(M)} = (\mathbf{X}_M^\top \mathbf{X}_M + n^{-1}I)^{-1} \mathbf{X}_M^\top \mathbf{y}$. Neglecting the constant c_M , simple algebra shows that ℓ_0 selector in 3 with $\kappa = \frac{1}{2} \ln(n) + \ln(1/\theta - 1)$ approximately maximizes $p(M \mid \mathbf{y}, \theta)$. It follows then a one-to-one correspondence between a choice of prior inclusion probability in our Bayesian spike-and-slab regression and a choice of ℓ_0 penalty in $\hat{S}^{sd}$.

Now, assume one has external information $\mathbf{Z}$ on the likelihood of variables to be truly associated with $\mathbf{y}$. In Bayesian spike-and-slab regression, this prior information would naturally be reflected in prior inclusion probabilities, and one would set

$$p(M \mid \theta) \propto \prod_{i=1}^p \text{Bern}(m_i; \theta(z_i)) I(M \in \mathcal{M}) \quad (7)$$

where $\theta(z_i)$ is an arbitrary function of z_i . In particular, consider the case where $\mathbf{Z}$ induces a partition of the set of variables in blocks $B_1, \dots, B_b$ such that the likelihood of being truly associated with $\mathbf{y}$ is thought to potentially vary across the B_j 's, as may occur in transfer learning. One would set block-dependent prior inclusion probabilities θ_j , $j = 1, \dots, b$,

$$p(M \mid \boldsymbol{\theta}) \propto \prod_{i=1}^p \text{Bern}(m_i; \theta_i) I(M \in \mathcal{M}), \quad (8)$$

where $\forall i \in B_j$, $\theta_i = \theta_j$ and $\boldsymbol{\theta} = (\theta_1, \dots, \theta_b)$. The BIC approximation for the model prior in (8) gives, up to rescaling,

$$\ln p(M \mid \mathbf{y}, \boldsymbol{\theta}) \approx \ln p(\mathbf{y} \mid \tilde{\boldsymbol{\beta}}^{(M)}) - \sum_{j=1}^b \left(\frac{1}{2} \ln(n) + \ln(1/\theta_j - 1) \right) |M_j| + c_M, \quad (9)$$

where $M_j \subseteq B_j$ are the selected parameters in block j and a model is denoted by $M = M_1 \cup \dots \cup M_b$.

In this work, we study the ℓ_0 selector analog to maximizing $p(M \mid \mathbf{y}, \boldsymbol{\theta})$ in (8). That is an externally-informed ℓ_0 selector that penalizes differently the blocks $B_1, \dots, B_b$ (2),

$$\hat{S}^{ei} \in \arg \max_{M \in \mathcal{M}} \left\{ \max_{\boldsymbol{\beta} \in \mathcal{L}_M} \ell(\mathbf{y}; \boldsymbol{\beta}) - \sum_{j=1}^b \kappa_j |M_j| \right\}, \quad (10)$$

where $\kappa_1 > 0, \dots, \kappa_b > 0$. Under the BIC approximation in (9), $\hat{S}^{ei}$ corresponds to the posterior mode with block prior inclusion probabilities $\theta_j = (e^{\kappa_j - \ln(n)/2} + 1)^{-1}$. In the case that $b = 1$, $\hat{S}^{ei}$ recovers $\hat{S}^{sdsd}$.

2.2 An algorithm for the transfer of selected variables

Consider now a transfer learning setting where in each source dataset $k = 1, \dots, K$ the set of variables $\hat{S}^{(k)}$ was selected and one wants to leverage these selected sets to improve selection in the target dataset. The simplistic approach of forcing the inclusion in the target model of the variables selected in the sources datasets is very prone to negative transfers as sets of variables associated to target and source outcomes may be similar but not identical. Given the connection between ℓ_0 penalties and prior inclusion probabilities (9), one may instead let a ℓ_0 penalty differ between variables that were selected in the source datasets and those that were not.

More formally, in our framework, one can form a vector of external information on the variables in the target linear model $\mathbf{z}$ such that, for each $i = 1, \dots, p$, $z_i = 1$ if $i \in \cup_{k=1}^K \hat{S}^{sd(k)}$ and $z_i = 0$ otherwise. The external information $\mathbf{z}$ then naturally induces a partition in two

blocks, a first block that gathering variables that were selected at least once in the source datasets, and a second block gathering the rest of the variables. Given said partition, the externally-informed $\hat{S}^{ei}$ may be used to perform selection in the target.

Algorithm 1

- (i) Obtain $\hat{S}^{(k)}$ for $k = 1, \dots, K$.
- (ii) Obtain $\hat{S}^{ei}$ by solving (10) with $B_1 = \{i : i \in \cup_{k=1}^K \hat{S}^{sd(k)}\}$ and $B_2 = V \setminus B_1$.

Algorithm 1 offers a principled frequentist procedure to transfer selected variables from source to target datasets. In Section ?? we discuss how to choose block penalties in $\hat{S}^{ei}$ in practice to mitigate negative transfer. We observe that our algorithm does not require access to the source datasets but only sets of selected variables, ensuring wider applicability and higher levels of privacy and communication efficiency. The construction of blocks in Algorithm 1 also allows the domains to differ between the target and source datasets, and between source datasets.

2.3 General assumptions of theoretical results

This work discusses the properties of $\hat{S}^{ei}$ and Algorithm 1 when the number of samples and the number of variables go to infinity. Although not explicitly denoted, s_j , p_j and $\beta_{\min,j}^*$ are to be thought of as functions of n , and so are s , p and $\beta_{\min}^*$. We study the asymptotic regimes where

$$(A1) \quad n \rightarrow \infty,$$

$$(A2) \quad \text{For all } j, p_j - s_j \rightarrow \infty.$$

Similarly, Assumption A2 is a mild assumption that can be relaxed but simplifies the exposition. Results are provided for both cases where the $s_j \geq 1$ are fixed and $s_j \rightarrow \infty$. No assumption is made about the asymptotic regime linking n and p , but the main interest is

in $n = o(p)$ settings. In the study of the properties of $\hat{S}^{ei}$, we assume

(A3) External information $\mathbf{Z}$ partitions the set of variables into a constant number of blocks b .

Assumption A3 can be slightly relaxed in our results to allow b to grow slowly with p , but b is assumed constant for simplicity. The assumption reflects real-world settings. It is immediately satisfied in our transfer learning algorithm, Algorithm 1. Another example arises when regressing on multimodal data (e.g., multi-omics, clinical history vs. genomic markers) or when meta-information suggests varying sparsity or signal strength across the set of variables (e.g. functional annotations on genes). Assumptions A1–A3 describe the general framework, in each result the needed assumptions are precisely specified.

3 Properties of externally-informed penalization

In this section we discuss the properties of the externally-informed model selector $\hat{S}^{ei}$ (10) in the linear model (1). Throughout the section our goal will be to show that $\hat{S}^{ei}$ is equal to the data-generating truth S with probability 1, as n grows, and that this occurs under milder conditions or at a faster rate than when using a model selector $\hat{S}^{sd}$ that penalizes equally across all blocks. In Section 3.1, we discuss the properties of $\hat{S}^{ei}$ and compare them to those of $\hat{S}^{sd}$ for, in both cases, penalties set by an oracle that knows the true pattern of sparsity and signal strength. We discuss the benefits of $\hat{S}^{ei}$ over $\hat{S}^{sd}$ in relation to the relevance of the external information for variable selection, that is to what extent the partition it induces discriminates between truly active and inactive signals. In Section ??, we propose data-based methods to set block penalties in $\hat{S}^{ei}$ that do not require an oracle.

3.1 Oracle properties of externally-informed penalization

Consider the well-known reformulation of $\hat{S}^{ei}$ in linear regression (1). For any model $M \in \mathcal{M}$, let $\tilde{\boldsymbol{\beta}}^{(M)} = (\mathbf{X}_M^\top \mathbf{X}_M)^{-1} \mathbf{X}_M^\top \mathbf{y} \in \mathbb{R}^{|M|}$ be the MLE under model M , and denote

$$C(M) := \frac{1}{2} \|\mathbf{X}_M \tilde{\boldsymbol{\beta}}^{(M)}\|^2 - \sum_{j=1}^b \kappa_j |M_j| \quad \text{and} \quad NC(M) := \frac{e^{C(M)}}{\sum_{M' \in \mathcal{M}} e^{C(M')}}. \quad (11)$$

Lemma 3.1. $\hat{S}^{ei}$ satisfies $\hat{S}^{ei} \in \arg \max_{M \in \mathcal{M}} NC(M)$

By Lemma 3.1(i), $\hat{S}^{ei}$ selects $M \in \mathcal{M}$ with the largest normalized score $NC(M)$. By the BIC approximation in (9), $C(M)$ and $NC(M)$ can be understood as, respectively, the unscaled and scaled pseudo-posterior probability for model M in a Bayesian variable selection framework. Lemma 3.2 reproduced from ? proves that the expected sum of the normalized scores for $M \neq S$ bounds the probability of an incorrect model selection ($\hat{S}^{ei} \neq S$).

Lemma 3.2. (i) $P(\hat{S}^{ei} \neq S) \leq 2 \sum_{M \in \mathcal{M} \setminus \{S\}} \mathbb{E}(NC(M))$.

(ii) For any $M, M' \in \mathcal{M}$, such that $M \neq M'$, $NC(M) \leq (1 + e^{C(M') - C(M)})^{-1}$.

Using Lemma 3.2 (i) we show the variable selection consistency of $\hat{S}^{ei}$ by guaranteeing the vanishing, in expectation, of the sum of the normalized scores. We thus show selection consistency in a strong sense, that is L_1 convergence to S . Using Lemma 3.2 (ii) with $M' = S$, we bound $\mathbb{E}(NC(M))$ for $M \neq S$ by bounding the expectation of a simple function the pairwise comparison of the unscaled scores of M and the data-generating S .

We now state two conditions that are sufficient for asymptotically recovering S .

(A4) For each block j , there exists $f_j \rightarrow \infty$ such that for every sufficiently large n ,

$$\kappa_j = \ln(p_j - s_j) + f_j$$

(A5) For each block j , there exists $g_j \rightarrow \infty$ such that for every sufficiently large n ,

$$\sqrt{\frac{(1-\gamma)n\rho(\mathbf{X})}{6}}\beta_{\min,j}^* - \sqrt{\kappa_j} = \sqrt{\ln(s_j)} + g_j.$$

where $\gamma := \frac{1}{2}(1 + \max_j \ln(p_j - s_j)/\kappa_j) \in (\frac{1}{2}, 1)$.

Assumptions A4 is a requirement on the strength of block penalties and Assumptions A5 a requirement on signal strength, commonly called a betamin assumption. We note that in our proof one could take a betamin condition that is slightly less strict than Assumption A5, but Assumption A5 is presented here to facilitate interpretation and comparison to literature.

We can now state the main theorem on the strong variable selection consistency of $\hat{S}^{ei}$. The result holds for either fixed or diverging $p_j - s_j$ and s_j .

Theorem 3.3. *Under Assumptions A1, A3, A4 and, A5, it holds that*

$$\lim_{n \rightarrow \infty} \sum_{M \in \mathcal{M} \setminus \{S\}} \mathbb{E}(NC(M)) = 0 \quad \text{and} \quad \lim_{n \rightarrow \infty} P(\hat{S}^{ei} = S) = 1.$$

A bound on the convergence rate of $P(\hat{S}^{ei} \neq S)$ follows from the proof of Theorem 3.3, and is given in Theorem 3.4 below. In the statement, $\delta < 1$ should be understood as being arbitrarily close to 1.

Theorem 3.4. *Assume A1, A3 A4, A5. Then, there exists a constant $c > 0$ such that for all sufficiently large n and any $\delta \in (0, 1)$,*

$$P(\hat{S}^{ei} \neq S) \leq c^{ei} \sum_{j=1}^b e^{-\frac{\delta}{2}[\kappa_j - \ln(p_j - s_j)]} + e^{-\frac{\delta}{2}[(\sqrt{\frac{(1-\gamma)n\rho(\mathbf{X})}{6}}\beta_{\min,j}^* - \sqrt{\kappa_j})^2 - \ln(s_j)]}. \quad (12)$$

Moreover, suppose that for all $j = 1, \dots, b$,

$$\lim_{n \rightarrow \infty} \frac{\sqrt{(1-\gamma)n\rho(\mathbf{X})/6} \beta_{\min,j}^*}{\sqrt{\ln(p_j - s_j)} + \sqrt{\ln(s_j)}} > 1, \quad (13)$$

and the κ_j are set at the oracle values such that

$$\sqrt{\kappa_j^*} = \frac{1}{2} \sqrt{\frac{(1-\gamma)n\rho(\mathbf{X})}{6}} \beta_{\min,j}^* + \frac{1}{2} \sqrt{\frac{6}{(1-\gamma)n\rho(\mathbf{X})}} \frac{1}{\beta_{\min,j}^*} (\ln(p_j - s_j) - \ln(s_j)). \quad (14)$$

Then Assumptions A4 and A5 hold. Moreover, if Assumption A1 holds too, then for all sufficiently large n and any $\delta \in (0, 1)$,

$$P(\hat{S}^{ei} \neq S) \leq 2c \sum_{j=1}^b e^{-\frac{\delta}{2} \left[\frac{(1-\gamma)n\rho(\mathbf{X})}{24} \beta_{\min,j}^*{}^2 - \ln \max\{p_j - s_j, s_j\} \right]}. \quad (15)$$

♠ Paul: I replaced the factor in b before the sum by a constant to simplify the expression. Dependence on b being fixed is already apparent with the sum going from 1 to b . In (12) the first term of each summand decreases exponentially in κ_j , while the second term decreases exponentially in $(\sqrt{\frac{(1-\gamma)n\rho(\mathbf{X})}{6}} \beta_{\min,j}^* - \sqrt{\kappa_j})^2$. The choice $\kappa_j = \kappa_j^*$ ensures that both terms are equal and hence approximates the values minimizing (12). We refer to these ideal κ_j^* as *oracle penalties* because they depend on quantities s_j and $\beta_{\min,j}$ that are unknown in practice. XXX eq:oraclepenalty is an upper bound on the convergence rate xxxx for oracle choice, that eases interpretation and is tight under the slightly stricter betamin condition ... XXX.

The results of Theorem 3.4 are tight: when applied to non-informed ℓ_0 selector $\hat{S}^{sdsd}$, they nearly match the minimax results available in the literature for the orthonormal case where $\rho(\mathbf{X}) = 1$. In ?, it is shown that, in this case, the choice $\sqrt{\tilde{\kappa}} = (1/2)n/2\beta_{\min}^* + (1/2)\sqrt{2/n \ln(p/s - 1)}/\beta_{\min}^*$ is exactly minimax for the Hamming loss and asymptotically minimax for the probability of incorrect recovery, when all signals are parameters all positive or null. The minimax rate for the Hamming loss is then shown to have one term decreasing exponentially in $\tilde{\kappa} - \ln(p - s)$ and one term decreasing exponentially in $(\sqrt{n/2}\beta_{\min}^* - \sqrt{\tilde{\kappa}})^2 - \ln(s)$, similarly to our rate in (12). In Supplement XXX, we discuss sharp minimax results for $\hat{S}^{ei}$ in the orthonormal case.

We now contrast our sufficient Assumptions A4 and A5 to assumptions on κ_j and $\beta_{\min,j}$ that are necessary for consistent variable selection with $\hat{S}^{ei}$. For every $j = 1, \dots, b$, tight

necessary assumptions follow from considering subsets of models close to S ,

$$\mathcal{O}_j = \{M \in \mathcal{M} \mid M = S \cup \{i\} \text{ where } i \in B_j \setminus S_j\}, \text{ and} \quad (16)$$

$$\mathcal{U}_j = \{M \in \mathcal{M} \mid M = S \setminus \{i\} \text{ where } \beta_i = \beta_{\min,j}^*\}. \quad (17)$$

That is $\mathcal{O}_j$ is the set of model that over-fit by only one variable from block B_j and $\mathcal{U}_j$ is set of models that under-fits by only one of the variables associated to $\beta_{\min,j}^*$. The asymptotic recovery of S implies that, $\max_{M \in \mathcal{O}_j} NC(M)/NC(S) < 1$ and $\max_{M \in \mathcal{U}_j} NC(M)/NC(S) < 1$ with probability going to 1 as n grows, for every $j = 1, \dots, b$. We obtain bound on those two ratios that account for the dependence between models through the next two quantities

$$\underline{\lambda}_j := \lambda_{\min}(C^j) \text{ where } C_{k,l}^j = \text{corr}(Z_{M^k}, Z_{M^l}) \quad \forall M^k, M^l \in \mathcal{O}_j, \text{ and} \quad (18)$$

$$\bar{\lambda} := \lambda_{\max}\left(n^{-1} \mathbf{X}_S^\top \mathbf{X}_S\right), \quad (19)$$

where, for any $M_k \in \mathcal{M}$, $Z_{M^k}^2 = \|\mathbf{X}_{M^k} \tilde{\beta}^{(M^k)}\|^2 - \|\mathbf{X}_S \tilde{\beta}^{(S)}\|^2$. That is, small $\underline{\lambda}_j$ indicates that truly inactive variables in block j are highly correlated with each other, whereas $\underline{\lambda}_j = 1$ that they are uncorrelated. Similarly, $\bar{\lambda}$ relates to the level of correlation between active variables across all blocks.

Proposition 3.5. (i) *If for some $j = 1, \dots, b$, $\lim_{n \rightarrow \infty} \kappa_j / (\underline{\lambda}_j^2 \ln(p_j - s_j)) < 1$, then*

$$\lim_{n \rightarrow \infty} P\left(\max_{M \in \mathcal{O}_j} \frac{NC(M)}{NC(S)} < 1\right) = 0 \quad \text{and} \quad \lim_{n \rightarrow \infty} P(\hat{S}^{sdb} = S) = 0.$$

(ii) *Assume A1. If for some $j = 1, \dots, b$, $\kappa_j \rightarrow \infty$ and $\lim_{n \rightarrow \infty} \sqrt{n} \bar{\lambda} \beta_{\min,j}^* - \sqrt{2\kappa_j} < \infty$*

$$\text{then } \lim_{n \rightarrow \infty} P\left(\max_{M \in \mathcal{U}_j} \frac{NC(M)}{NC(S)} < 1\right) < 1 \text{ and } \lim_{n \rightarrow \infty} P(\hat{S}^{ei} = S) < 1.$$

(iii) *If for some $j \in \{1, \dots, b\}$, $\underline{\lambda}_j^2 \ln(p_j - s_j) \rightarrow \infty$ and $\lim_{n \rightarrow \infty} \sqrt{n} \bar{\lambda} \beta_{\min,j}^* -$*

$$\underline{\lambda}_j \sqrt{2 \ln(p_j - s_j)} < \infty, \text{ then, under Assumption A1 and A2, } \lim_{n \rightarrow \infty} P(\hat{S}^{ei} = S) < 1.$$

Proposition 3.5 (i) shows that the block penalties cannot be too small compared to the logarithm of the number of inactive parameters in the blocks. Proposition 3.5 (ii) further shows that the signals cannot be too small with respect to block penalties. Following from

Proposition 3.5 (i) and (ii), (iii) yields a necessary betamin assumption for consistency that is

$$\lim_{n \rightarrow \infty} \sqrt{n \bar{\lambda} \beta_{\min, j}^*} - \underline{\lambda}_j \sqrt{2 \ln(p_j - s_j)} = \infty \quad \text{for all } j = 1, \dots, b. \quad (20)$$

It also follows that Assumption A4 and A5 are tight. Indeed, it holds that $\underline{\lambda}_j \in [0, 1]$ and $\underline{\lambda}_j = 1$ corresponds to the worst case. Then, in that worst case, necessary assumption (??) matches sufficient assumption A4. Necessary assumption (20) resembles assumption A5 where $\rho(\mathbf{X})$ is replaced by $\bar{\lambda}$ and up to a constant factor, a factor logarithmic in s_j and factor $1 - \gamma$ which is bounded $(0, 1/2)$ if one sets f_j such that $\ln(p_j - s_j) = O(f_j)$. That is, assumption A5 and assumption (20) imply the same scaling of $(n, p_j, s_j, \beta_{\min, j}^*)$ up to a factor logarithmic in s_j . In Supplement XXX, we discuss how our sufficient and necessary assumptions, when applied to non-informed ℓ_0 selector $\hat{S}^{sd}$, improve on the known limits on variable selection in linear regression ??.

3.2 Benefits of externally-informed penalties

This section examines the benefits of externally-informed selector $\hat{S}^{ei}$ in relation to the relevance of the external information. We compare, for oracle-set penalties, the properties of $\hat{S}^{ei}$ to those of the standard ℓ_0 selector S , the best-known selector in absence of external information . We first discuss two types of benefits, softening the conditions for model selection consistency and improving the associated convergence rates. We then proceed with examples contemplating different scenarios of sparsity regime and informativeness of the blocks.

Assumptions A4–A5 give ranges of penalties that are sufficient (but not necessary) for asymptotic support recovery. For simplicity, the discussion is restricted to choices of penalty κ_j such that, in Assumption A4, $\ln(p_j - s_j) = O(f_j)$ for all j , so that γ is bounded in $(0, 1/2)$. For the standard selector $\hat{S}^{sd}$, the single penalty κ is essentially required to satisfy,

for some sequences f, g growing at arbitrarily slow rates, and up to constants

$$\sqrt{\ln(p-s)} + f \leq \sqrt{\kappa} \leq \sqrt{n\rho(\mathbf{X})\beta_{\min}^*} + \sqrt{\ln(s)} + g. \quad (21)$$

For a block selector $\hat{S}^{ei}$, the ranges for the κ_j 's essentially are, for some sequences f_j, g_j growing at arbitrarily slow rates and up to constants

$$\sqrt{\ln(p_j - s_j)} + f_j \leq \sqrt{\kappa_j} \leq \sqrt{n\rho(\mathbf{X})\beta_{\min,j}^*} - \sqrt{\ln(s_j)} + g_j. \quad (22)$$

The ranges in (22) imply that $\hat{S}^{sdi}$ requires milder conditions for selection consistency than $\hat{S}^{sd}$. Intuitively, if there exist two blocks such that the ranges in (22) do not overlap, then a global penalty κ cannot possibly satisfy (22) for all j and consistent selection is essentially not possible. For example, this occurs if the global smallest active signal $\beta_{\min}^*$ is in block b and satisfies $\beta_{\min}^* - \sqrt{(n\rho(\mathbf{X}))^{-1} \ln(s_b)} < \sqrt{\ln(p_1 - s_1)}$. Although not discussed here for brevity, block penalties also lead to similar improvements on the necessary assumptions, relative to those for standard ℓ_0 penalties. Corollary 3.6 gives conditions under which consistent selection is possible with $\hat{S}^{ei}$ but not with $\hat{S}^{sdsd}$.

Corollary 3.6. Assume A1, A2, A3, A4, A5 and $\underline{\lambda}^2 \ln(p-s) \rightarrow \infty$. If $\lim_{n \rightarrow \infty} \sqrt{n\bar{\lambda}}\beta_{\min}^* - \underline{\lambda}\sqrt{\ln(p-s)} < \infty$ then $P(\hat{S}^{sd} = S) \not\rightarrow 1$ and $P(\hat{S}^{ei} = S) \rightarrow 1$.

We now discuss differences in the probability of correct selection, when consistent selection is possible for both procedures. Assume that block penalties are set to their oracle values κ_j^* in (14) and similarly that the single penalty takes its oracle value κ^* defined in an analogous way. Let OR^{ei} be the oracle convergence rate for κ_j^* in (??) and OR_{orth} that for τ^* . Then, up to constant factors,

$$\frac{OR^{ei}}{OR} = \sum_{j=1}^b e^{-\frac{\delta}{2} \left[\frac{n\rho(\mathbf{X})}{24} ((1-\gamma)\beta_{\min,j}^* - (1-\gamma')\beta_{\min}^*)^2 + \ln \max\{p-s, s\} - \ln \max\{p_j-s_j, s_j\} \right]},$$

where $\gamma = \frac{1}{2}(1 + \max_j \ln(p_j - s_j)/\kappa_j^*)$ and $\gamma' = \frac{1}{2}(1 + \ln(p-s)/\kappa^*)$. Since $\beta_{\min,j}^* \geq \beta_{\min}^*$ and $\ln \max\{p-s, s\} \geq \ln \max\{p_j-s_j, s_j\}$, equation (??) shows that for every n large enough

$OR_{orth}^b \leq OR_{orth}$ for any partition in blocks, i.e. the oracle convergence rate in (??) for κ_j^* is never worse than for κ^* . The magnitude of the gain depends on how informative the blocks are. For any sparse setting where $s_j < p_j - s_j$ for every j , assuming that $\beta_{\min,b}^* = \beta_{\min}^*$ without loss of generality and that $\gamma = \gamma'$ for simplicity, it holds that

$$\frac{OR^{ei}}{OR} = \frac{p_b - s_b}{p - s} + \sum_{j < b} \frac{p_j - s_j}{p - s} e^{-\frac{n\rho(\mathbf{X})(1-\gamma)}{24}(\beta_{\min,j}^{*2} - \beta_{\min}^{*2})} \quad (23)$$

There are then two sources of improvement in convergence rates: in any block j , a smaller number of truly zero means, that is $p_j - s_j < p - s$, or a larger smallest signal, $\beta_{\min,j}^* > \beta_{\min}^*$. Depending on the sparsity of the block, one of the two effects prevails. In blocks where $\beta_{\min,j}^* \approx \beta_{\min}^*$, improvement comes mainly from having a smaller number of truly zero means. In those blocks, the oracle sets $\kappa_j^* < \kappa^*$, so that smaller signals are detected. If $\beta_{\min,j}^* \gg \beta_{\min}^*$, the oracle sets $\kappa_j^* > \kappa^*$ and the probability of false positives is reduced.

We illustrate benefits of $\hat{S}^{ei}$ with four concrete examples contemplating different scenarios of sparsity regimes and informativeness of the blocks, summarized in Table 1. We assume an orthonormal design such that $\rho(\mathbf{X}) = 1$ for simplicity and focus on a sparse setting where $\ln(s) = o(\ln(p - s))$, with two blocks ($b = 2$). The smallest non-zero $|\beta_i^*|$, $\beta_{\min}^*$ is assumed to be located in block 2, without loss of generality.

Table 1: Sparsity and block assumptions in Examples 1–4

Example	$p - s$	$p_1 - s_1$	$p_2 - s_2$	s_1	s_2
1	$3n/2$	$3n/2 - \sqrt{n}$	$\sqrt{n}$	$3 \ln(n)/2$	$3 \ln(n)/2$
2	$e^{n/20}$	$e^{n/20} - n^2$	n^2	$3 \ln(n)/2$	$3 \ln(n)/2$
3	n	$n - \ln(n)$	$\ln(n)$	$3 \ln(n)/2$	$3 \ln(n)/2$
4	n	$n/2$	$n/2$	$3 \ln(n)/2$	$3 \ln(n)/2$

In Examples 1 to 3, external information is discriminative as it singles out a block B_1 that is sparser than block B_2 . In Example 1 the blocks are moderately discriminative, in that

$(p_1 - s_1)/(p_2 - s_2)$ is a power of n . In Example 2 they are highly discriminative, since $(p_1 - s_1)/(p_2 - s_2)$ is exponential in n and in Example 3 they are highly discriminative in that β^* is sparse overall, but it is non-sparse within block 2 ($s_2 > p_2 - s_2$). Example 4 is a non-discriminative random guess where each block has half of the inactive parameters.

Figures ?? and ?? plot the ranges of penalty values κ and (κ_1, κ_2) ensuring selection consistency with $\hat{S}^{sd}$ and $\hat{S}^{sdb}$ given in (??) and (??) in the four examples, assuming $\beta_{\min,1}^* = 2/3$, $\beta_{\min,2}^* = \beta_{\min}^* = 1/10$, $f = g = \ln(\ln(n))$ and $f_j = g_j = \ln(\ln(n))$ for all j . In Example 2, asymptotic recovery with $\hat{S}^{sd}$ is not possible in the range of values of n considered while it is with $\hat{S}^{sdb}$. In the other examples, asymptotic recovery with $\hat{S}^{sd}$ is possible, but it requires larger n than with $\hat{S}^{sdb}$. Example 1 shows that the gain can be large even when blocks are moderately discriminative, and Example 3 when some blocks are non-sparse. The gain in terms of the value of n making recovery possible is close to null in Example 4 when external information is non-discriminative.

Figure ?? plots the ratio OR^{ei}/OR in (23) for Examples 1 to 3 where blocks are discriminative with γ set to $3/4$ for simplicity. The smallest signals in each block are such that $\beta_{\min}^* = \beta_{\min,2}^* = 1/5$ and $\beta_{\min,1}^* = 1.3\beta_{\min}^*$ to guarantee the recovery is possible with both $\hat{S}^{sdsd}$ and $\hat{S}^b$. Figure ?? shows that in the three examples, the convergence is much faster with $\hat{S}^b$. The more discriminative the blocks are, the larger the gain. ♠ Paul: I have made simplifications assumptions on γ in the discussions of the oracle convergence rates but in reality κ^* depends on γ and γ depends on κ^* .

3.3 Convergence with no betamin condition

We now derive a convergence result for pseudo-posterior probabilities $NC(M)$ defined in (38), $M \in \mathcal{M}$, under no requirement on signal strength other than minimal technical ones. The result generalizes Theorem 3.3 and is key to the proofs of Section 3.4. It shows that if

one sets sufficiently large penalties, all inactive and all small active signals are discarded. On the other hand, all large signals (relative to the specified penalties) are retained. We next formally define small and large signals. For fixed penalties κ_j 's, and for γ and g_j as defined in Assumption A5, let

$$\begin{aligned} S_j^S(\kappa) &:= \left\{ \beta_i^* \in S_j \mid \sqrt{n\lambda} |\beta_i^*| = o(\sqrt{\kappa_j}) \right\} \\ S_j^L(\kappa) &:= \left\{ \beta_i^* \in S_j \mid \sqrt{\frac{(1-\gamma)n\rho(\mathbf{X})}{6}} |\beta_i^*| - \sqrt{\kappa_j} = \sqrt{\ln(s_j)} + g_j \right\} \\ S_j^I(\kappa) &:= S_j \setminus (S_j^L(\kappa) \cup S_j^S(\kappa)). \end{aligned} \quad (24)$$

The subset $S_j^S(\kappa)$ gathers signals in S_j that are small with respect to the penalty κ_j , $S_j^L(\kappa)$ those that are large in that they satisfy Assumption A5, and $S_j^I(\kappa)$ those that are neither large nor small. Moreover, let $\mathcal{T}(\kappa)$ be the set of models that contain all large signals in $S^L(\kappa) = \cup_{j=1}^b S_j^L(\kappa)$, and neither truly inactive parameters in $V \setminus S$ nor small signals in $S^S(\kappa) = \cup_{j=1}^b S_j^S(\kappa)$. That is,

$$\mathcal{T}(\kappa) := \left\{ M \in \mathcal{M} \mid M = S^L(\kappa) \cup R, R \in \mathcal{P}(S^I(\kappa)) \right\}, \quad (25)$$

where for a set A , $\mathcal{P}(A)$ denotes the power set of A . We show that, if one sets sufficiently large penalties, pseudo-posterior probabilities concentrate on $\mathcal{T}(\kappa)$ and require the next mild technical assumption to do so,

(A6) For each block j , $|S^I(\kappa)| = O(1)$ and $|S_j^S(\kappa)| = O(p_j - s_j)$.

Theorem 3.7. Assume A1, A3, A4, and A6, for every $j = 1, \dots, b$, then,

$\lim_{n \rightarrow \infty} \sum_{M \in \mathcal{M} \setminus \mathcal{T}(\kappa)} \mathbb{E}(NC(M)) = 0$ and $\lim_{n \rightarrow \infty} P(\hat{S}^{ei} \in \mathcal{T}(\kappa)) = 1$.

Theorem 3.7 shows, assuming only sufficiently large penalties as given in Assumption A4, that posterior model pseudo-probabilities concentrate on $\mathcal{T}(\kappa)$. That is, all inactive and small active signals are discarded, large signals are retained and intermediate signals in $S^I(\kappa)$ may or may not be retained. Assumption $|S^I(\kappa)| = O(1)$ was made for simplicity, in

fact $|S^I(\kappa)|$ can be allowed to increase moderately.

3.4 Data analysis with externally-informed penalization

♠ Paul: I changed this part and kept only $\hat{S}^{A,ei}$ because I realised the comparison of the properties of $\hat{S}^{EB,ei}$ compared to those of $\hat{S}^{EB,sd}$ was more subtle than I thought and I constructed an example where $\hat{S}^{EB,sd}$ greatly outperforms $\hat{S}^{EB,ei}$ in terms of probability of recovery, but $\hat{S}^{A,sd}$ does not outperform $\hat{S}^{A,ei}$ too much. Keeping only one also saves space.

We propose a data analysis methods to set block penalties that does not require oracle values. The proposed method adapts to the relevance of the external-information which, as discussed in XXX, is driven by varying sparsity levels across blocks. Exploiting the connection between block ℓ_0 penalties and Bayesian variable selection, we use an empirical Bayes approach to estimate the proportion of truly active variables in each block and set penalties. Relying on such connection also allows the use of fast Bayesian computational methods that overcome the intractability of ℓ_0 penalties. Despite their Bayesian motivation, the methods is fully data-dependent and does not require any prior distribution.

Recall, by the BIC approximation in (9), for any model M , $NC(M)$ can be understood as the pseudo-posterior probability for model M in a Bayesian variable selection framework where the κ_j 's are a suitable function of the block prior inclusion probability θ_j . In that Bayesian framework, θ_j is the prior belief on s_j/p_j , the proportion of active coefficients in block B_j . A related standard empirical Bayes strategy to set $\hat{\theta}_j$ consists in maximizing the marginal likelihood of $\mathbf{y}$ given $\boldsymbol{\theta}$, i.e.

$$\hat{\boldsymbol{\theta}} := \arg \max_{\boldsymbol{\theta}} p(\mathbf{y} \mid \boldsymbol{\theta}) = \arg \max_{\boldsymbol{\theta}} \int p(\mathbf{y} \mid \boldsymbol{\beta}, M) dP(\boldsymbol{\beta}, M \mid \boldsymbol{\theta}).$$

The empirical Bayes estimator $\hat{\boldsymbol{\theta}}$ satisfies the property in the next result related to Proposition 3.1 in ?.

Lemma 3.8. For $p(M \mid \theta)$ in (8), the empirical Bayes estimator satisfies

$$\hat{\theta}_j = \frac{1}{p_j} \sum_{i \in B_j} \sum_{M \in \mathcal{M}: i \in M} P(M \mid y, \hat{\theta}).$$

Lemma and the interpretation, for any model M , of $NC(M)$ as the pseudo-posterior probability for model M , motivates estimating the number of active variables in block j by

$$\hat{s}_j := \sum_{i \in B_j} \sum_{M \in \mathcal{M}: i \in M} NC(M) \quad (26)$$

for a given set block penalties $\kappa_1, \dots, \kappa_b$. That is, $\hat{s}_j \approx p_j \hat{\theta}_j$.

The next asymptotic bounds on the expectation of $\hat{s}_j/p_j$ follow from Theorem 3.7.

Proposition 3.9. Assume A1, A3, A4, and A6. Then

$$\frac{|S_j^L(\kappa)|}{p_j} \leq \lim_{n \rightarrow \infty} \mathbb{E} \left(\frac{\hat{s}_j}{p_j} \right) \leq \frac{s_j - |S_j^S(\kappa)|}{p_j} \quad \text{for all } j = 1, \dots, b.$$

An immediate consequence is that, if the betamin condition in Assumption A5 also holds, then $\hat{s}_j/p_j \xrightarrow{L_1} s_j/p_j$ because $S_j = S_j^L(\kappa)$ and $S_j^S(\kappa) = \emptyset$ for all j . The proposition also shows that, when Assumption A7 is not met, $\hat{s}_j/p_j$ is asymptotically downward biased by at least $|S_j^S(\kappa)|/p_j$, but it is guaranteed to be larger than the proportion of signals that satisfy Assumption A5.

We propose a two-step procedure that uses $\hat{s}_j$ in (26) to set adaptive block penalties leveraging the difference in sparsity across the blocks induced by the external information. In the first step of the procedure, we use a standard (non-block-based) penalty κ° and estimate the number of active signals s_j in each block j with $\hat{s}_j$. In the second step, we use the estimated s_j to set block penalties. The variable selection consistency of the proposed procedure follows from the results of Section 3.1. While in Section 3.1 the gains in conditions for consistency were driven by how $p_j - s_j$ compared to $p - s$, here they are driven by how $p_j - |S_j^L(\kappa^\circ)|$ compares to $p - |S^L(\kappa^\circ)|$. This occurs because recovery of

small signals in Step 1 is not guaranteed. Our choice of block penalty is motivated by the joint consideration of the BIC approximation in (9) and the sufficient Assumption A6. It is natural to set $\theta_j \approx (p_j - s_j + 1)^{-1}$: this choice yields block penalties that are sufficiently large for Theorem 3.3 to hold, where f_j in Assumption A6 takes value $\ln(n)/2$ for all j .

Algorithm 2

- (i) Set $\kappa_j = \kappa^\circ = \ln(p) + \frac{1}{2} \ln(n)$ for $j = 1, \dots, b$. Compute $\hat{s}_j/p_j$ in (26) for $j = 1, \dots, b$.
- (ii) Obtain $\hat{S}^{A,ei}$ solving (10) with $\kappa_j^A = \ln(p_j - \hat{s}_j) + \frac{1}{2} \ln(n)$.

We refer to $\hat{S}^{A,ei}$ as the block adaptive selector. Step 1 approximates posterior model probabilities under equal prior inclusion probabilities $\theta_j = (p + 1)^{-1}$ in (8) across blocks, and estimates the proportion of truly active coefficients with $\hat{s}_j/p_j$. Step 1 can then be approximated by any Bayesian computational method for posterior model probabilities. In particular, in the examples of Section ??, we use an MCMC algorithm. Step 2 selects a model using the block penalties $\kappa_j^A = \ln(p_j - \hat{s}_j) + \frac{1}{2} \ln(n)$. Any fast computational method for the exact or approximate resolution of the ℓ_0 problem may be used for Step 2. Under mild assumptions κ_j^A leads to variable selection consistency. In Supplement XXX, we discuss the properties of an alternative choice for block penalties in step 2 that corresponds to directly setting prior inclusion probabilities with empirical Bayes, that is $\theta_j \approx \hat{\theta}_j$.

Theorem 3.10 below shows the consistency of $\hat{S}^{A,ei}$ under betamin assumption,

- (A9) For each block j , there exists $d_j \rightarrow \infty$ such that for all sufficiently large n ,

$$\sqrt{\frac{(1 - \xi)n\rho(\mathbf{X})}{2}}\beta_{\min,j}^* - \sqrt{\ln(p_j - |S_j^L(\kappa^\circ)|) + \frac{1}{2} \ln(n)} = \sqrt{\ln(s_j)} + d_j.$$

where $\xi = \frac{1}{2}(1 + \max_j \ln(p_j - s_j)/(\ln(p_j - s_j) + 0.5 \ln(n)))$.

Theorem 3.10. Assume A1, A3, A6, and A9, then $\lim_{n \rightarrow \infty} P(\hat{S}^{A,ei} = S) = 1$, and there

exists a constant $c > 0$ such that for all sufficiently large n and any $\delta \in (0, 1)$,

$$P(\hat{S}^{A,ei} \neq S) \leq c \sum_{j=1}^b e^{-\frac{\delta}{2} [\kappa_j^A - \ln(p_j - s_j)]} + e^{-\frac{\delta}{2} [(\sqrt{\frac{(1-\xi)n\rho(\mathbf{X})}{6}} \beta_{\min,j}^* - \sqrt{\kappa_j^A})^2 - \ln(s_j)]}. \quad (27)$$

Our proposed selector $\hat{S}^{A,ei}$ has similar but slightly stricter sufficient betamin condition than the oracle selector with penalties κ_j^* . Assumption A9 resembles the betamin assumption in (13) with $\ln(p_j - s_j)$ replaced by $\ln(p_j - |S_j^L(\kappa^\circ)|) + \frac{1}{2} \ln(n)$. By definition, $\hat{S}^{A,ei}$ has slower convergence rates than the oracle selector. It approximates better the oracle selector when signals are small. Indeed, when $\sqrt{n\rho(\mathbf{X})/6} \beta_{\min,j}^* \approx \ln(p_j/s_j - 1)$, the oracle penalty κ_j^* in (14) satisfies $\kappa_j^* \approx \ln(p_j/s_j - 1)$ and κ_j^A is larger but not too far from the oracle block penalties, in particular when $\ln(\sqrt{n}s_j) = o(\ln(p_j/s_j - 1))$. When $\sqrt{n\rho(\mathbf{X})/6} \beta_{\min,j}^* \gg \ln(p_j/s_j - 1)$, we have $\kappa_j^* \approx \sqrt{n\rho(\mathbf{X})/24} \beta_{\min,j}^*$ and then, $\kappa_j^* \gg \kappa_j^A$.

We compare next $\hat{S}^{A,ei}$ to a fairer competitor, its non-informed counterpart $\hat{S}^{sdA,sd}$ that sets in Step 2 a single common penalty $\kappa^A = \ln(p - \hat{s}^{sd}) + \frac{1}{2} \ln(n)$. By Theorem 3.10, $\hat{S}^{sdA}$ is variable selection consistent under the betamin assumption

$$\sqrt{\frac{(1-\xi')n\rho(\mathbf{X})}{2}} \beta_{\min}^* - \sqrt{\ln(p - |S^L(\kappa^\circ)|) + \frac{1}{2} \ln(n)} = \sqrt{\ln(s)} + d, \quad (28)$$

where $\xi' = \frac{1}{2}(1 + \ln(p - s)/(\ln(p - s) + 0.5 \ln(n)))$ and $d \rightarrow \infty$. We have that for any j , $\ln(p - |S^L(\kappa^\circ)|) \geq \ln(p_j - |S_j^L(\kappa^\circ)|)$ and $\ln(s) \geq \ln(s_j)$. The sufficient assumption for consistency is then milder for $\hat{S}^{A,ei}$ than for $\hat{S}^{A,sd}$, in particular in less sparse blocks where $p_j - |S_j^L(\kappa^\circ)| \ll p - |S^L(\kappa^\circ)|$. Although not developed here, necessary betamin assumptions are also milder for $\hat{S}^{A,ei}$ than for $\hat{S}^{A,sd}$.

Assume now that sufficient assumptions for both $\hat{S}^{A,ei}$ and $\hat{S}^{A,sd}$ to be consistent are met and denote their convergence rate in (??), $CR^{A,ei}$ and $CR^{A,sd}$ respectively. Assume without loss of generality that the smallest signal is in block b , $\beta_{\min,b}^* = \beta_{\min}^*$. In a small signal setting such that

$$(\beta_{\min}^* - \sqrt{\kappa_b^A})^2 - \ln(s_b) = \min_{j=1,\dots,b} \{(\beta_{\min,j}^* - \sqrt{\kappa_j^A})^2 - \ln(s_j), \kappa_j^A - \ln(p_j - s_j)\} \quad \text{and} \quad (\beta_{\min}^* - \sqrt{\kappa^A})^2 - \ln(s) = \min$$

we have

$$CR^{A,ei} \sim ce^{-\frac{\delta}{2}((\beta_{\min}^* - \sqrt{\kappa_b^A})^2 - \ln(s_b))} \quad \text{and} \quad ; CR^{A,sd} \sim ce^{-\frac{\delta}{2}((\beta_{\min}^* - \sqrt{\kappa^A})^2 - \ln(s))}$$

and then

$$\frac{CR^{A,ei}}{CR^{A,ds}} = e^{-\frac{\delta}{2}((\beta_{\min}^* - \sqrt{\kappa_b^A})^2 - (\beta_{\min}^* - \sqrt{\kappa^A})^2 + \ln(s/s_b))},$$

and $CR^{A,ei} < CR^{A,ds}$ since $\kappa^A > \kappa_b^A$ and $s > s_b$. The less sparse block b is, the smaller κ_b^A is compared to κ^A , the smaller $CR^{A,ei}$ is compared to $CR^{A,ds}$.

In a large signal setting such that

$$\kappa_b^A - \ln(p_b - s_b) = \min_{j=1, \dots, b} \left\{ (\beta_{\min,j}^* - \sqrt{\kappa_j^A})^2 - \ln(s_j), \kappa_j^A - \ln(p_j - s_j) \right\} \quad \text{and} \quad \kappa^A - \ln(p - s) = \min \left\{ (\beta_{\min}^* - \sqrt{\kappa^A})^2 \right\}$$

we have

$$\frac{CR^{A,ei}}{CR^{A,ds}} = e^{-\frac{\delta}{2}(\kappa_b^A - \kappa^A) + \ln((p-s)/(p_b-s_b))},$$

and $CR^{A,ei} > CR^{A,ds}$.

That is, $\hat{S}^{A,ei}$ recovers most of the benefits of oracle block penalties. It has milder sufficient conditions for variable selection consistency than its non-informed counterpart $\hat{S}^{A,sd}$. It also has faster convergence rates in small signal settings by setting lower penalties in the blocks that are less sparse. The more informative is the block partition, that is one or several blocks being significantly less sparse than the global set of variables, the greater the gains. However, in large signals settings, its convergence rate is slower than that of $\hat{S}^{A,sd}$. This is because the strength of the smallest signal cannot be reliably estimated and, unlike the oracle penalties κ_j^* , the κ_j^A do not grow with the $\beta_{\min,j}^*$. We show in our simulation study that this loss in probability of recovery that $\hat{S}^{A,ei}$ suffers in high signals settings is small when it is compared to $\hat{S}^{A,sd}$.

4 Transfer learning for variable selection

In this section we use the results of Section 3.1 to obtain the properties of the transfer learning Algorithm 1. This algorithm transfers the sets of variable selected in the source datasets, $\hat{S}^{(k)}$ $k = 1, \dots, K$, to ease variable selection in the target dataset. The set of variables in the target dataset V is partitioned in two blocks, $B_1 = \cup_{k=1}^K \hat{S}^{(k)}$ and $B_2 = V \setminus \cup_{k=1}^K \hat{S}^{(k)}$ and an externally-informed selector to select variables in the target dataset. In particular, we derive the properties Algorithm 1 where the externally-informed selector is taken to be $\hat{S}^{A,ei}$. For simplicity we assumed the set of variables to be same in the target and source datasets. This assumption can be relaxed in our results at the expense of more complicated expressions.

Consider the

The properties of Algorithm 1 with externally-informed selector $\hat{S}^{A,ei}$ follow from Theorem ??.

Consider the betamin assumption (A10) For each block j , there exists $d_j \rightarrow \infty$ such that for all sufficiently large n ,

$$\sqrt{\frac{(1-\xi)n\rho(\mathbf{X})}{6}}\beta_{\min,1}^* - \sqrt{\ln\left(\left|\cup_{k=1}^K (S^{(k)} \setminus S_1^L(\kappa^\circ))\right|\right) + \frac{1}{2}\ln(n)} = \sqrt{\ln\left(\left|\cup_{k=1}^K (\hat{S}^{(k)} \cap S)\right|\right) + d_1}, \text{ and } \sqrt{\frac{(1-\xi)n\rho(\mathbf{X})}{6}}\beta_{\min,1}^* + \sqrt{\ln\left(\left|\cup_{k=1}^K (S^{(k)} \setminus S_1^L(\kappa^\circ))\right|\right) + \frac{1}{2}\ln(n)} = \sqrt{\ln\left(\left|\cup_{k=1}^K (\hat{S}^{(k)} \cap S)\right|\right) + d_1},$$

where $\xi = \frac{1}{2}(1 + \max_j \ln(p_j - s_j)/(\ln(p_j - s_j) + 0.5 \ln(n)))$.

By definition of B_1 and B_2 , we have

$$s_1 = |B_1 \cap S| = |\cup_{k=1}^K (\hat{S}^{(k)} \cap S)|, \quad (29)$$

$$s_2 = |B_2 \cap S| = |\cap_{k=1}^K (S \setminus \hat{S}^{(k)})|, \quad (30)$$

$$p_1 - |S_1^L(\kappa^\circ)| = |B_1 \setminus S_1^L(\kappa^\circ)| = |\cup_{k=1}^K (\hat{S}^{(k)} \setminus S_1^L(\kappa^\circ))|, \quad (31)$$

$$p_2 - |S_2^L(\kappa^\circ)| = |B_2 \cap S_2^L(\kappa^\circ)^C| = |\cap_{k=1}^K (\hat{S}^{(k)} \cup S_2^L(\kappa^\circ))^C|. \quad (32)$$

$$(33)$$

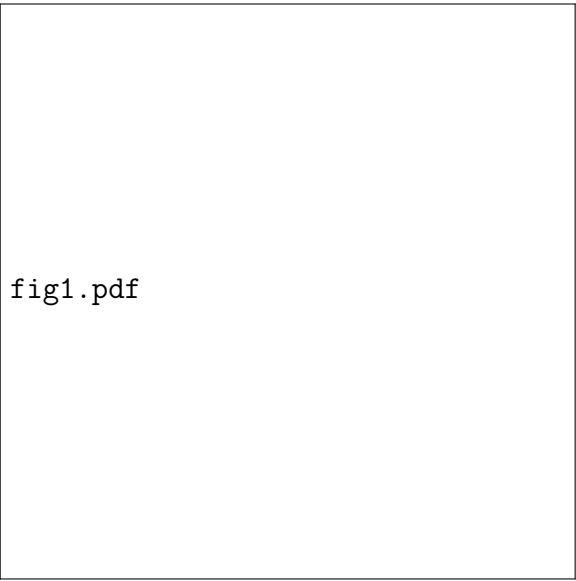


Figure 1: Consistency comparison in fitting surrogate model in the tidal power example.

Table 2: D-optimality values for design X under five different scenarios.

one	two	three	four	five
1.23	3.45	5.00	1.21	3.41
1.23	3.45	5.00	1.21	3.42
1.23	3.45	5.00	1.21	3.43

- hey

5 Methods

Don't take any of these section titles seriously. They're just for illustration.

6 Verifications

This section will be just long enough to illustrate what a full page of text looks like, for margins and spacing.

? offer some guidance about key ideas about statistical ideas. On an unrelated note, spreadsheets are important to use correctly (?). Log-linear models are an attractive way to model categorical data (?).

7 Disclosure statement

The authors have the following conflicts of interest to declare (or replace with a statement that no conflicts of interest exist).

8 Data Availability Statement

Deidentified data have been made available at the following URL: XX.

SUPPLEMENTARY MATERIAL

Title: Brief description. (file type)

R-package for MYNEW routine: R-package MYNEW containing code to perform the diagnostic methods described in the article. The package also contains all datasets used as examples in the article. (GNU zipped tar file)

HIV data set: Data set used in the illustration of MYNEW method in Section 6 (.txt file).

9 Tightness of conditions for variable selection consistency in linear regression

Sufficient and necessary assumptions for variable selection consistency for a (non-informed) oracle selector that knows s were derived by [? and ?](#). We compare those to our assumptions for the standard ℓ_0 selector $\hat{S}^{sd}$. Specifically, we compare the scaling of n implied by the assumptions under six regimes of interest of $(p, s, \beta_{\min}^*)$. For simplicity, the effect of the conditioning of $\mathbf{X}$ is disregarded, i.e. $\rho(\mathbf{X})$, $\bar{\lambda}$ and $\underline{\lambda}$ are considered fixed and bounded away from 0. As discussed in Section [??](#) Section [??](#), [?](#) and [?](#) derive sufficient and necessary assumptions in a setting that slightly differ from the one considered in this thesis. They study a slightly weaker notion of consistency and necessary conditions are given for the harder task of recovering S when the latter is chosen uniformly at random among subsets of size s . Although their assumptions and mines are then not entirely comparable, the comparison gives an idea of the tightness of our results.

Theorem [3.3](#) of this thesis shows variable selection consistency with $\hat{S}^{ei}$ under Assumptions [A4](#) and [A5](#). By [\(15pp.83\)](#) in the proof of Theorem [3.4](#), an assumption slightly less stringent than [A5](#), and easier to analyse, is sufficient together with [A4](#). That assumption is: for each block j , there exists $g_j \rightarrow \infty$ such that for every sufficiently large n ,

$$\frac{(1 - \gamma)n\rho(\mathbf{X})}{6} \beta_{\min,j}^{*2} - \kappa_j = \ln(s_j) + g_j.$$

where $\gamma := \frac{1}{2}(1 + \max_j \ln(p_j - s_j)/\kappa_j) \in (\frac{1}{2}, 1)$

Consider assumptions [A5](#) and [A15](#) for the standard ℓ_0 selector $\hat{S}^{sd}$ with single penalty κ . Their combination implies an assumption on the scaling of $(n, p, s, \beta_{\min}^*)$. To simplify the analysis of that assumption, we assume $\kappa = (1 + \varepsilon) \ln(p - s)$ for some fixed $\varepsilon > 0$. This choice guarantees that κ meets assumption [A4](#) and that $1 - \gamma$ is constant and bounded away from 0. A sufficient assumption on $(n, p, s, \beta_{\min}^*)$ that follows from assumptions [A5](#)

and (9) is then that there exists $t \rightarrow \infty$, growing at an arbitrarily slow rate, such that:

$$n = \frac{12(1+\varepsilon)}{\varepsilon} \frac{(1+\varepsilon) \ln(p-s) + \ln(s)}{\rho(\mathbf{X})\beta_{\min}^{*2}} + t \quad (34)$$

Theorem 1 in ? shows a sufficient assumption on $(n, p, s, \beta_{\min}^*)$ for the optimal selector that knows s to be variable selection consistent in random Gaussian designs is:

$$n > (c_1 + 2048) \max \left\{ \log \binom{p-s}{s}, \frac{\log(p-s)}{\rho(\mathbf{X})\beta_{\min}^{*2}} \right\} \quad (35)$$

for some $c_1 > 0$.

An immediate initial conclusion is that for any regime of $(n, p, s, \beta_{\min}^*)$ such that $\ln(s) = O(\beta_{\min}^{*2})$, (34) is less stringent than (35). It is also the case when $\beta_{\min}^{*2} = \Theta(1)$ and $s < p/2$. Table ?? gives, for six regimes of interest, the scalings of n implied by (34) and (35). In regimes 1, 2 and 5 the scalings implied by our sufficient assumptions match those implied by the sufficient assumption of ?. Our assumption is tighter in regimes 3 and 6, with the difference being particularly large in regime 3. In that regime, results by ? imply a sample size growing linearly in the dimension, while our results shows a logarithmic growth in dimension is sufficient. In regime 4 where $s = o(p)$, our assumption is loser by a term of order $s \ln(s)$. Overall, our results refine the known limits on variable selection in the standard non-informed setting and show a well calibrated standard ℓ_0 penalty performs as well as an optimal selector in the vast majority of regimes.

In this thesis we establish the necessity of assumption (20) which, when applied to $\hat{S}^{sd}$ ($\hat{S}^{ei}$ with $b = 1$), implies

$$\lim_{n \rightarrow \infty} \sqrt{n\bar{\lambda}\beta_{\min}^*} - \underline{\lambda}\sqrt{\ln(p-s)n} = \frac{\underline{\lambda}\sqrt{\ln(p-s)}}{\bar{\lambda}\beta_{\min}^{*2}} + t' \quad (36)$$

where $t' \rightarrow \infty$ and $1/t = o(\beta_{\min}^{*2})$. In other words, t must grow faster than $\beta_{\min}^{*2}$ shrinks. For comparison, Theorem 1 in ? gives, for the *harder* problem of recovering S when S is chosen at random in $\mathcal{M}$, the necessary assumption

$$n > \max \{f_1(p, s, \lambda), \dots, f_k(p, s, \lambda), s\} \quad (37)$$

where

$$f_m(p, s, \lambda) := \frac{\ln \binom{p-s+m}{m} - 1}{\frac{1}{2} \ln \left(1 + m \beta_{\min, j}^2 \left(1 - \frac{m}{p-s+m} \right) \right)} \quad \text{for } m = 1, \dots, s$$

Table ?? gives, for six regimes of interest, the scalings of n implied by (36) and (37).

Observe first that the scalings implied by our necessary assumption match those implied by our sufficient assumption in all regimes but 4 and 6. In regimes 4 and 6, where $s = o(p)$, our sufficient assumption is stricter by linear and logarithmic terms in the sparsity s . In regimes 1, 2 and 4, the scalings implied by our necessary assumption and that of ? match. In regime 5, our necessary assumption is tighter than that of ?. In regimes 3 and 6, the necessary assumptions of ? are stricter than mine and also stricter than our sufficient assumption. This may be explained by the harder nature of the recovery problem considered by ?.

10 Proofs

10.0.1 Proof strategy

We state a well-known reformulation of $\hat{S}^{ei}$ in linear regression (1). For any model $M \in \mathcal{M}$, let $\tilde{\boldsymbol{\beta}}^{(M)} = (\mathbf{X}_M^\top \mathbf{X}_M)^{-1} \mathbf{X}_M^\top \mathbf{y} \in \mathbb{R}^{|M|}$ be the MLE under model M , and denote

$$C(M) := \frac{1}{2} \|\mathbf{X}_M \tilde{\boldsymbol{\beta}}^{(M)}\|^2 - \sum_{j=1}^b \kappa_j |M_j| \quad \text{and} \quad NC(M) := \frac{e^{C(M)}}{\sum_{M' \in \mathcal{M}} e^{C(M')}}. \quad (38)$$

By Lemma 3.1, $\hat{S}^{ei}$ selects $M \in \mathcal{M}$ with the largest normalized score $NC(M)$. By the BIC approximation in (9), $C(M)$ and $NC(M)$ can be understood as, respectively, the unscaled and scaled pseudo-posterior probability for model M in a Bayesian variable selection framework. Proposition 1 in ?, reproduced below as Lemma 3.2 (i), proves that the expectation of those pseudo-posterior probabilities bounds the probability of an incorrect model selection ($\hat{S}^{ei} \neq S$).

we use Lemma 3.2 (i) to show the variable selection consistency of $\hat{S}^{ei}$ by guaranteeing the vanishing, in expectation, of the sum of the normalized scores. Such vanishing bears

resemblance to what is known as *strong selection consistency* property (?): the concentration of pseudo-posterior model probabilities on the truth S . We prove, however, L_1 convergence instead of convergence in probability. To bound $\mathbb{E}(NC(M))$ for $M \neq S$, by Lemma 3.2 (ii) with $M' = S$, it is enough to bound the expectation of a simple function of $C(S) - C(M)$, the pairwise comparison between each M and the data-generating S . This is achieved by noting that, directly by the definition in (38), $C(S) - C(M) = \frac{1}{2}L_{SM} + \Delta_{MS}$, where for any two models $M, T \subseteq V$, we denote

$$\Delta_{MT} := \sum_{j=1}^b \kappa_j (|M_j| - |T_j|) \quad \text{and} \quad L_{TM} := \|\mathbf{X}_T \tilde{\boldsymbol{\beta}}^{(T)}\|^2 - \|\mathbf{X}_M \tilde{\boldsymbol{\beta}}^{(M)}\|^2. \quad (39)$$

To lower bound $C(S) - C(M)$, we use that L_{SM} can be expressed in terms of chi-squared variables. The idea is to take the union model $Q_S = S \cup M$, and to note that $L_{SM} = L_{Q_S M} - L_{Q_S S}$. The next lemma from ? is used to bound $L_{Q_S M}$, $L_{Q_S S}$, and hence also L_{SM} .

Lemma 10.1. *Let M, Q be any two nested models such that $M \subseteq Q$. Then*

$$L_{QM} = \|\mathbf{X}_Q \tilde{\boldsymbol{\beta}}^{(Q)}\|^2 - \|\mathbf{X}_M \tilde{\boldsymbol{\beta}}^{(M)}\|^2 \sim \chi_{|Q \setminus M|}^2(\mu_{QM}),$$

where $\chi_k^2(\mu)$ denotes, when $\mu > 0$, the noncentral chi-squared distribution with k degrees of freedom and noncentrality parameter μ and, when $\mu = 0$, the chi-squared distribution with k degrees of freedom χ_k^2 . The parameter μ_{QM} is given by

$$\mu_{QM} := \|(I_n - P_M) \mathbf{X}_{Q \setminus M} \boldsymbol{\beta}_{Q \setminus M}^*\|^2 \quad (40)$$

where $P_M = \mathbf{X}_M (\mathbf{X}_M^\top \mathbf{X}_M)^{-1} \mathbf{X}_M^\top$.

If $M \supset S$ (over-fitted), then $Q_S = M$ and $\beta_{Q_S \setminus S}^* = \beta_{M \setminus S}^* = 0$, because any parameter outside the true support S is by definition 0. Moreover, $-L_{SM} = L_{Q_S S}$ and by Lemma 10.1, $-L_{SM} \sim \chi_{|Q_S \setminus S|}^2$ since $\beta_{Q_S \setminus S}^* = 0$. It also holds that $\Delta_{MS} > 0$. Using properties of the chi-squared distribution, we show that if one sets large enough κ_j (and thus Δ_{MS}), then $C(S) - C(M)$ is also large and $NC(M)$ vanishes. If $M \subset S$ (under-fitted), then $Q_S = S$, $L_{SM} = L_{Q_S M}$, and by Lemma 10.1 $L_{SM} \sim \chi_{|Q_S \setminus M|}^2(\mu_{Q_S M})$, which has expectation

$|Q_S \setminus M| + \mu_{Q_{SM}}$. If the noncentrality parameter $\mu_{Q_{SM}}$ is large enough, then L_{SM} and $C(S) - C(M)$ are also large, and, by the properties of the noncentral chi-squared distribution, $NC(M)$ vanishes. Lemma 10.2 below shows that large $\mu_{Q_{SM}}$ can be achieved by setting a betamin condition and an eigenvalue condition on $\mathbf{X}$ involving the following quantity,

$$\rho(\mathbf{X}) = \min_{M \in \mathcal{M}: M \not\supseteq S} \lambda_{\min} \left(\frac{1}{n} \mathbf{X}_{S \setminus M}^\top (I_n - P_M) \mathbf{X}_{S \setminus M} \right). \quad (41)$$

The quantity $\rho(\mathbf{X})$ is nonnegative and relates to how distinguishable the other models $M \in \mathcal{M}$ are from S (?). More specifically, $\frac{1}{n} \mathbf{X}_{S \setminus M}^\top (I_n - P_M) \mathbf{X}_{S \setminus M}$ is the sample covariance matrix of the residuals when regressing $\mathbf{X}_{S \setminus M}$ on $\mathbf{X}_M$. In an orthonormal case where $\mathbf{X}^\top \mathbf{X} = n\mathbb{I}_p$, then $\rho(\mathbf{X}) = 1$.

Lemma 10.2. *For any $M \in \mathcal{M}$, let $Q_S = S \cup M$. Then the non-centrality parameter (40) satisfies*

$$\mu_{Q_{SM}} \geq n \rho(\mathbf{X}) \sum_{j=1}^b |S_j \setminus M_j| \beta_{\min, j}^{*2}.$$

If $M \neq S$ is such that $M \not\supseteq S$ and $M \not\subset S$, simultaneously large enough κ_j and $\mu_{Q_{SM}}$ guarantee that $C(S) - C(M)$ is also large, and that $NC(M)$ vanishes.

11 BibTeX

We encourage you to use BibTeX. If you have, please feel free to use the package natbib with any bibliography style you're comfortable with. The .bst file agsm has been included here for your convenience.

12 Proofs

The appendix is organized as follows. Section 13 presents results auxiliary to the rest of the proofs, as well as the proofs of that auxiliary results. Section 14 gathers the proofs of Section ??, Section 15 the proofs of Section 3.1 and Section 16 the proofs of Section 3.4.

13 Auxiliary results

In this section we collect some technical results auxiliary to the rest of the proofs.

13.1 Tail bounds

Lemma 13.1. (i) If $\mathbf{y} \sim N_p(0, n^{-1}I_p)$, $p > 1$, and $a \geq \sqrt{2 \ln(p)/n}$,

$$P\left(\max_{i \in \{1, \dots, p\}} |y_i| > a\right) \leq \frac{e^{-\frac{n}{2}\left(a^2 - \frac{2 \ln(p)}{n}\right)}}{\sqrt{\pi \ln(p)}}.$$

(ii) If $y \sim N(\mu, \sigma^2)$ and $a \leq |\mu|$,

$$P(|y| > a) = P\left(z < \frac{|\mu| - a}{\sigma}\right) + P\left(z < -\frac{a + |\mu|}{\sigma}\right)$$

Proof. **Part (i)** By the union bound and the identical distribution of the y_i 's,

$$P\left(\max_{i \in \{1, \dots, p\}} |y_i| > a\right) \leq \sum_{i=1}^p P(|\sqrt{n}y_i| > \sqrt{na}) = p P(|z| > \sqrt{na})$$

where $z \sim N(0, 1)$. By symmetry and using the standard tail bound for standard normal,

$P(z \geq \delta) \leq (2\pi)^{-1/2} \delta^{-1} e^{-\delta^2/2}$ for $\delta \geq 0$, we obtain that

$$P\left(\max_{i \in \{1, \dots, p\}} |y_i| > a\right) \leq p \, 2 P(z > \sqrt{na}) \leq \frac{1}{\sqrt{\pi}} \frac{\sqrt{2}}{\sqrt{na}} p e^{-\frac{n}{2}a^2}.$$

Since $a \geq \sqrt{2 \ln(p)/n}$, we have that $\sqrt{2}/\sqrt{na} \leq 1/\sqrt{\ln(p)}$. Taking $a^2 = a^2 - \frac{2 \ln(p)}{n} + \frac{2 \ln(p)}{n}$,

we get

$$P\left(\max_{i \in \{1, \dots, p\}} |y_i| > a\right) \leq \frac{1}{\sqrt{\pi \ln(p)}} e^{-\frac{n}{2}\left(a^2 - \frac{2 \ln(p)}{n}\right)}.$$

Part (ii) We have

$$P(|y| > a) = P(y > a) + P(y < -a) = P\left(z > \frac{a - \mu}{\sigma}\right) + P\left(z < -\frac{a + \mu}{\sigma}\right)$$

where $z \sim N(0, 1)$. If $\mu > 0$, $P(z > \frac{a - \mu}{\sigma}) = P(z > \frac{a - |\mu|}{\sigma}) = P(z < \frac{|\mu| - a}{\sigma})$ by symmetry of the standard Gaussian, and $P(z < -\frac{a + \mu}{\sigma}) = P(z < -\frac{a + |\mu|}{\sigma})$, then $P(|y_i| > a) = P(z < \frac{|\mu| - a}{\sigma}) + P(z < -\frac{a + |\mu|}{\sigma})$. If $\mu \leq 0$, then $P(z > \frac{a - \mu}{\sigma}) = P(z > \frac{a + |\mu|}{\sigma}) = P(z < -\frac{a + |\mu|}{\sigma})$, by

symmetry of the standard Gaussian, and $P(z < -\frac{a+\mu}{\sigma}) = P(z < \frac{|\mu|-a}{\sigma})$. Then for any μ , $P(|y_i| > a) = P(z < \frac{|\mu|-a}{\sigma}) + P(z < -\frac{a+\mu}{\sigma})$. $\square$

Lemma 13.2. *For any $T \subseteq S$ and $M \in \mathcal{M}$ such that $T \not\subseteq M$, let $Q_T = M \cup T$. The non-centrality parameter defined in (40) satisfies:*

$$\mu_{Q_TM} \geq n \rho(\mathbf{X}) \sum_{j=1}^b |T_j \setminus M_j| \min_{i \in T_j \setminus M_j} \beta_i^{*2}. \quad (13pp.42)$$

Proof. The non-centrality parameter μ_{Q_TM} , as defined in (40), satisfies

$$\begin{aligned} \mu_{Q_TM} &= \|(I_n - P_M) \mathbf{X}_{Q_T \setminus M} \boldsymbol{\beta}_{Q_T \setminus M}^*\|^2 \\ &= \|(I_n - P_M) \mathbf{X}_{T \setminus M} \boldsymbol{\beta}_{T \setminus M}^*\|^2 \\ &= n \boldsymbol{\beta}_{T \setminus M}^{*\top} \left(\frac{1}{n} \mathbf{X}_{T \setminus M}^\top (I_n - P_M) \mathbf{X}_{T \setminus M} \right) \boldsymbol{\beta}_{T \setminus M}^* \\ &\geq n \lambda_{\min} \left(\frac{1}{n} \mathbf{X}_{T \setminus M}^\top (I_n - P_M) \mathbf{X}_{T \setminus M} \right) \|\boldsymbol{\beta}_{T \setminus M}^*\|^2, \end{aligned}$$

where the second equality follows from observing that $Q_T \setminus M = T \setminus M$. Since T is a subset of S , by reordering columns, $\mathbf{X}_{S \setminus M} = [\mathbf{X}_{T \setminus M}, \mathbf{X}_{S \setminus (M \cup T)}]$ and therefore $\frac{1}{n} \mathbf{X}_{T \setminus M}^\top (I_n - P_M) \mathbf{X}_{T \setminus M}$ is a principal submatrix of $\frac{1}{n} \mathbf{X}_{S \setminus M}^\top (I_n - P_M) \mathbf{X}_{S \setminus M}$. Hence, Cauchy's interlacing theorem gives that $\lambda_{\min} \left(\frac{1}{n} \mathbf{X}_{T \setminus M}^\top (I_n - P_M) \mathbf{X}_{T \setminus M} \right) \geq \lambda_{\min} \left(\frac{1}{n} \mathbf{X}_{S \setminus M}^\top (I_n - P_M) \mathbf{X}_{S \setminus M} \right)$. Finally, by definition of $\rho(\mathbf{X})$ in (41) we have that $\lambda_{\min} \left(\frac{1}{n} \mathbf{X}_{S \setminus M}^\top (I_n - P_M) \mathbf{X}_{S \setminus M} \right) \geq \rho(\mathbf{X})$, and further noting that $\|\boldsymbol{\beta}_{T \setminus M}^*\|^2 \geq \sum_{j=1}^b |T_j \setminus M_j| \min_{i \in T_j \setminus M_j} \beta_i^{*2}$ gives the desired result. $\square$

Lemma 13.3. *Let $W \sim \chi_\nu^2(\mu)$ with $\mu \geq 0$, then for any $w > \mu + \nu$*

$$P(W > w) \leq e^{-\left(\frac{w+\mu}{2} - \sqrt{2w(2\mu+\nu)-2\mu\nu-\nu^2}\right)}.$$

Moreover, assume w , ν and μ are functions of n such that w is increasing, $\nu = o(w)$, and $\mu = o(w)$. Then, for any $\phi \in (0, 1)$ and n large enough

$$P(W > w) \leq e^{-\phi \frac{w}{2}}.$$

Proof. By ?, Lemma 8.1 we have that for any $x > 0$

$$P(W > (\nu + \mu) + 2\sqrt{(\nu + 2\mu)x} + 2x) \leq e^{-x}.$$

The function $f : x \mapsto (\nu + \mu) + 2\sqrt{(\nu + 2\mu)x} + 2x$ is one-to-one between $\mathbb{R}^+$ and $(\nu + \mu, \infty)$.

Hence, we have that for any $w > \mu + \nu$,

$$P(W > w) \leq e^{-f^{-1}(w)} = e^{-\left(\frac{w+\mu}{2} - \sqrt{2w(2\mu+\nu) - 2\mu\nu - \nu^2}\right)}.$$

Observe that

$$\frac{w + \mu}{2} - \sqrt{2w(2\mu + \nu) - 2\mu\nu - \nu^2} = \frac{w}{2} \left(1 + \frac{\mu}{w} - \sqrt{\frac{8(2\mu + \nu)}{w} \left(1 - \frac{2\nu\mu - \nu^2}{2(2w\mu + w\nu)} \right)} \right).$$

Since $\nu = o(w)$ and $\mu = o(w)$ by assumption, we have $\frac{w+\mu}{2} - \sqrt{2w(2\mu + \nu) - 2\mu\nu - \nu^2} = \frac{w}{2}(1 + o(1))$. Therefore, for any $\phi \in (0, 1)$ and every n large enough.

$$P(W > w) \leq e^{-\phi \frac{w}{2}}.$$

□

Lemma 13.4. *Let $W \sim \chi_\nu^2(\mu)$ with $\mu > 0$. For any $w < \mu$,*

$$P(W < w) \leq \frac{e^{-\frac{1}{2}(\sqrt{\mu} - \sqrt{w})^2}}{(\mu/w)^{\nu/4}}.$$

Proof. The result follows directly from ?, Lemma S2. □

Lemma 13.5. *Let $W \sim \chi_\eta^2(\mu)$ with $\mu \geq 0$. Assume that h, c, η and μ are functions of n such that h is positive and increasing, c is positive, $\eta = o(c \log(h))$, and $\mu = o(c \log(h))$. Let $\bar{u}, \underline{u}$ in $(0, 1)$ such that $1 > \bar{u} > \underline{u} \geq \left(1 + h^\psi e^{-(\eta+\mu)/c}\right)^{-1}$ where $\psi \in (0, 1)$, then for every n large enough, we have*

$$\int_{\underline{u}}^{\bar{u}} P\left(W > c \log\left(\frac{h}{1/u - 1}\right)\right) du \leq \frac{1}{h^\psi} \left(\bar{u} - \underline{u} + \log\left(\frac{\bar{u}}{\underline{u}}\right)\right).$$

Proof. For any $u \in [\underline{u}, \bar{u}]$, we have $c \log\left(\frac{h}{1/u - 1}\right) \geq c \log\left(\frac{h}{1/\underline{u} - 1}\right)$. Since $\underline{u} \geq (1 + h^\psi e^{-(\eta+\mu)/c})^{-1}$ by assumption we also have that, for any $u \in (\underline{u}, \bar{u})$,

$$c \log\left(\frac{h}{1/u - 1}\right) \geq c(1 - \psi) \log(h) + \eta + \mu.$$

It follows,

$$\frac{\eta}{c \log\left(\frac{h}{1/u-1}\right)} \leq \frac{\eta}{c(1-\psi) \log(h) + \eta + \mu} \quad \text{and} \quad \frac{\mu}{c \log\left(\frac{h}{1/u-1}\right)} \leq \frac{\mu}{c(1-\psi) \log(h) + \eta + \mu}.$$

By assumption $\eta = o(c \log(h))$ and $\mu = o(c \log(h))$, then for any $u \in (\underline{u}, \bar{u})$, $\eta = o\left(c \log\left(\frac{h}{1/u-1}\right)\right)$ and $\mu = o\left(c \log\left(\frac{h}{1/u-1}\right)\right)$. By Lemma 13.3, for any $\psi \in (0, 1)$ and every n large enough,

$$\int_{\underline{u}}^{\bar{u}} P\left(W > c \log\left(\frac{h}{1/u-1}\right)\right) du < \frac{1}{h^\psi} \int_{\underline{u}}^{\bar{u}} (1/u - 1)^\psi du. \quad (13\text{pp.43})$$

Applying the change of variables $v = 1/u - 1$ to the integral on the right-hand side above gives

$$\int_{\underline{u}}^{\bar{u}} (1/u - 1)^\psi du = \int_{1/\bar{u}-1}^{1/\underline{u}-1} \frac{v^\psi}{(v+1)^2} dv. \quad (13\text{pp.44})$$

Rewrite $v^\psi = (v - 1 + 1)^\psi$. Since $\bar{u} < 1$, we have that for any $v > 1/\bar{u} - 1$, $v - 1 > -1$. Note that for any $x \geq -1$ and $r \in [0, 1]$ $(1+x)^r \leq 1+rx$. Then, for any $v > 1/\bar{u} - 1$, $v^\psi = (v - 1 + 1)^\psi \leq 1 + \psi(v - 1) \leq 1 + \psi(v + 1)$. Applying this last inequality to the right-hand side in (13pp.44) gives

$$\int_{\underline{u}}^{\bar{u}} (1/u - 1)^\psi du < \int_{1/\bar{u}-1}^{1/\underline{u}-1} \frac{1}{(v+1)^2} + \frac{\psi}{v+1} dv = \bar{u} - \underline{u} + \psi \log\left(\frac{\bar{u}}{\underline{u}}\right). \quad (13\text{pp.45})$$

The result follows inputting the bound from (13pp.45) in (13pp.43) and using that $\psi < 1$ and $\log(\bar{u}/\underline{u}) \geq 0$ ($\underline{u} \leq \bar{u}$). $\square$

Lemma 13.6. *Let $W \sim \chi_\eta^2(\mu)$ with $\mu \geq 0$. Assume that h , c , η and μ are functions of n such that h is positive and increasing, c is positive, $\eta = o(c \log(h))$, and $\mu = o(c \log(h))$. Then for any $\alpha \in (0, 1)$ and every large enough n , we have*

$$\int_0^1 P\left(W > c \log\left(\frac{h}{1/u-1}\right)\right) du = o(h^{-\alpha}).$$

Proof. Since a probability is bounded by 1, for any $a \in (0, 1)$,

$$\int_0^1 P\left(W > c \log\left(\frac{h}{1/u-1}\right)\right) du \leq 2a + \int_a^{1-a} P\left(W > c \log\left(\frac{h}{1/u-1}\right)\right) du. \quad (13\text{pp.46})$$

Take $a = \left(1 + h^\psi e^{-(\eta+\mu)/c}\right)^{-1}$ for some $\psi \in (\alpha, 1)$. By Lemma 13.5 with $\underline{u} = a$ and $\bar{u} = 1 - a$, we have that

$$\int_0^1 P\left(W > c \log\left(\frac{h}{1/u - 1}\right)\right) du \leq \frac{2}{1 + h^\psi e^{-(\eta+\mu)/c}} + \frac{1 - 2a + \log(h^\psi e^{-(\eta+\mu)/c})}{h^\psi}.$$

Since $1 + h^\psi e^{-(\eta+\mu)/c} > h^\psi e^{-(\eta+\mu)/c}$ and $1 - 2a - \frac{\eta+\mu}{c} \leq \log(h^\psi)$ for every n large enough, we have

$$\int_0^1 P\left(W > 2 \log\left(\frac{h}{1/u - 1}\right)\right) du \leq h^{-\psi} (2e^{(\eta+\mu)/c} + 2 \log(h^\psi))$$

We have that

$$\frac{h^{-\psi} 2e^{(\eta+\mu)/c}}{h^{-\alpha}} = e^{-(\psi-\alpha) \log(h) + \log(2) + (\eta+\mu)/c} = e^{-(\psi-\alpha) \log(h) \left(1 - \frac{\eta+\mu}{c(\psi-\alpha) \log(h)} - \frac{\log(2)}{(\psi-\alpha) \log(h)}\right)} \quad (13\text{pp.47})$$

and similarly that

$$\frac{h^{-\psi} 2 \log(h^\psi)}{h^{-\alpha}} = e^{-(\psi-\alpha) \log(h) \left(1 - \frac{\log(\psi \log(h))}{(\psi-\alpha) \log(h)} - \frac{\log(2)}{(\psi-\alpha) \log(h)}\right)}. \quad (13\text{pp.48})$$

Since $\alpha < \psi$ as stated above, and by assumption h is increasing, $\eta = o(c \log(h))$ and $\mu = o(c \log(h))$, both expressions in (13pp.47) and (13pp.48) vanish as n grows. Hence,

$$\int_0^1 P\left(W > 2 \log\left(\frac{h}{1/u - 1}\right)\right) du = o(h^{-\alpha}).$$

□

13.2 Bounds related to model normalized scores

Let $NC(M)$ be the normalized score for model M defined in (38), $\mathcal{M}$ the set of models under consideration, and μ_{QM} be the noncentrality parameter for any two nested models $Q \supseteq M$ defined in Lemma 10.1. Lemma 13.7 generalizes Lemma 3.2 and shows that the probability of not selecting a set of models is bounded above by the expected sum of the normalized scores of the models outside the set. Lemma 13.8 provides a bound on the expected normalized score of any $M \in \mathcal{M}$ based on the pairwise comparison $C(T) - C(M)$ (cf (38)) for some $T \subseteq S$. It essentially shows that, if the block penalties diverge and the

signals in $M \setminus T$ are small, $NC(M)$ is small. Lemma 13.9 gives, for any $T \subseteq S$ and $M \in \mathcal{M}$, upper and lower bounds on $\mu_{Q_T T}$ where $Q_T = M \cup T$.

Lemma 13.7. *For $\hat{S}^{ei}$ as in (10) and any k models $M^1, \dots, M^k$*

$$P(\hat{S}^{ei} \notin \{M^1, \dots, M^k\}) \leq (k+1) \sum_{M \in \mathcal{M} \setminus \{M^1, \dots, M^k\}} \mathbb{E}(NC(M)).$$

Proof. Suppose that $NC(M^1) + \dots + NC(M^k) > \frac{k}{k+1}$ then for any $M \notin \{M^1, \dots, M^k\}$, we have that

$$NC(M) = 1 - \sum_{M' \neq M} NC(M') < 1 - \sum_{M' \in \{M^1, \dots, M^k\}} NC(M') < \frac{1}{k+1}.$$

In addition, if $NC(M^1) + \dots + NC(M^k) > \frac{k}{k+1}$ then necessarily $\max_{l=1, \dots, k} NC(M_l) > \frac{1}{k+1} > NC(M)$ for any $M \notin \{M^1, \dots, M^k\}$, and therefore $\hat{S}^{ei} \in \{M^1, \dots, M^k\}$. Consequently,

$$P(\hat{S}^{ei} \notin \{M^1, \dots, M^k\}) \leq P\left(NC(M^1) + \dots + NC(M^k) \leq \frac{k}{k+1}\right).$$

Moreover, we have

$$P\left(NC(M^1) + \dots + NC(M^k) \leq \frac{k}{k+1}\right) = P\left(\sum_{M \in \mathcal{M} \setminus \{M^1, \dots, M^k\}} NC(M) \geq \frac{1}{k+1}\right).$$

The result follows from the Markov's inequality applied to the right-hand side above. $\square$

Lemma 13.8. *For any $T \subseteq S$ and $M \in \mathcal{M} \setminus \{T\}$, denote $Q_T = M \cup T$ and $A_T := \gamma \Delta_{MT} + \frac{1-\gamma}{6} \mu_{Q_T M}$ (cf (39) and (40)). Suppose that, for some $\gamma \in (1/2, 1]$, it holds that $A_T > 0$, $|M \setminus T| = o(A_T)$, and $\mu_{Q_T T} = o(A_T)$. For any $\psi \in (0, 1)$ and every n large enough,*

$$\mathbb{E}(NC(M)) \leq e^{-\psi A_T}.$$

Proof. Since $M \in \mathcal{M} \setminus \{T\}$, by Lemma 3.2 (ii), $NC(M) \leq (1 + e^{C(T)-C(M)})^{-1} \in [0, 1]$.

The first step of the proof is to use that for any random variable $Z \geq 0$ we have $\mathbb{E}(Z) = \int_0^\infty P(Z > u) du$, so that

$$\begin{aligned} \mathbb{E}(NC(M)) &\leq \int_0^1 P\left((1 + e^{C(T)-C(M)})^{-1} \geq u\right) du \\ &= \int_0^1 P\left(C(T) - C(M) \leq \ln\left(\frac{1}{u} - 1\right)\right) du \\ &= \int_0^1 P\left(-\frac{1}{2}L_{TM} \geq \Delta_{MT} - \ln\left(\frac{1}{u} - 1\right)\right) du. \end{aligned}$$

The second step of the proof is to use the union bound to upper bound the probability in the integrand above. Let $Q_T = M \cup T$ and recall that $L_{TM} = L_{Q_TM} - L_{Q_T T}$. For any γ

$$-\frac{1}{2}L_{TM} - (\Delta_{MT} - \ln(\frac{1}{u} - 1)) = \left(\frac{1}{2}L_{Q_T T} - \ln\left(\frac{e^{\gamma\Delta_{MT}}}{\frac{1}{u} - 1}\right) \right) - \left(\frac{1}{2}L_{Q_TM} + (1 - \gamma)\Delta_{MT} \right).$$

Observe that for any random variables U, V , and any $\epsilon, \gamma' \geq 0$, the event $\{U - V \geq 0\}$ implies $\{U \geq \gamma'\epsilon\} \cup \{V < \gamma'\epsilon\}$. Let $U = \frac{1}{2}L_{Q_T T} - \ln\left(\frac{e^{\gamma\Delta_{MT}}}{\frac{1}{u} - 1}\right)$ and $V = \frac{1}{2}L_{Q_TM} + (1 - \gamma)\Delta_{MT}$. Take $\epsilon = \mu_{Q_TM}$ and $\gamma' = \frac{1}{6}(1 - \gamma)$, and observe that $A_T := \gamma\Delta_{MT} + \frac{1-\gamma}{6}\mu_{Q_TM} = \gamma\Delta_{MT} + \gamma'\mu_{Q_TM}$.

We then have

$$\begin{aligned} \{U \geq \gamma'\epsilon\} &= \left\{ \frac{1}{2}L_{Q_T T} \geq \ln\left(\frac{e^{\gamma\Delta_{MT}}}{\frac{1}{u} - 1}\right) + \gamma'\mu_{Q_TM} \right\} = \left\{ \frac{1}{2}L_{Q_T T} \geq \ln\left(\frac{e^{A_T}}{\frac{1}{u} - 1}\right) \right\} \\ \{V < \gamma'\epsilon\} &= \left\{ \frac{1}{2}L_{Q_TM} < -(1 - \gamma)\Delta_{MT} + \gamma'\mu_{Q_TM} \right\} = \left\{ \frac{1}{2}L_{Q_TM} < \gamma'(\mu_{Q_TM} - 6\Delta_{MT}) \right\}. \end{aligned}$$

By the union bound we have that

$$\mathbb{E}(NC(M)) \leq \int_0^1 P\left(\frac{1}{2}L_{Q_T T} \geq \ln\left(\frac{e^{A_T}}{\frac{1}{u} - 1}\right)\right) du + P\left(\frac{1}{2}L_{Q_TM} < \gamma'(\mu_{Q_TM} - 6\Delta_{MT})\right). \quad (13pp.49)$$

The third and final step of the proof is to upper bound each of the terms in the right-hand side of (13pp.49). The intuition is that both T and M are nested within Q_T , and therefore $L_{Q_T T}$ and L_{Q_TM} follow chi-squared distributions. We first bound the first term. If $M \subset T$ then $Q_T = T$, $L_{Q_T T} = 0$, and this term is zero. Suppose now that $M \not\subset T$. Then, by Lemma 10.1, $L_{Q_T T} \sim \chi_{|Q_T \setminus T|}^2(\mu_{Q_T T})$ with $|Q_T \setminus T| = |M \setminus T|$. By assumption, $A_T > 0$, $|M \setminus T| = o(\ln(e^{A_T}))$ and $\mu_{Q_T T} = o(\ln(e^{A_T}))$, then by Lemma 13.6, for $\alpha \in (\psi, 1)$, and every n large enough,

$$\int_0^1 P\left(L_{Q_T T} > 2 \ln\left(\frac{e^{A_T}}{1/u - 1}\right)\right) du < e^{-\alpha A_T}. \quad (13pp.50)$$

We now bound the second term in (13pp.49). If $M \supset T$, then $Q_T = M$, $L_{Q_TM} = 0$, $\mu_{Q_TM} = 0$, and this term is zero. Alternatively, if $M \not\supset T$ then, by Lemma 10.1, $L_{Q_TM} \sim \chi_{|Q_T \setminus M|}^2(\mu_{Q_TM})$ with $|Q_T \setminus M| = |T \setminus M|$. Clearly, when $\mu_{Q_TM} \leq 6\Delta_{MT}$ this term is also zero, so suppose that $\mu_{Q_TM} > 6\Delta_{MT}$. We have

$$P(L_{Q_TM} < 2\gamma'(\mu_{Q_TM} - 6\Delta_{MT})) \leq P(L_{Q_TM} < 2\gamma'\mu_{Q_TM})$$

and, by Lemma 13.4 and using that $\gamma' \in (0, \frac{1}{12})$, we obtain that

$$P(L_{Q_TM} < 2\gamma'\mu_{Q_TM}) \leq (2\gamma')^{\frac{|T \setminus M|}{4}} e^{-\frac{1}{2}(1-\sqrt{2\gamma'})^2 \mu_{Q_TM}} \leq \left(\frac{1}{6}\right)^{\frac{|T \setminus M|}{4}} e^{-\frac{1}{2}(1-\sqrt{2\gamma'})^2 \mu_{Q_TM}}.$$

Since $\mu_{Q_TM} > 6\Delta_{MT}$, we get

$$A_T = \gamma'\mu_{Q_TM} + \gamma\Delta_{MT} < \frac{1-\gamma}{6}\mu_{Q_TM} + \frac{\gamma}{6}\mu_{Q_TM} = \frac{1}{6}\mu_{Q_TM} \leq \frac{1}{2}(1-\sqrt{2\gamma'})^2 \mu_{Q_TM},$$

where the last inequality follows from the fact that $\gamma' \in (0, \frac{1}{12})$. We then get

$$P(L_{Q_TM} < 2\gamma'(\mu_{Q_TM} - 6\Delta_{MT})) \leq \left(\frac{1}{4}\right)^{\frac{|T \setminus M|}{6}} e^{-A_T} \leq e^{-A_T} \leq e^{-\alpha A_T} \quad (13pp.51)$$

Summing the bounds in (13pp.50) and (13pp.51) gives that for every n large enough $\mathbb{E}(NC(M)) < 2e^{-\alpha A_T}$. Since $\psi < \alpha$, for every n large enough, we have that $\mathbb{E}(NC(M)) < e^{-\psi A_T}$ as claimed. $\square$

Lemma 13.9. *For any $T \subseteq S$ and $M \in \mathcal{M}$, let $Q_T = M \cup T$, and $\mu_{Q_T T}$ as defined in (40), then,*

$$\mu_{Q_T T} \leq n \bar{\lambda} \sum_{j=1}^b |(S_j \cap M_j) \setminus T_j| \max_{i \in (S_j \cap M_j) \setminus T_j} \beta_i^{*2} \quad \text{where} \quad \bar{\lambda} := \lambda_{\max}\left(\frac{1}{n} \mathbf{X}_S^\top \mathbf{X}_S\right) \quad (13pp.52)$$

and

$$\mu_{Q_T T} \geq n \lambda_{\min}\left(\frac{1}{n} \mathbf{X}_S^\top \mathbf{X}_S\right) \sum_{j=1}^b |(S_j \cap M_j) \setminus T_j| \min_{i \in (S_j \cap M_j) \setminus T_j} \beta_i^{*2} \quad (13pp.53)$$

Proof. Using the definition of $\mu_{Q_T T}$ in (40), we have that

$$\begin{aligned} \mu_{Q_T T} &= \beta_{Q_T \setminus T}^*{}^\top \mathbf{X}_{Q_T \setminus T}^\top (I_n - P_T) \mathbf{X}_{Q_T \setminus T} \beta_{Q_T \setminus T}^* \\ &= \beta_{M \setminus T}^*{}^\top \mathbf{X}_{M \setminus T}^\top (I_n - P_T) \mathbf{X}_{M \setminus T} \beta_{M \setminus T}^* \\ &= \beta_{(S \cap M) \setminus T}^*{}^\top \mathbf{X}_{(S \cap M) \setminus T}^\top (I_n - P_T) \mathbf{X}_{(S \cap M) \setminus T} \beta_{(S \cap M) \setminus T}^* \end{aligned} \quad (13pp.54)$$

where the second equality follows from $Q_T \setminus T = M \setminus T$ and the third equality from $\beta_{M \setminus S}^* = 0$.

We start by showing the upper bound in (13pp.52). Denote for any square matrix A , its largest eigenvalue $\lambda_{\max}(A)$. By (13pp.54), we have that

$$\mu_{Q_T T} \leq n \lambda_{\max}\left(\frac{1}{n} \mathbf{X}_{(S \cap M) \setminus T}^\top (I_n - P_T) \mathbf{X}_{(S \cap M) \setminus T}\right) \|\beta_{(S \cap M) \setminus T}^*\|^2.$$

Let $B := \frac{1}{n} \mathbf{X}_{(S \cap M) \setminus T}^\top (I_n - P_T) \mathbf{X}_{(S \cap M) \setminus T}$, $C := \frac{1}{n} \mathbf{X}_{(S \cap M) \cup T}^\top \mathbf{X}_{(S \cap M) \cup T}$ and $D := \frac{1}{n} \mathbf{X}_T^\top \mathbf{X}_T$. D is a principal submatrix of C and B is the Schur complement of D of C . The inverse B^{-1} is then a principal submatrix of C^{-1} , and by Cauchy's interlacing theorem we have that $\lambda_{\min}(B^{-1}) \geq \lambda_{\min}(C^{-1})$ and then $\lambda_{\max}(B) \leq \lambda_{\max}(C)$. Since $T \subseteq S$ by assumption, we also have that $(S \cap M) \cup T \subseteq S$, then by interlacing again $\lambda_{\max}(C) \leq \lambda_{\max}\left(\frac{1}{n} \mathbf{X}_S^\top \mathbf{X}_S\right) = \bar{\lambda}$. The upper bound in (13pp.52) follows from the latter inequality and also observing that $\|\beta_{(S \cap M) \setminus T}^*\|_2^2 \leq \sum_{j=1}^b |(S_j \cap M_j) \setminus T_j| \max_{i \in (S_j \cap M_j) \setminus T_j} \beta_i^{*2}$.

We now derive the lower bound in (13pp.53). By (13pp.54), we have that

$$\mu_{Q_T T} \geq n \lambda_{\min} \left(\frac{1}{n} \mathbf{X}_{(S \cap M) \setminus T}^\top (I_n - P_T) \mathbf{X}_{(S \cap M) \setminus T} \right) \|\beta_{(S \cap M) \setminus T}^*\|_2^2.$$

Recall that B^{-1} is a principal submatrix of C^{-1} , hence by interlacing $\lambda_{\max}(B^{-1}) \leq \lambda_{\max}(C^{-1})$ and $\lambda_{\min}(B) \geq \lambda_{\min}(C)$. Since $T \subseteq S$ by assumption, we have that $(S \cap M) \cup T \subseteq S$, and hence $\lambda_{\min}(C) \geq \lambda_{\min}(\frac{1}{n} \mathbf{X}_S^\top \mathbf{X}_S)$. The bound in (13pp.53) is obtained by using the latter inequality and noting that also $\|\beta_{(S \cap M) \setminus T}^*\|_2^2 \geq \sum_{j=1}^b |(S_j \cap M_j) \setminus T_j| \min_{i \in (S_j \cap M_j) \setminus T_j} \beta_i^{*2}$. $\square$

14 Proofs of Section ??

14.1 Proof of Proposition ??

First note that by discarding constant term

$$\arg \max_{M \in \mathcal{M}} \left\{ \max_{\beta \in \mathcal{L}_M} \ell(\mathbf{y}; \beta) - \sum_{j=1}^b \kappa_j |M_j| \right\} = \arg \min_{M \in \mathcal{M}} \left\{ \min_{\beta \in \mathcal{L}_M} \left\{ \frac{1}{2} \|\mathbf{y} - \sqrt{n} \beta\|^2 \right\} + \sum_{j=1}^b \kappa_j |M_j| \right\}$$

We also have $\min_{\beta \in \mathcal{L}_M} \frac{1}{2} \|\mathbf{y} - \sqrt{n} \beta\|^2 = \frac{1}{2} \|\mathbf{y} - \sqrt{n} \tilde{\beta}_M\|^2 = \frac{1}{2} \|\mathbf{y}\|^2 - \frac{1}{2} \|\sqrt{n} \tilde{\beta}_M\|^2$. The maximization in (10) can be then replaced with:

$$\arg \max_{M \in \mathcal{M}} \left\{ \frac{1}{2} \|\sqrt{n} \tilde{\beta}_M\|^2 - \sum_{j=1}^b \kappa_j |M_j| \right\}, \quad M = M_1 \cup \dots \cup M_b. \quad (14\text{pp.55})$$

Under the assumptions of (??), we can write

$$\frac{1}{2} \|\sqrt{n} \tilde{\beta}_M\|^2 - \sum_{j=1}^b \kappa_j |M_j| = \sum_{j=1}^b \sum_{i \in M_j} \left(\frac{n}{2} \tilde{\beta}_i^2 - \kappa_j \right).$$

Then (14pp.55) can be maximized with respect to each M_j by including $i \in \hat{S}_j$ whenever $n\tilde{\beta}_i^2 \geq 2\kappa_j$.

14.2 Proof of Lemma ??

Denote by $\hat{\beta}$ the minimizer of the problem (??). The MLE under the full model is $\tilde{\beta} = \frac{1}{\sqrt{n}}\mathbf{y}$ and the optimized function in (??) can be then rewritten as

$$\frac{1}{2}\mathbf{y}^\top \mathbf{y} + \frac{n}{2} \sum_{j=1}^b \sum_{i \in B_j} \left(\beta_i^2 - 2\tilde{\beta}_i \beta_i + 2\frac{\lambda_j}{n} |\beta_i| \right)$$

which we optimize with respect to each β_i separately. For any $i \in B_j$ we have $\hat{\beta}_i = 0$ if and only if $|\tilde{\beta}_i| \leq \lambda_j/n$. Similarly for (??), denoting $\hat{\beta}$ the minimizer of the problem, for any $i \in B_j$ we have $\hat{\beta}_i = 0$ if and only if $|\tilde{\beta}_i| \leq \lambda_j/(\lambda_i^\circ |n|)$. If $\beta^\circ = \tilde{\beta}$, then for any $i \in B_j$ $\hat{\beta}_i = 0$ if and only if $|\tilde{\beta}_i| \leq \sqrt{\lambda_j/n}$. If β° is a LASSO estimate with penalization λ° , then for any $i \in B_j$ $\hat{\beta}_i = 0$ if and only if $\tilde{\beta}_i^2 + \text{sign}(\lambda^\circ/n - \tilde{\beta}_i)\lambda^\circ/(n\tilde{\beta}_i) - \lambda_j/n \leq 0$. Equivalently, for any $i \in B_j$ $\hat{\beta}_i = 0$ if and only if $|\tilde{\beta}_i| \leq \frac{1}{2} \left(\lambda^\circ/n + \sqrt{(\lambda^\circ/n)^2 + 4\lambda_j/n} \right)$.

14.3 Proof of Proposition ??

14.3.1 Part (i)

By the union bound and by Lemma 13.1 (i),

$$P(\hat{S}^{ei} \not\subseteq S) \leq \sum_{j=1}^b P\left(\max_{i \in B_j \setminus S_j} |y_i/\sqrt{n}| > \tau_j\right) \leq \sum_{j=1}^b \frac{e^{-\frac{n}{2}\left(\tau_j^2 - \frac{2\ln(p_j - s_j)}{n}\right)}}{\sqrt{\pi \ln(p_j - s_j)}}. \quad (14pp.56)$$

By Assumption A4, the numerator on the right is bounded above by 1 for all sufficiently large n . It follows that

$$P(\hat{S}^{ei} \not\subseteq S) \leq \sum_{j=1}^b (\pi \ln(p_j - s_j))^{-1/2} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

14.3.2 Part (ii)

By independence, we have

$$P(\hat{S}^{ei} \subseteq S) = \prod_{j=1}^b P\left(\max_{i \in B_j \setminus S_j} |y_i/\sqrt{n}| \leq \tau_j\right).$$

Consider j such that $\lim_{n \rightarrow \infty} \frac{\sqrt{n}\tau_j}{\sqrt{2\ln(p_j - s_j)}} < 1$. In particular, there exists $c < 1$ such that, for all sufficiently large n , we have that $\tau_j \leq c\sqrt{2\ln(p_j - s_j)/n}$. Then, for any such n ,

$$\begin{aligned} P\left(\max_{i \in B_j \setminus S_j} |y_i/\sqrt{n}| \leq \tau_j\right) &\leq P\left(\max_{i \in B_j \setminus S_j} |y_i| \leq c\sqrt{2\ln(p_j - s_j)}\right) \\ &\leq P\left(\frac{1}{\sqrt{2\ln(p_j - s_j)}} \max_{i \in B_j \setminus S_j} y_i \leq c\right). \end{aligned}$$

On the other hand, results from extreme value theory, ? Example 4.4.1, show that

$$\frac{1}{\sqrt{2\ln(p_j - s_j)}} \max_{i \in B_j \setminus S_j} y_i \xrightarrow{P} 1$$

and so

$$P\left(\frac{1}{\sqrt{2\ln(p_j - s_j)}} \max_{i \in B_j \setminus S_j} y_i \leq c\right) \rightarrow 0,$$

which implies that $P(\hat{S}^{ei} \subseteq S) \rightarrow 0$.

14.4 Proof of Lemma ??

Consider the events $A := \{\min_{i \in \{1, \dots, s\}} |\mu_i| - \max_{i \in \{1, \dots, s\}} |y_i - \mu_i| > a\}$ and $B := \{\min_{i \in \{1, \dots, s\}} |y_i| > a\}$.

We first show that A implies B . A implies that, for all i , $(\min_{i \in \{1, \dots, s\}} |\mu_i|) - |y_i - \mu_i| > a$. By the triangle inequality, we have that $-|y_i - \mu_i| \leq |y_i| - |\mu_i|$ and therefore that

$$\text{for all } i, \quad (\min_{i \in \{1, \dots, s\}} |\mu_i|) - |y_i - \mu_i| \leq (\min_{i \in \{1, \dots, s\}} |\mu_i|) + |y_i| - |\mu_i| < |y_i|.$$

Then, A implies that, for all i , $|y_i| > a$, that is B . By the monotonicity of the probability P

$$P\left(\min_{i \in \{1, \dots, s\}} |\mu_i| - \max_{i \in \{1, \dots, s\}} |y_i - \mu_i| > a\right) \leq P\left(\min_{i \in \{1, \dots, s\}} |y_i| > a\right).$$

14.5 Proof of Lemma ??

Since the y_i 's are independent and identically distributed we have $P(\min_{i \in \{1, \dots, s\}} |y_i| > a) = P(|y_i| > a)^s = \exp(s \ln P(|y_i| > a))$ for any $i \in \{1, \dots, s\}$. Using that $\ln(1+x) < x$ for $x \in (-1, 0)$, we have

$$P(\min_{i \in \{1, \dots, s\}} |y_i| > a) \leq \exp(s(P(|y_i| > a) - 1)). \quad (14\text{pp.57})$$

To get a bound on $P(\min_{i \in \{1, \dots, s\}} |y_i| > a)$ it is then enough to bound $P(|y_i| > a)$ for any i .

By Lemma 13.1 (ii), for any $a < |\mu|$ we further have

$$\begin{aligned} P(|y_i| > a) &= P\left(z < \frac{|\mu| - a}{\sigma}\right) + 1 - P\left(z < \frac{a + |\mu|}{\sigma}\right) \\ &= P(z < 0) + P\left(0 < z < \frac{|\mu| - a}{\sigma}\right) + 1 - P(z < 0) - P\left(0 < z < \frac{a + |\mu|}{\sigma}\right) \\ &= 1 + P\left(0 < z < \frac{|\mu| - a}{\sigma}\right) - P\left(0 < z < \frac{a + |\mu|}{\sigma}\right). \end{aligned} \quad (14pp.58)$$

From ?, for any $\delta > 0$, $\frac{1}{2} \left(1 - e^{-\delta^2/2}\right)^{\frac{1}{2}} \leq P(0 < z < \delta) \leq \frac{1}{2} \left(1 - e^{-2\delta^2/\pi}\right)^{\frac{1}{2}}$. Thus, for any $\gamma, \delta > 0$, we have that

$$P(0 < z \leq \gamma) - P(0 < z \leq \delta) \leq \frac{1}{2} \left(1 - e^{-2\gamma^2/\pi}\right)^{\frac{1}{2}} - \frac{1}{2} \left(1 - e^{-\delta^2/2}\right)^{\frac{1}{2}}.$$

Applying the above to (14pp.58), we get

$$P(|y_i| > a) \leq 1 + \frac{1}{2} \left(\left(1 - e^{-2(|\mu| - a)^2/\pi\sigma^2}\right)^{\frac{1}{2}} - \left(1 - e^{-(a + |\mu|)^2/2\sigma^2}\right)^{\frac{1}{2}} \right).$$

Using that $x^{\frac{1}{2}} - y^{\frac{1}{2}} = \frac{x - y}{x^{\frac{1}{2}} + y^{\frac{1}{2}}}$ for $x, y > 0$, we get

$$P(|y_i| > a) \leq 1 - \frac{e^{-2(|\mu| - a)^2/\pi\sigma^2} - e^{-(a + |\mu|)^2/2\sigma^2}}{2 \left(\left(1 - e^{-2(|\mu| - a)^2/\pi\sigma^2}\right)^{\frac{1}{2}} + \left(1 - e^{-(a + |\mu|)^2/2\sigma^2}\right)^{\frac{1}{2}} \right)}. \quad (14pp.59)$$

Finally, inputting in the bound in (14pp.59) in (14pp.57) gives the desired inequality.

14.6 Proof of Proposition ??

14.6.1 Part (i)

By the union bound,

$$P(\hat{S}^{ei} \not\subseteq S) \leq \sum_{j=1}^b P\left(\min_{i \in S_j} |y_i/\sqrt{n}| \leq \tau_j\right).$$

By Lemma ??, for each j ,

$$P\left(\min_{i \in S_j} |y_i/\sqrt{n}| \leq \tau_j\right) \leq P\left(\max_{i \in S_j} |y_i/\sqrt{n} - \beta_i| \geq \beta_{\min, j}^* - \tau_j\right).$$

By Lemma 13.1 (i)

$$P(\hat{S}^{ei} \not\supseteq S) \leq \sum_{j=1}^b \frac{e^{-\frac{n}{2}((\beta_{\min,j}^* - \tau_j)^2 - \frac{2\ln(s_j)}{n})}}{\sqrt{\pi \ln(s_j)}}. \quad (14\text{pp.60})$$

By Assumption A5, the nominator on the right is bounded above by 1 for all sufficiently large n . It follows that, by Assumption A3

$$P(\hat{S}^{ei} \not\supseteq S) \leq \sum_{j=1}^b (\pi \ln(s_j))^{-1/2} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

14.6.2 Part (ii)

Take any $j = 1, \dots, b$ satisfying $\lim_{n \rightarrow \infty} \sqrt{n}(\beta_{\min,j}^* - \tau_j) / \sqrt{(\pi/2) \ln(s_j)} \leq 1$. Consider first the case $\beta_{\min,j}^* \leq \tau_j$. Then $\lim_{n \rightarrow \infty} \sqrt{n}(\beta_{\min,j}^* - \tau_j) < \infty$, and by Lemma ??, $\lim_{n \rightarrow \infty} P(\hat{S}^{ei} \supseteq S) < 1$. We now consider the case $\beta_{\min,j}^* > \tau_j$. By Lemma ??:

$$P\left(\min_{i \in S_j} \left| \frac{y_i}{\sqrt{n}} \right| > \tau_j\right) \leq \exp \left\{ -\frac{s_j}{2} \frac{e^{-\frac{2n}{\pi}(\beta_{\min,j}^* - \tau_j)^2} - e^{-\frac{n}{2}(\tau_j + \beta_{\min,j}^*)^2}}{\left(1 - e^{-2n(\beta_{\min,j}^* - \tau_j)^2/\pi}\right)^{\frac{1}{2}} + \left(1 - e^{-n(\tau_j + \beta_{\min,j}^*)^2/2}\right)^{\frac{1}{2}}} \right\}.$$

Let $a_n = 2 \left(1 - e^{-2n(\beta_{\min,j}^* - \tau_j)^2/\pi}\right)^{\frac{1}{2}} + 2 \left(1 - e^{-n(\tau_j + \beta_{\min,j}^*)^2/2}\right)^{\frac{1}{2}}$ and note that $a_n \in (0, 4]$.

Thus, to show that $P\left(\min_{i \in S_j} \left| \frac{y_i}{\sqrt{n}} \right| > \tau_j\right)$ is bounded away from 1, it is enough to show that

$$\lim_{n \rightarrow \infty} \frac{s_j}{4} (e^{-2n(\beta_{\min,j}^* - \tau_j)^2/\pi} - e^{-n(\tau_j + \beta_{\min,j}^*)^2/2}) > 0.$$

Since $\lim_{n \rightarrow \infty} \sqrt{n}(\beta_{\min,j}^* - \tau_j) / \sqrt{(\pi/2) \ln(s_j)} \leq 1$, we have that $\lim_{n \rightarrow \infty} s_j e^{-2n(\beta_{\min,j}^* - \tau_j)^2/\pi} \geq$

1. To conclude, we show that

$$\lim_{n \rightarrow \infty} s_j e^{-n(\tau_j + \beta_{\min,j}^*)^2/2} \leq c < 1$$

for some $c \in (0, 1)$. Take c such that $\frac{s_j}{p_j} \leq c$. Such c exists by our assumption $\frac{s_j}{p_j} < 1$. We equivalently need $\sqrt{n}(\tau_j + \beta_{\min,j}^*) \geq \sqrt{2 \ln(s_j/c)}$ for all n sufficiently large. To show that, note that, by assumption $\tau_j < \beta_{\min,j}^*$, and $s_j/c \leq p_j$, which gives

$$\frac{\tau_j + \beta_{\min,j}^*}{\sqrt{\frac{2 \ln(s_j/c)}{n}}} \geq \frac{2\tau_j}{\sqrt{\frac{2 \ln(p_j)}{n}}} = \frac{\sqrt{\frac{2 \ln(p_j - s_j)}{n}}}{\sqrt{\frac{2 \ln(p_j)}{n}}} \frac{2\tau_j}{\sqrt{\frac{2 \ln(p_j - s_j)}{n}}} = \sqrt{\frac{\ln(p_j - s_j)}{\ln(p_j)}} \frac{2\tau_j}{\sqrt{\frac{2 \ln(p_j - s_j)}{n}}}.$$

Since by assumption τ_j satisfies $\lim_{n \rightarrow \infty} \sqrt{n}\tau_j / \sqrt{2\ln(p_j - s_j)} \geq 1$, the second term on the right converges to something ≥ 2 , and it is enough to show that the first term converges to something $> 1/2$. Using that $\frac{s_j}{p_j} < c$, we get

$$\sqrt{\frac{\ln(p_j - s_j)}{\ln(p_j)}} = \sqrt{\frac{\ln(p_j) + \ln(1 - \frac{s_j}{p_j})}{\ln(p_j)}} \geq \sqrt{\frac{\ln(p_j) + \ln(1 - c)}{\ln(p_j)}} \rightarrow 1,$$

which concludes the proof.

14.7 Proof of Lemma ??

We have

$$\frac{\beta_{\min,j}^*}{\sqrt{\frac{2\ln(p_j - s_j)}{n}} + \sqrt{\frac{\pi \ln(s_j)}{2n}}} = \frac{\beta_{\min,j}^* - \tau_j + \tau_j}{\sqrt{\frac{2\ln(p_j - s_j)}{n}} + \sqrt{\frac{\pi \ln(s_j)}{2n}}}$$

and (??) implies that we have either $\lim_{n \rightarrow \infty} \sqrt{n}\tau_j / \sqrt{2\ln(p_j - s_j)} < 1$ or else $\lim_{n \rightarrow \infty} \sqrt{n}(\beta_{\min,j}^* - \tau_j) / \sqrt{(\pi/2)\ln(s_j)} < 1$.

If $\lim_{n \rightarrow \infty} \sqrt{n}\tau_j / \sqrt{2\ln(p_j - s_j)} < 1$ by Proposition ?? (ii), $\lim_{n \rightarrow \infty} P(\hat{S}^{ei} \subseteq S) < 1$ and recovery is not possible asymptotically. Suppose now that $\lim_{n \rightarrow \infty} \sqrt{n}\tau_j / \sqrt{2\ln(p_j - s_j)} \geq 1$, then it holds that $\lim_{n \rightarrow \infty} \sqrt{n}(\beta_{\min,j}^* - \tau_j) / \sqrt{\pi \ln(s_j)/2} < 1$. By Proposition ?? (ii), $\lim_{n \rightarrow \infty} P(\hat{S}^{ei} \supseteq S) < 1$ and recovery is not possible asymptotically.

14.8 Proof of Theorem ??

We start by showing the bound in (??). By the union bound,

$$P(S \neq \hat{S}^{ei}) \leq P(\hat{S}^{ei} \not\supseteq S) + P(\hat{S}^{ei} \not\subseteq S) \quad (14\text{pp.61})$$

By the union bound and by Lemma 13.1 (i), for n large enough,

$$P(\hat{S}^{ei} \not\subseteq S) \leq \sum_{j=1}^b P\left(\max_{i \in B_j \setminus S_j} |y_i| / \sqrt{n} > \tau_j\right) \leq \sum_{j=1}^b \frac{e^{-\frac{n}{2}\left(\tau_j^2 - \frac{2\ln(p_j - s_j)}{n}\right)}}{\sqrt{\pi \ln(p_j - s_j)}}. \quad (14\text{pp.62})$$

where the assumption of Lemma 13.1 (i) is met because Assumption A4 is assumed to hold. Moreover, by the union bound, $P(\hat{S}^{ei} \not\supseteq S) \leq \sum_{j=1}^b P(\min_{i \in S_j} |y_i| / \sqrt{n} \leq \tau_j)$. By

Assumption A5, $\beta_{\min,j} > \tau_j$ and by Lemma ??, for each j ,

$$P\left(\min_{i \in S_j} |y_i/\sqrt{n}| \leq \tau_j\right) \leq P\left(\max_{i \in S_j} |y_i/\sqrt{n} - \beta_i| \geq \beta_{\min,j}^* - \tau_j\right).$$

By Assumption A5, $\beta_{\min,j} - \tau_j \geq \sqrt{2 \ln(s_j)/n}$ and by Lemma 13.1 (i)

$$P(\hat{S}^{ei} \not\subseteq S) \leq \sum_{j=1}^b \frac{e^{-\frac{n}{2} \left((\beta_{\min,j}^* - \tau_j)^2 - \frac{2 \ln(s_j)}{n} \right)}}{\sqrt{\pi \ln(s_j)}}. \quad (14\text{pp.63})$$

Inputting the bounds in (14pp.56) and (14pp.60) into (14pp.61) gives

$$P(S \neq \hat{S}^{ei}) \leq \sum_{j=1}^b \frac{e^{-\frac{n}{2} \left(\tau_j^2 - \frac{2 \ln(p_j - s_j)}{n} \right)}}{\sqrt{\pi \ln(p_j - s_j)}} + \sum_{j=1}^b \frac{e^{-\frac{n}{2} \left((\beta_{\min,j}^* - \tau_j)^2 - \frac{2 \ln(s_j)}{n} \right)}}{\sqrt{\pi \ln(s_j)}}. \quad (14\text{pp.64})$$

which shows (??).

We continue with the second part of the theorem and show that if (??) holds then τ_j^* satisfies Assumptions A4 and A5. Under (??), there exists $c > 1$ such that $\beta_{\min,j}^* = c(\sqrt{\frac{2 \ln(p_j - s_j)}{n}} + \sqrt{\frac{2 \ln(s_j)}{n}})$. Denote $a = \sqrt{\frac{2 \ln(p_j - s_j)}{n}}$ and $b = \sqrt{\frac{2 \ln(s_j)}{n}}$. We have

$$\tau_j^* = \frac{c}{2}(a+b) + \frac{(a^2 - b^2)}{2c(a+b)} = \frac{c}{2}(a+b) + \frac{a-b}{2c} = \frac{c^2+1}{2c}a + \frac{c^2-1}{2c}b.$$

Since $\frac{c^2+1}{2c} > 1$, $\frac{c^2-1}{2c} > 0$ and $b \geq 0$, $\tau_j^* > a = \sqrt{\frac{2 \ln(p_j - s_j)}{n}}$ and so Assumption A4 holds for the τ_j^* . To show that the upper bound in Assumption A5 also holds, observe that

$$\beta_{\min,j}^* - \tau_j^* = c(a+b) - \frac{c^2+1}{2c}a - \frac{c^2-1}{2c}b = \frac{c^2-1}{2c}a + \frac{c^2+1}{2c}b > b = \sqrt{\frac{2 \ln(s_j)}{n}}.$$

We proceed with the proof of the bound in (??). Since Assumptions A4 and A5 hold for the τ_j^* 's, for all sufficiently large n , (14pp.64) holds for these oracle penalties. Moreover, by assumption $p_j - s_j > 1$ and $s_j > 1$, then $\sqrt{\pi \ln(p_j - s_j)} > 1$, and $\sqrt{\pi \ln(s_j)} > 1$ and we get the bound:

$$P(S \neq \hat{S}^{ei}) \leq \sum_{j=1}^b e^{-\frac{n}{2} \left(\tau_j^{*2} - \frac{2 \ln(p_j - s_j)}{n} \right)} + \sum_{j=1}^b e^{-\frac{n}{2} \left((\beta_{\min,j}^* - \tau_j^*)^2 - \frac{2 \ln(s_j)}{n} \right)}.$$

Simple algebra shows that the τ_j^* satisfies $\tau_j^{*2} - \frac{2 \ln(p_j - s_j)}{n} = (\beta_{\min,j}^* - \tau_j^*)^2 - \frac{2 \ln(s_j)}{n}$ for all j .

It follows that

$$P(S \neq \hat{S}^{ei}) \leq 2 \sum_{j=1}^b e^{-\frac{n}{2} \left(\tau_j^{*2} - \frac{2 \ln(p_j - s_j)}{n} \right)}. \quad (14\text{pp.65})$$

For convenience, denote $d = \frac{1}{n\beta_{\min,j}^*} \ln(p_j/s_j - 1)$ such that $\tau_j^* = \beta_{\min,j}^*/2 + d$. Then

$$e^{-\frac{n}{2}\left(\tau_j^{*2} - \frac{2\ln(p_j-s_j)}{n}\right)} = e^{-\left[\frac{n}{8}\beta_{\min,j}^{*2} + \frac{n}{2}d^2 - \frac{1}{2}(\ln(p_j-s_j) + \ln(s_j))\right]}.$$

By considering separately the two possible maxima in $\ln \max\{p_j - s_j, s_j\}$, we get that

$$\frac{e^{-\frac{n}{2}\left(\tau_j^{*2} - \frac{2\ln(p_j-s_j)}{n}\right)}}{e^{-\left[\frac{n}{8}\beta_{\min,j}^{*2} - \ln \max\{p_j-s_j, s_j\}\right]}} = e^{-\left[\frac{n}{2}d^2 + \frac{1}{2}|\ln(p_j-s_j) - \ln(s_j)|\right]} < 1.$$

From (14pp.65), we then have

$$P(\hat{S}^{ei} \neq S) \leq 2 \sum_{j=1}^b e^{-\left[\frac{n}{8}\beta_{\min,j}^{*2} - \ln \max\{p_j-s_j, s_j\}\right]}.$$

which proves (??).

14.9 Proof of Corollary ??

Since $\hat{S}^{sd}$ is $\hat{S}^{ei}$ with $b = 1$, the assumptions of Lemma ?? for $\hat{S}^{sd}$ are met and $\lim_{n \rightarrow \infty} P(\hat{S}^{sd} = S) < 1$. Since Assumptions A4 and A5 hold, by Propositions ?? (i) and ?? (i), $\lim_{n \rightarrow \infty} P(\hat{S}^{ei} \subseteq S) = 1$ and $\lim_{n \rightarrow \infty} P(\hat{S}^{ei} \supseteq S) = 1$, and then $\lim_{n \rightarrow \infty} P(\hat{S}^{ei} = S) = 1$.

14.10 Proof of Proposition ??

By the union bound,

$$P(\hat{S}^{ei} \not\subseteq S) \leq \sum_{j=1}^b P\left(\min_{i \in S_j} |y_i/\sqrt{n}| \leq \tau_j\right).$$

By Lemma ??, for each j ,

$$P\left(\min_{i \in S_j} |y_i/\sqrt{n}| \leq \tau_j\right) \leq P\left(\max_{i \in S_j} |y_i/\sqrt{n} - \beta_i| \geq \beta_{\min,j}^* - \tau_j\right).$$

By Lemma 13.1 (i), we have;

$$P(\hat{S}^{ei} \not\subseteq S) \leq \sum_{j=1}^b \frac{e^{-\frac{n}{2}\left((\beta_{\min,j}^* - \tau_j)^2 - \frac{2\ln(s_j)}{n}\right)}}{\sqrt{\pi \ln(s_j)}}. \quad (14pp.66)$$

Under Assumptions A3bis and A5bis, the right-hand side vanishes, which proves part (i).

We now prove part (ii). By Proposition ?? (i), since A4 is assumed to hold, we also have $\lim_{n \rightarrow \infty} P(\hat{S}^{ei} \subseteq S) = 1$. This implies that $\lim_{n \rightarrow \infty} P(\hat{S}^{ei} = S) = 1$.

14.11 Proof of Lemma ??

Take any j , satisfying (??). Denote by $i_o \in B_j$ an entry such that $|\beta_{i_o}^*| = \beta_{\min,j}^*$. The event $\hat{S}^{ei} \supseteq S$ implies that $|y_{i_o}|/\sqrt{n} > \tau_j$. It follows that

$$P(\hat{S}^{ei} \supseteq S) \leq P(|y_{i_o}| > \sqrt{n}\tau_j). \quad (14pp.67)$$

Consider first the case where $\tau_j > \beta_{\min,j}^*$. Using that $|y_{i_o}| = |y_{i_o} - \sqrt{n}\beta_{\min,j}^* + \sqrt{n}\beta_{\min,j}^*| \leq |y_{i_o} - \sqrt{n}\beta_{\min,j}^*| + \sqrt{n}\beta_{\min,j}^*$, we get that

$$P(|y_{i_o}| > \sqrt{n}\tau_j) \leq P(|y_{i_o} - \sqrt{n}\beta_{\min,j}^*| > \sqrt{n}(\tau_j - \beta_{\min,j}^*)) = 2P(z > \sqrt{n}(\tau_j - \beta_{\min,j}^*))$$

where $z \sim N(0, 1)$ and the equality follows from the symmetry of the standard Gaussian distribution. Since $\tau_j > \beta_{\min,j}^*$, $P(z > \sqrt{n}(\tau_j - \beta_{\min,j}^*)) < 1/2$, $P(|y_{i_o}| > \sqrt{n}\tau_j) < 1$ and $P(\hat{S}^{ei} \supseteq S) < 1$ too by (14pp.67).

Consider next the case where $\tau_j \leq \beta_{\min,j}^*$. By Lemma 13.1 (ii), we have

$$P(|y_{i_o}| > \sqrt{n}\tau_j) = P(z > \sqrt{n}(\tau_j - \beta_{\min,j}^*)) + P(z < -\sqrt{n}(\tau_j + \beta_{\min,j}^*))$$

where $z \sim N(0, 1)$. Since $\tau_j \rightarrow \infty$, $\sqrt{n}(\tau_j + \beta_{\min,j}^*) \rightarrow \infty$ and we have that $P(z < -\sqrt{n}(\tau_j + \beta_{\min,j}^*)) \rightarrow 0$. Further, by (??) we have that $\lim_{n \rightarrow \infty} P(z > \sqrt{n}(\tau_j - \beta_{\min,j}^*)) < 1$ and hence we get $\lim_{n \rightarrow \infty} P(\hat{S}^{ei} \supseteq S) < 1$, as claimed.

14.12 Proof of Lemma ??

First, we re-write

$$\sqrt{n}\beta_{\min,j}^* - \sqrt{2\ln(p_j - s_j)} = \sqrt{n}\beta_{\min,j}^* - \sqrt{n}\tau_j + \sqrt{2\ln(p_j - s_j)}\left(\frac{\sqrt{n}\tau_j}{\sqrt{2\ln(p_j - s_j)}} - 1\right).$$

Condition (??) implies that there exists $c \in \mathbb{R}^+$ such that

$$\lim_{n \rightarrow \infty} \sqrt{n}\beta_{\min,j}^* - \sqrt{n}\tau_j + \sqrt{2\ln(p_j - s_j)}\left(\frac{\sqrt{n}\tau_j}{\sqrt{2\ln(p_j - s_j)}} - 1\right) \leq c \quad (14pp.68)$$

Consider the case $\lim_{n \rightarrow \infty} \frac{\sqrt{n}\tau_j}{\sqrt{2\ln(p_j - s_j)}} - 1 < 0$. Then by Proposition ?? (ii), $\lim_{n \rightarrow \infty} P(\hat{S}^{ei} \subseteq S) < 1$ and $\lim_{n \rightarrow \infty} P(\hat{S}^{ei} = S) < 1$. Now consider the case $\lim_{n \rightarrow \infty} \frac{\sqrt{n}\tau_j}{\sqrt{2\ln(p_j - s_j)}} - 1 \geq 0$. It

follows that, by Assumption A2 $\tau_j \rightarrow \infty$ and, by (14pp.68), $\lim_{n \rightarrow \infty} \sqrt{n} \beta_{\min, j}^* - \sqrt{n} \tau_j \leq c$. We can then apply Lemma ??, and we have that $\lim_{n \rightarrow \infty} P(\hat{S}^{ei} \supseteq S) < 1$ and $\lim_{n \rightarrow \infty} P(\hat{S}^{ei} = S) < 1$.

14.13 Proof of Corollary ??

Observe that the conditions of Lemma ?? hold for $\hat{S}^{sd}$ ($\hat{S}^{ei}$ for $b = 1$), and then $\lim_{n \rightarrow \infty} P(\hat{S}^{sd} = S) < 1$. Since Assumptions A4 and A5bis hold, by Proposition ?? (i) and Proposition ??, $\lim_{n \rightarrow \infty} P(\hat{S}^{ei} \subseteq 1) = 1$ and $\lim_{n \rightarrow \infty} P(\hat{S}^{ei} \supseteq S) = 1$, and then $\lim_{n \rightarrow \infty} P(\hat{S}^{ei} = S) = 1$.

15 Proofs of Section 3.1

15.1 Proof of Lemma 3.1

First note that by discarding constant terms

$$\arg \max_{M \in \mathcal{M}} \left\{ \max_{\beta \in \mathcal{L}_M} \ell(\mathbf{y}; \beta) - \sum_{j=1}^b \kappa_j |M_j| \right\} = \arg \min_{M \in \mathcal{M}} \left\{ \min_{\beta \in \mathcal{L}_M} \left\{ \frac{1}{2} \|\mathbf{y} - \mathbf{X}\beta\|^2 \right\} + \sum_{j=1}^b \kappa_j |M_j| \right\}.$$

We also have that

$$\min_{\beta \in \mathcal{L}_M} \frac{1}{2} \|\mathbf{y} - \mathbf{X}\beta\|^2 = \frac{1}{2} \|\mathbf{y} - \mathbf{X}_M \tilde{\beta}^{(M)}\|^2 = \frac{1}{2} \|\mathbf{y}\|^2 - \frac{1}{2} \|\mathbf{X}_M \tilde{\beta}^{(M)}\|^2,$$

where in the last equality we used that $\tilde{\beta}^{(M)} = (\mathbf{X}_M^T \mathbf{X}_M)^{-1} \mathbf{X}_M^T \mathbf{y}$. The maximization in (10) can be then replaced with the maximization of $C(M) = \frac{1}{2} \|\mathbf{X}_M \tilde{\beta}^{(M)}\|^2 - \sum_{j=1}^b \kappa_j |M_j|$ which is equivalent to maximizing $NC(M)$.

15.2 Proof of Lemma 3.2

Part (i) follows directly from Lemma 13.7 by taking $\{M^1, \dots, M^k\} = \{S\}$. Part (ii) follows from

$$NC(M) = \left(1 + \sum_{N \neq M} e^{C(N) - C(M)} \right)^{-1} < \left(1 + e^{C(M') - C(M)} \right)^{-1}.$$

15.3 Proof of Lemma 10.1

This result follows directly from Lemma S7 in ? taking $\phi^* = 1$.

15.4 Proof of Lemma 10.2

This result follows directly Lemma 13.2 by taking $T = S$, and observing that $\min_{i \in S_j \setminus M_j} \beta_i^{*2} \geq \beta_{\min, j}^{*2}$.

15.5 Proof of Theorem 3.3

The proof strategy is to first use Lemma 13.8 with $T = S$ to show that for every $M \neq S$, $\mathbb{E}(NC(M)) \leq e^{-\psi A_S}$ for every large enough n and any $\psi \in (0, 1)$, where $A_S = \gamma \Delta_{MS} + \frac{1-\gamma}{6} \mu_{Q_S M}$ (cf (39) and (40)), $\gamma \in (1/2, 1)$ is defined in Assumption A5 and $Q_S = M \cup S$. Assumption A4 and the fact that $M \setminus S \subseteq S^C$ ensure the assumptions of Lemma 13.8 are met. The second step is to obtain a lower bound for A_S , which gives a new upper bound for $\mathbb{E}(NC(M))$. The final step is to use these bounds to get an upper-bound on $\sum_{M \in \mathcal{M} \setminus \{S\}} \mathbb{E}(NC(M))$ that asymptotically vanishes under Assumptions A4 and A5. We then use Lemma 3.2 to conclude on the vanishing of $P(\hat{S}^{ei} \neq S)$.

First, to show that $\mathbb{E}(NC(M)) \leq e^{-\psi A_S}$ for any $M \in \mathcal{M} \setminus \{S\}$, we show that A_S satisfies the conditions of Lemma 13.8, taking $T = S$. That is, we wish to show that, $A_S > 0$, $|M \setminus S| = o(A_S)$, and $\mu_{Q_S S} = o(A_S)$. Observe that Δ_{MS} , defined in (39), can be rewritten as $\Delta_{MS} = \sum_{j=1}^b (|M_j \setminus S_j| - |S_j \setminus M_j|) \kappa_j$. By Lemma 10.2, for every $n \in \mathbb{N}$ we have

$$\begin{aligned} A_S &= \gamma \Delta_{MS} + \frac{1-\gamma}{6} \mu_{Q_S M} \\ &\geq \gamma \sum_{j=1}^b |M_j \setminus S_j| \kappa_j + \sum_{j=1}^b |S_j \setminus M_j| \left(\frac{1-\gamma}{6} n \rho(\mathbf{X}) \beta_{\min, j}^{*2} - \gamma \kappa_j \right) \end{aligned} \tag{15pp.69}$$

Since $M \neq S$, $|M \setminus S| \neq 0$ or $|S \setminus M| \neq 0$, then by Assumptions A4 and A5, for every n large enough, $A_S > 0$. We immediately have $\mu_{Q_S S} = o(A_S)$ because $\beta_{Q_S \setminus S}^* = \beta_{M \setminus S}^* = 0$ (any parameter outside the true support S is by definition 0) and hence $\mu_{Q_S S} = 0$. If $|M \setminus S| = 0$, $|M \setminus S| = o(A_S)$ also immediately. Consider now the case $|M \setminus S| \neq 0$. By Assumption A5,

the last term in (15pp.69) is nonnegative, and hence

$$\frac{|M \setminus S|}{A_S} = \frac{|M \setminus S|}{\gamma \Delta_{MS} + \frac{1-\gamma}{6} \mu_{QSM}} \leq \left[\gamma \sum_{j=1}^b \frac{|M_j \setminus S_j|}{|M \setminus S|} \kappa_j \right]^{-1} \leq \left[\gamma \min_{j=1, \dots, b} \kappa_j \right]^{-1}$$

where the last inequality follows from $\sum_{j=1}^b \frac{|M_j \setminus S_j|}{|M \setminus S|} = 1$. By Assumption A4 we have that $\min_j \kappa_j \rightarrow \infty$ as $n \rightarrow \infty$, and hence $|M \setminus S| = o(A_S)$. Thus, by Lemma 13.8, for any $\psi \in (0, 1)$ and all n large enough, $\mathbb{E}(NC(M)) \leq e^{-\psi A_S}$.

For the second step of the proof, let A_S^* be the lower bound for A_S given in (15pp.69). That is

$$A_S^* := \gamma \sum_{j=1}^b |M_j \setminus S_j| \kappa_j + \sum_{j=1}^b |S_j \setminus M_j| \left(\frac{1-\gamma}{6} n \rho(\mathbf{X}) \beta_{\min, j}^{*2} - \gamma \kappa_j \right).$$

By (15pp.69), we have, for all n large enough,

$$\mathbb{E}(NC(M)) \leq e^{-\psi A_S^*}. \quad (15pp.70)$$

Assumption A5 implies there exist $g'_j \rightarrow \infty$ such that

$$\frac{(1-\gamma)n\rho(\mathbf{X})}{6} \beta_{\min, j}^{*2} - \kappa_j = \ln(s_j) + g'_j. \quad (15pp.71)$$

Let $\delta \in (0, 1)$ and denote $\bar{m}_j = \max \left\{ \frac{2 \ln(p_j - s_j)}{f_j}, \frac{2 \ln(s_j)}{g'_j} \right\}$, where f_j is given in Assumption A4. Take $\psi = \max_{j=1, \dots, b} \frac{\xi + \delta + \bar{m}_j}{1 + \bar{m}_j}$ for some $\xi \in (0, 1 - \delta)$ then $\psi \in (0, 1)$ and we have, for every $j = 1, \dots, b$,

$$\psi > \frac{\delta + \frac{2 \ln(p_j - s_j)}{f_j}}{1 + \frac{2 \ln(p_j - s_j)}{f_j}} = \frac{\delta f_j / 2 + \ln(p_j - s_j)}{f_j / 2 + \ln(p_j - s_j)} \quad (15pp.72)$$

$$\psi > \frac{\delta + \frac{2 \ln(s_j)}{g'_j}}{1 + \frac{2 \ln(s_j)}{g'_j}} = \frac{\delta g'_j / 2 + \ln(s_j)}{g'_j / 2 + \ln(s_j)} \geq \frac{\delta g'_j / 2 + \ln(s_j)}{g'_j + \ln(s_j)}. \quad (15pp.73)$$

In Assumption A5, γ is defined as $\gamma := \frac{1}{2}(1 + \max_j \ln(p_j - s_j)/\kappa_j)$, we then have

$$\gamma \kappa_j \geq \frac{1}{2} \left(1 + \frac{\ln(p_j - s_j)}{\kappa_j} \right) \kappa_j = \ln(p_j - s_j) + \frac{1}{2} (\kappa_j - \ln(p_j - s_j)) = \ln(p_j - s_j) + \frac{1}{2} f_j.$$

Hence, by (15pp.72), we have

$$\psi \gamma \kappa_j \geq \psi \left(\ln(p_j - s_j) + \frac{1}{2} f_j \right) \geq \ln(p_j - s_j) + \delta \frac{1}{2} f_j. \quad (15pp.74)$$

Further,

$$\psi \left(\frac{1-\gamma}{6} n \rho(\mathbf{X}) \beta_{\min,j}^*{}^2 - \gamma \kappa_j \right) \geq \psi \left(\ln(s_j) + g'_j \right) \geq \ln(s_j) + \delta \frac{1}{2} g'_j. \quad (15\text{pp.75})$$

where the first inequality follows from (15pp.71) and the second inequality from (15pp.73).

In (15pp.70), $\psi A_S^* = \sum_{j=1}^b |M_j \setminus S_j| \psi \gamma \kappa_j + \sum_{j=1}^b |S_j \setminus M_j| \psi \left(\frac{1-\gamma}{6} n \rho(\mathbf{X}) \beta_{\min,j}^*{}^2 - \gamma \kappa_j \right)$. Then by (15pp.74) and (15pp.75), we get

$$\mathbb{E}(NC(M)) \leq \exp \left\{ - \sum_{j=1}^b |M_j \setminus S_j| (\ln(p_j - s_j) + \delta \frac{f_j}{2}) - \sum_{j=1}^b |S_j \setminus M_j| (\ln(s_j) + \delta \frac{g'_j}{2}) \right\}. \quad (15\text{pp.76})$$

For the final step of the proof, denote $\mathcal{S} = \sum_{M \in \mathcal{M} \setminus \{S\}} \mathbb{E}(NC(M))$ for convenience. By (15pp.76) we have

$$\mathcal{S} \leq \sum_{M \in \mathcal{M} \setminus \{S\}} e^{-\sum_{j=1}^b |M_j \setminus S_j| \left(\ln(p_j - s_j) + \delta \frac{f_j}{2} \right) - \sum_{j=1}^b |S_j \setminus M_j| \left(\ln(s_j) + \delta \frac{g'_j}{2} \right)}.$$

Observe that if $|M_j \setminus S_j| = 0$ and $|S_j \setminus M_j| = 0$ for all j , then $M = S$ and the summand in the right-hand side above is 1. Then by adding and subtracting 1 we get

$$\mathcal{S} \leq \sum_{M \in \mathcal{M}} e^{-\sum_{j=1}^b |M_j \setminus S_j| \left(\ln(p_j - s_j) + \delta \frac{f_j}{2} \right) - \sum_{j=1}^b |S_j \setminus M_j| \left(\ln(s_j) + \delta \frac{g'_j}{2} \right)} - 1.$$

We can split the sum in the right-hand side above into sums over the models that have the same number of inactive variables and missing the same number of truly active variables in every block. That is, the models M such that for all j , $|M_j \setminus S_j| = u_j$ and $|S_j \setminus M_j| = w_j$ with $u_j \in \{0, \dots, p_j - s_j\}$ and $w_j \in \{0, \dots, s_j\}$. Denote those sums

$$S_{\mathbf{w}}^{\mathbf{u}} = \sum_{M \in \mathcal{M} : \forall j |M_j \setminus S_j| = u_j, |S_j \setminus M_j| = w_j} e^{-\sum_{j=1}^b u_j \left(\ln(p_j - s_j) + \delta \frac{f_j}{2} \right) - \sum_{j=1}^b w_j \left(\ln(s_j) + \delta \frac{g'_j}{2} \right)}.$$

We get

$$\mathcal{S} \leq -1 + \sum_{w_1=0}^{s_1} \cdots \sum_{w_b=0}^{s_b} \sum_{u_1=0}^{p_1-s_1} \cdots \sum_{u_b=0}^{p_b-s_b} S_{\mathbf{w}}^{\mathbf{u}}. \quad (15\text{pp.77})$$

The number of models having, for all j , u_j inactive parameters and missing w_j out of the s_j

active parameters is $\prod_{j=1}^b \binom{p_j - s_j}{u_j} \binom{s_j}{w_j}$. We thus have

$$\begin{aligned} S_{\mathbf{w}}^{\mathbf{u}} &= \left(\prod_{j=1}^b \binom{p_j - s_j}{u_j} \binom{s_j}{w_j} \right) e^{-\sum_{j=1}^b u_j \left(\ln(p_j - s_j) + \delta \frac{f_j}{2} \right) - \sum_{j=1}^b w_j \left(\ln(s_j) + \delta \frac{g'_j}{2} \right)} \\ &= \prod_{j=1}^b \binom{p_j - s_j}{u_j} e^{-u_j \left(\ln(p_j - s_j) + \delta \frac{f_j}{2} \right)} \binom{s_j}{w_j} e^{-w_j \left(\ln(s_j) + \delta \frac{g'_j}{2} \right)}. \end{aligned}$$

Inputting the expression above in (15pp.77) gives

$$\begin{aligned} \mathcal{S} &\leq -1 + \sum_{w_1=0}^{s_1} \cdots \sum_{w_b=0}^{s_b} \sum_{u_1=0}^{p_1-s_1} \cdots \sum_{u_b=0}^{p_b-s_b} \prod_{j=1}^b \binom{p_j - s_j}{u_j} e^{-u_j \left(\ln(p_j - s_j) + \delta \frac{f_j}{2} \right)} \binom{s_j}{w_j} e^{-w_j \left(\ln(s_j) + \delta \frac{g'_j}{2} \right)} \\ &\leq -1 + \prod_{j=1}^b \left(1 + \sum_{u_j=1}^{p_j-s_j} \binom{p_j - s_j}{u_j} e^{-u_j \left(\ln(p_j - s_j) + \delta \frac{f_j}{2} \right)} \right) \left(1 + \sum_{w_j=1}^{s_j} \binom{s_j}{w_j} e^{-w_j \left(\ln(s_j) + \delta \frac{g'_j}{2} \right)} \right) \end{aligned}$$

where the second inequality follows from first factorizing over terms in u_j and w_j and then taking the term in 0 out of every sum. A standard bound on binomial coefficient for $1 \leq k \leq n$ is

$$\binom{n}{k} \leq \left(\frac{ne}{k} \right)^k \leq (ne)^k = e^{k(\ln(n)+1)}. \quad (15pp.78)$$

Then

$$\mathcal{S} \leq -1 + \prod_{j=1}^b \left(1 + \sum_{u_j=1}^{p_j-s_j} e^{-u_j \left(\delta \frac{f_j}{2} - 1 \right)} \right) \left(1 + \sum_{w_j=1}^{s_j} e^{-w_j \left(\delta \frac{g'_j}{2} - 1 \right)} \right). \quad (15pp.79)$$

Denote

$$d_j = e^{1 - \delta \frac{f_j}{2}}, \quad h_j = e^{1 - \delta \frac{g'_j}{2}}$$

where both expressions go to zero as n increases since $f_j \rightarrow \infty$ and $g'_j \rightarrow \infty$. For every j , by the properties of geometric sums, we have

$$\begin{aligned} 1 + \sum_{u_j=1}^{p_j-s_j} e^{-u_j \left(\delta \frac{f_j}{2} - 1 \right)} &= \frac{1 - d_j^{p_j-s_j+1}}{1 - d_j} \\ 1 + \sum_{w_j=1}^{s_j} e^{-w_j \left(\delta \frac{g'_j}{2} - 1 \right)} &= \frac{1 - h_j^{s_j+1}}{1 - h_j}. \end{aligned}$$

Since both expressions converge to 1 as n grows, we get that

$$\lim_{n \rightarrow \infty} \mathcal{S} = \lim_{n \rightarrow \infty} \sum_{M \in \mathcal{M} \setminus \{S\}} \mathbb{E}(NC(M)) = 0.$$

By Lemma 3.2, $P(\hat{S}^{ei} \neq S) \leq 2\mathcal{S}$ and then $\lim_{n \rightarrow \infty} P(\hat{S}^{ei} = S) = 1$.

15.6 Proof of Theorem 3.4

First, we prove the upper bound on $P(\hat{S}^{ei} \neq S)$ in (12). We assume here A1, A3, A4, and A5. Under the same assumptions, in the proof of Theorem 3.3, the following bound was shown in (15pp.79):

$$\sum_{M \in \mathcal{M} \setminus \{S\}} \mathbb{E}(NC(M)) \leq -1 + \prod_{j=1}^b \left(1 + \sum_{u_j=1}^{p_j-s_j} e^{-u_j \left(\delta \frac{f_j}{2} - 1 \right)} \right) \left(1 + \sum_{w_j=1}^{s_j} e^{-w_j \left(\delta \frac{g'_j}{2} - 1 \right)} \right). \quad (15pp.80)$$

Denote $S(u_j) = \sum_{u_j=1}^{p_j-s_j} e^{-u_j \left(\delta \frac{f_j}{2} - 1 \right)}$, $S(w_j) = \sum_{w_j=1}^{s_j} e^{-w_j \left(\delta \frac{g'_j}{2} - 1 \right)}$, $d_j = e^{1-\delta \frac{f_j}{2}}$, and $h_j = e^{1-\delta \frac{g'_j}{2}}$. For every j , we have, by the properties of geometric sums:

$$\begin{aligned} S(u_j) &= d_j \frac{1 - d_j^{p_j-s_j}}{1 - d_j} \\ S(w_j) &= h_j \frac{1 - h_j^{s_j}}{1 - h_j}. \end{aligned}$$

Developing the product in the right-hand side in (15pp.80) and reordering the resulting terms gives

$$\sum_{M \in \mathcal{M} \setminus \{S\}} \mathbb{E}(NC(M)) \leq -1 + 1 + \sum_{j=1}^b [S(u_j) + S(w_j)] + \mathcal{R}$$

where all the terms in $\mathcal{R}$ are product of two or more of the sums $S(u_1), \dots, S(u_b), S(w_1), \dots, S(w_b)$.

Given that $\delta > 0$, $f_j \rightarrow \infty$ and $g'_j \rightarrow \infty$ by assumption, and hence $d_j \rightarrow 0$ and $h_j \rightarrow 0$, the $S(u_j)$ and $S(w_j)$ are smaller than 1 for all sufficiently large n for all j . Then each of the $2^{2b} - 2b - 1$ terms in $\mathcal{R}$ is bounded above by $\sum_{j=1}^b [S(u_j) + S(w_j)]$ and we get, for every n large enough,

$$\sum_{M \in \mathcal{M} \setminus \{S\}} \mathbb{E}(NC(M)) \leq (2^{2b} - 2b) \sum_{j=1}^b \left[d_j \frac{1 - d_j^{p_j-s_j}}{1 - d_j} + h_j \frac{1 - h_j^{s_j}}{1 - h_j} \right]. \quad (15pp.81)$$

We have $d_j \rightarrow 0$ and $h_j \rightarrow 0$, then for every n large enough $\frac{1-d_j^{p_j-s_j}}{1-d_j} \rightarrow 1$ and $\frac{1-h_j^{s_j}}{1-h_j} \rightarrow 1$ for all j . Since $r > 1$, we get, for every n large enough,

$$r > \max_{j=1, \dots, b} \left\{ \frac{1 - d_j^{p_j-s_j}}{1 - d_j}, \frac{1 - h_j^{s_j}}{1 - h_j} \right\}. \quad (15pp.82)$$

By Lemma 3.2, (15pp.81) and (15pp.82), we then obtain:

$$P(\hat{S}^{ei} \neq S) \leq 2 \sum_{M \in \mathcal{M} \setminus \{S\}} \mathbb{E}(NC(M)) \leq 2(2^{2b} - 2b)r e \sum_{j=1}^b e^{-\delta \frac{f_j}{2}} + e^{-\delta \frac{g'_j}{2}}$$

By the definition of f_j in Assumption A4, that of g'_j in (15pp.71), and the fact that $2e < 6$, we get, for every n large enough, that

$$P(\hat{S}^{ei} \neq S) \leq (2^{2b} - 2b)6r \sum_{j=1}^b e^{-\frac{\delta}{2} [\kappa_j - \ln(p_j - s_j)]} + e^{-\frac{\delta}{2} \left[\frac{(1-\gamma)n\rho(\mathbf{X})}{6} \beta_{\min,j}^*{}^2 - \kappa_j - \ln(s_j) \right]}. \quad (15pp.83)$$

We have $\frac{(1-\gamma)n\rho(\mathbf{X})}{6} \beta_{\min,j}^*{}^2 - \kappa_j > \left(\sqrt{\frac{(1-\gamma)n\rho(\mathbf{X})}{6}} \beta_{\min,j}^* - \sqrt{\kappa_j} \right)^2$ and then:

$$P(\hat{S}^{ei} \neq S) \leq (2^{2b} - 2b)6r \sum_{j=1}^b e^{-\frac{\delta}{2} [\kappa_j - \ln(p_j - s_j)]} + e^{-\frac{\delta}{2} \left[\left(\sqrt{\frac{(1-\gamma)n\rho(\mathbf{X})}{6}} \beta_{\min,j}^* - \sqrt{\kappa_j} \right)^2 - \ln(s_j) \right]}$$

which proves (12).

Second, we prove that, if for all $j = 1, \dots, b$, it holds that

$$\lim_{n \rightarrow \infty} \frac{\sqrt{(1-\gamma)n\rho(\mathbf{X})/3} \beta_{\min,j}^*}{\sqrt{2 \ln(p_j - s_j)} + \sqrt{2 \ln(s_j)}} > 1 \quad (15pp.84)$$

then κ_j^* defined in (14) satisfies Assumptions A4 and A5. The proof is essentially the same as the proof of the second part of Theorem ??, replacing τ_j^* by $\sqrt{\kappa_j^*}$. Under (15pp.84), for every j , there exists a sequence c such that $\lim_{n \rightarrow \infty} c > 1$ and $\sqrt{\frac{(1-\gamma)n}{6}} \beta_{\min,j}^* = c \left(\sqrt{\ln(p_j - s_j)} + \sqrt{\ln(s_j)} \right)$. Denote $a = \sqrt{\ln(p_j - s_j)}$ and $b = \sqrt{\ln(s_j)}$. Proceeding as in the proof of Theorem ?? shows that:

$$\sqrt{\kappa_j^*} = \frac{c^2 + 1}{2c} a + \frac{c^2 - 1}{2c} b \geq \left(\frac{c^2 + 1}{2c} + 1 - 1 \right) a = \left(1 + \frac{(c-1)^2}{2c} \right) \sqrt{\ln(p_j - s_j)}$$

which implies Assumption A5 since $\lim_{n \rightarrow \infty} c > 1$. We also have

$$\frac{\sqrt{(1-\gamma)n\rho(\mathbf{X})}}{6} \beta_{\min,j}^* - \sqrt{\kappa_j^*} = \frac{c^2 - 1}{2c} a + \frac{c^2 + 1}{2c} b \geq \left(1 + \frac{(c-1)^2}{2c} \right) \sqrt{\ln(s_j)}.$$

which implies Assumption A4 since $\lim_{n \rightarrow \infty} c > 1$.

Finally, we prove (15). Since Assumptions A4 and A5 hold for the κ_j^* , and because we assume A1, by the first part of the theorem, for any $r > 1$ and every n large enough,

$$P(\hat{S}^{ei} \neq S) \leq 6(2^{2b} - 2b)r \sum_{j=1}^b e^{-\frac{\delta}{2} [\kappa_j^* - \ln(p_j - s_j)]} + e^{-\frac{\delta}{2} \left[\left(\sqrt{\frac{(1-\gamma)n\rho(\mathbf{X})}{6}} \beta_{\min,j}^* - \sqrt{\kappa_j^*} \right)^2 - \ln(s_j) \right]}.$$

Simple algebra shows that the κ_j^* satisfy $\kappa_j^* - \ln(p_j - s_j) = \left(\sqrt{\frac{(1-\gamma)n\rho(\mathbf{X})}{6}} \beta_{\min,j}^* - \sqrt{\kappa_j^*} \right)^2 - \ln(s_j)$ for all j . We then have

$$P(\hat{S}^{ei} \neq S) \leq 12(2^{2b} - 2b)r \sum_{j=1}^b e^{-\frac{\delta}{2} [\kappa_j^* - \ln(p_j - s_j)]}. \quad (15\text{pp.85})$$

Proceeding as in the proof of Theorem ??, denote

$d = \frac{1}{2} \sqrt{\frac{6}{(1-\gamma)n\rho(\mathbf{X})}} \frac{1}{\beta_{\min,j}^*} \left(\ln(p_j - s_j) - \ln(s_j) \right)$ such that $\sqrt{\kappa_j^*} = \frac{1}{2} \sqrt{\frac{(1-\gamma)n\rho(\mathbf{X})}{6}} \beta_{\min,j}^* + d$. Then

$$e^{-\frac{\delta}{2} (\kappa_j^* - \ln(p_j - s_j))} = e^{-\frac{\delta}{2} \left[\frac{(1-\gamma)n\rho(\mathbf{X})}{24} \beta_{\min,j}^{*2} + d^2 - \frac{1}{2} (\ln(p_j - s_j) + \ln(s_j)) \right]}.$$

By considering separately the two possible maxima in $\ln \max\{p_j - s_j, s_j\}$, we get that

$$\frac{e^{-\frac{\delta}{2} (\kappa_j^* - \ln(p_j - s_j))}}{e^{-\frac{\delta}{2} \left[\frac{(1-\gamma)n\rho(\mathbf{X})}{24} \beta_{\min,j}^{*2} - \ln \max\{p_j - s_j, s_j\} \right]}} = e^{-\frac{\delta}{2} \left[d^2 + \frac{1}{2} |\ln(p_j - s_j) - \ln(s_j)| \right]} < 1. \quad (15\text{pp.86})$$

It follows that, by (15pp.85) and (15pp.86),

$$P(\hat{S}^{ei} \neq S) \leq 12(2^{2b} - 2b)r \sum_{j=1}^b e^{-\frac{\delta}{2} \left[\frac{(1-\gamma)n\rho(\mathbf{X})}{24} \beta_{\min,j}^{*2} - \ln \max\{p_j - s_j, s_j\} \right]}.$$

which proves (15).

15.7 Proof of Proposition ??

The event $\hat{S}^{ei} = S$ (correct recovery of S) requires the event $\max_{M \in \mathcal{O}_j} \frac{NC(M)}{NC(S)} < 1$ where NC is the normalized criterion defined in (38). That is S is preferred to any model in $\mathcal{O}_j$ i.e. over-fitting S by 1 variable in block j .

$$P(\hat{S}^{ei} = S) \leq P\left(\max_{M \in \mathcal{O}_j} \frac{NC(M)}{NC(S)} < 1\right) = P\left(\max_{M \in \mathcal{O}_j} e^{C(M) - C(S)} < 1\right)$$

Using that $C(S) - C(M) = L_{SM}/2 + \Delta_{MS}$ for L_{SM} and Δ_{MS} defined in (39), we obtain

$$P(\hat{S}^{ei} = S) \leq P\left(\max_{M \in \mathcal{O}_j} e^{-(\frac{1}{2}L_{SM} + \Delta_{MS})} < 1\right) = P\left(\max_{M \in \mathcal{O}_j} L_{MS} < 2\kappa_j\right). \quad (15\text{pp.87})$$

Taking square root on both sides of the inequality in the right-most probability in (15pp.87), we have

$$P(\hat{S}^{ei} = S) \leq P\left(\max_{M \in \mathcal{O}_j} \sqrt{L_{MS}} < \sqrt{2\kappa_j}\right). \quad (15\text{pp.88})$$

By Lemma 10.1, for every $M \in \mathcal{O}_j$, $L_{MS} \in \chi_1^2$ and there exists $Z_M \sim N(0, 1)$ such that $\sqrt{L_{MS}} = |Z_M|$. Since for every $M \in \mathcal{O}_j$, $|Z_M| \geq Z_M$, we have

$$P(\hat{S}^{ei} = S) \leq 1 - P\left(\max_{M \in \mathcal{O}_j} Z_M \geq \sqrt{2\kappa_j}\right). \quad (15\text{pp.89})$$

The set $\mathcal{O}_j$ has cardinality $p_j - s_j$, then by Theorem 3.4 in ? and our Assumption A2, for any $\varepsilon > 0$,

$$P\left(\max_{M \in \mathcal{O}_j} Z_M \geq \underline{\lambda}_j \sqrt{2 \ln(p_j - s_j)(1 - \varepsilon)}\right) \rightarrow 1,$$

where $\underline{\lambda}_j$ is as defined prior to the statement of Proposition ??. If $\lim_{n \rightarrow \infty} \frac{\kappa_j}{\underline{\lambda}_j^2 \ln(p_j - s_j)} < 1$, then $\lim_{n \rightarrow \infty} P\left(\max_{M \in \mathcal{O}_j} Z_M \geq \sqrt{2\kappa_j}\right) = 1$. Hence, by (15pp.89) we have that $\lim_{n \rightarrow \infty} P(\hat{S}^{ei} = S) = 0$, as claimed.

15.8 Proof of Lemma ??

The result follows directly from Lemma 13.9, with $M = S$.

15.9 Proof of Proposition ??

Let $M = S \setminus S^S(\kappa)$. Since $S^S(\kappa) \neq \emptyset$ by assumption and $S^S(\kappa) \subseteq S$, we have that $M \neq S$. With this notation, we aim at proving that $\lim_{n \rightarrow \infty} P(NC(M) < NC(S)) = \lim_{n \rightarrow \infty} P(\hat{S}^{ei} = S) = 0$. The event $\hat{S}^{ei} = S$ (correct selection of S) implies that $NC(M) < NC(S)$ (preferring S over M), and hence

$$P(\hat{S}^{ei} = S) \leq P(NC(M) < NC(S)) = P(L_{SM} > 2\Delta_{SM}), \quad (15\text{pp.90})$$

where we used the definition of NC in (38), and that $C(S) - C(M) = L_{SM}/2 + \Delta_{MS}$ for L_{SM} and Δ_{MS} defined in (39). It suffices then to show that the right-hand side in (15pp.90) converges to 0 as $n \rightarrow \infty$. To do this, we note that, by Lemma 10.1, we have $L_{SM} \sim \chi_{|S \setminus M|}^2(\mu_{SM})$. We then use the non-central chi-square bound in Lemma 13.3.

To apply Lemma 13.3, we first show that the degrees of freedom satisfy $|S \setminus M| = o(2\Delta_{SM})$ and also the non-centrality parameter is such that $\mu_{SM} = o(2\Delta_{SM})$. We have that $\Delta_{SM} =$

$\sum_{j=1}^b |S_j \setminus M_j| \kappa_j$ and then

$$\frac{|S \setminus M|}{2\Delta_{SM}} \leq \left[\sum_{j=1}^b \frac{2|S_j \setminus M_j|}{|S \setminus M|} \kappa_j \right]^{-1} \leq \left[2 \min_{j=1, \dots, b} \kappa_j \right]^{-1}$$

where the last inequality follows from $\sum_{j=1}^b \frac{|S_j \setminus M_j|}{|S \setminus M|} = 1$. By Assumption A4, $\kappa_j \rightarrow \infty$ for all $j = 1, \dots, b$, then the left-hand side above goes to 0 as n grows and $|S \setminus M| = o(2\Delta_{SM})$.

Further, by Lemma ??, we also have

$$\frac{\mu_{SM}}{2\Delta_{SM}} \leq \frac{\sum_{j=1}^b |S_j \setminus M_j| n \bar{\lambda} \max_{i \in S_j \setminus M_j} \beta_i^{*2}}{\sum_{j=1}^b |S_j \setminus M_j| 2\kappa_j}. \quad (15\text{pp.91})$$

Let $\tilde{r} := \sum_{j=1}^b n \bar{\lambda} \max_{i \in S_j \setminus M_j} \beta_i^{*2} / (2\kappa_j)$. We show next that $\tilde{r}$ is an upper bound on $\mu_{SM} / (2\Delta_{SM})$. By restricting the sum in $\tilde{r}$ to the j such that $|S_j \setminus M_j| \neq 0$ and multiplying the numerator and denominator of the summand by $|S_j \setminus M_j|$, we get the lower bound on $\tilde{r}$

$$\tilde{r} \geq \sum_{j=1, |S_j \setminus M_j| \neq 0}^b \frac{|S_j \setminus M_j| n \bar{\lambda} \max_{i \in S_j \setminus M_j} \beta_i^{*2}}{|S_j \setminus M_j| 2\kappa_j}. \quad (15\text{pp.92})$$

Note that for any collections $(\alpha_j, \delta_j) \in \mathbb{R} \times \mathbb{R} \setminus \{0\}$, $j = 1, \dots, b$, we have

$$\sum_{j=1}^b \frac{\alpha_j}{\delta_j} = \sum_{j=1}^b \frac{\alpha_j \frac{1}{\delta_j} (\delta_j + \sum_{l \neq j} \delta_l)}{\sum_{j=1}^b \delta_j} = \frac{\sum_{j=1}^b \alpha_j (1 + \sum_{l \neq j} \frac{\delta_l}{\delta_j})}{\sum_{j=1}^b \delta_j} \quad (15\text{pp.93})$$

Using the above in the right-hand side of (15pp.92) gives

$$\begin{aligned} \tilde{r} &\geq \frac{\sum_{j=1, |S_j \setminus M_j| \neq 0}^b |S_j \setminus M_j| n \bar{\lambda} \max_{i \in S_j \setminus M_j} \beta_i^{*2} \left(1 + \sum_{l \neq j} \frac{|S_l \setminus M_l| 2\kappa_l}{|S_j \setminus M_j| 2\kappa_j} \right)}{\sum_{j=1, |S_j \setminus M_j| \neq 0}^b |S_j \setminus M_j| 2\kappa_j} \\ &\geq \frac{\sum_{j=1}^b |S_j \setminus M_j| n \bar{\lambda} \max_{i \in S_j \setminus M_j} \beta_i^{*2}}{\sum_{j=1}^b |S \setminus M_j| 2\kappa_j} \end{aligned}$$

where last inequality follows from $\left(1 + \sum_{l \neq j} \frac{|S_l \setminus M_l| 2\kappa_l}{|S_j \setminus M_j| 2\kappa_j} \right) \geq 1$ for all j and that $\sum_{j=1}^b |S \setminus M_j| 2\kappa_j = \sum_{j=1, |S_j \setminus M_j| \neq 0}^b |S_j \setminus M_j| 2\kappa_j$. Then by (15pp.91)

$$\frac{\mu_{SM}}{2\Delta_{SM}} \leq \tilde{r} = \sum_{j=1}^b \frac{n \bar{\lambda} \max_{i \in S_j \setminus M_j} \beta_i^{*2}}{2\kappa_j}.$$

For every $j = 1, \dots, b$, $S_j \setminus M_j \subset S_j^S(\kappa)$, then by definition of the $S_j^S(\kappa)$, the right-hand side above goes to 0 as $n \rightarrow \infty$ and $\mu_{SM} = o(2\Delta_{SM})$. We can now use Lemma 13.3. For any $\phi \in (0, 1)$ and every n large enough,

$$P(L_{SM} > 2\Delta_{SM}) \leq e^{-\phi 2\Delta_{SM}} = e^{-\phi 2 \sum_{j=1}^b |S(\kappa)_j^S| \kappa_j}$$

where the right hand-side goes to 0 since for all $j = 1, \dots, b$, $\kappa_j \rightarrow \infty$. It follows by (15pp.90) that $\lim_{n \rightarrow \infty} P(NC(M)/NC(S) < 1) = \lim_{n \rightarrow \infty} P(\hat{S}^{ei} = S) = 0$ as claimed.

15.10 Proof of Proposition ??

Consider the model M under-fitting S by the variable in block j associated to $\beta_{\min,j}^*$. The event $\hat{S}^{ei} = S$ (correct recovery of S) requires the event $\frac{NC(M)}{NC(S)} < 1$ where NC is the normalized criterion in (38). That is S is preferred to model M .

$$P(\hat{S}^{ei} = S) \leq P\left(\frac{NC(M)}{NC(S)} < 1\right) = P\left(e^{C(M)-C(S)} < 1\right)$$

Using that $C(S) - C(M) = L_{SM}/2 + \Delta_{MS}$ for L_{SM} and Δ_{MS} defined in (39), and that $\Delta_{MS} = -\kappa_j$ we obtain

$$P(\hat{S}^{ei} = S) \leq P\left(e^{-(\frac{1}{2}L_{SM} + \Delta_{MS})} < 1\right) = P\left(L_{SM} > 2\kappa_j\right). \quad (15pp.94)$$

Taking square root on both sides of the inequality in the right-most probability in (15pp.94), we get

$$P(\hat{S}^{ei} = S) \leq P\left(\sqrt{L_{SM}} > \sqrt{2\kappa_j}\right). \quad (15pp.95)$$

By Lemma 10.1, $L_{SM} \in \chi_1^2(\mu_{SM})$ and there exists then $Z_M \sim N(\sqrt{\mu_{SM}}, 1)$ such that $\sqrt{L_{SM}} = |Z_M|$. We have

$$P(\hat{S}^{ei} = S) \leq P\left(|Z_M| > \sqrt{2\kappa_j}\right). \quad (15pp.96)$$

Consider first the case where $2\kappa_j > \mu_{SM}$. Using that $|Z_M| = |Z_M - \sqrt{\mu_{SM}} + \sqrt{\mu_{SM}}| \leq |Z_M - \sqrt{\mu_{SM}}| + \sqrt{\mu_{SM}}$, we get that

$$P\left(|Z_M| > \sqrt{2\kappa_j}\right) \leq P\left(|Z_M - \sqrt{\mu_{SM}}| > \sqrt{2\kappa_j} - \sqrt{\mu_{SM}}\right) = 2P\left(z > \sqrt{2\kappa_j} - \sqrt{\mu_{SM}}\right)$$

where $z \sim N(0, 1)$ and the equality follows from the symmetry of the standard Gaussian distribution. Since $2\kappa_j > \mu_{SM}$, $P\left(z > \sqrt{2\kappa_j} - \sqrt{\mu_{SM}}\right) < 1/2$, $P\left(|Z_M| > \sqrt{2\kappa_j}\right) < 1$ and $P(\hat{S}^{ei} = S) < 1$ too by (15pp.96).

Consider next the case where $2\kappa_j \leq \mu_{SM}$. By Lemma 13.1 (ii), we have

$$P(|Z_M| > \sqrt{2\kappa_j}) = P(z > \sqrt{2\kappa_j} - \sqrt{\mu_{SM}}) + P(z < -\sqrt{2\kappa_j} - \sqrt{\mu_{SM}})$$

where $z \sim N(0, 1)$. Since $\kappa_j \rightarrow +\infty$, $\sqrt{2\kappa_j} + \sqrt{\mu_{SM}} \rightarrow \infty$ and we have that $P(z < -\sqrt{2\kappa_j} - \sqrt{\mu_{SM}}) \rightarrow 0$. Further, by Lemma ??, $\sqrt{\mu_{SM}} \leq \sqrt{n\bar{\lambda}\beta_{\min,j}^*}$. It follows that, by the assumption in (??), $\lim_{n \rightarrow \infty} \sqrt{\mu_{SM}} - \sqrt{2\kappa_j} < \infty$ and we have that $\lim_{n \rightarrow \infty} P(z > \sqrt{2\kappa_j} - \sqrt{\mu_{SM}}) < 1$. Hence we get $\lim_{n \rightarrow \infty} P(\hat{S}^{ei} = S) < 1$, as claimed.

15.11 Proof of Corollary ??

First, we re-write

$$\sqrt{n\bar{\lambda}\beta_{\min,j}^*} - \underline{\lambda}_j \sqrt{2 \ln(p_j - s_j)} = \sqrt{n\bar{\lambda}\beta_{\min,j}^*} - \sqrt{2\kappa_j} + \underline{\lambda}_j \sqrt{2 \ln(p_j - s_j)} \left(\sqrt{\frac{\kappa_j}{\underline{\lambda}_j^2 \ln(p_j - s_j)}} - 1 \right).$$

Condition (??) implies that there exists $c \in \mathbb{R}^+$ such that

$$\lim_{n \rightarrow \infty} \sqrt{n\bar{\lambda}\beta_{\min,j}^*} - \sqrt{2\kappa_j} + \underline{\lambda}_j \sqrt{2 \ln(p_j - s_j)} \left(\sqrt{\frac{\kappa_j}{\underline{\lambda}_j^2 \ln(p_j - s_j)}} - 1 \right) \leq c \quad (15pp.97)$$

Consider the case $\lim_{n \rightarrow \infty} \sqrt{\frac{\kappa_j}{\underline{\lambda}_j^2 \ln(p_j - s_j)}} - 1 < 0$. Then by Proposition ??, we have $\lim_{n \rightarrow \infty} P(\hat{S}^{ei} = S) < 1$. Now consider the case $\lim_{n \rightarrow \infty} \sqrt{\frac{\kappa_j}{\underline{\lambda}_j^2 \ln(p_j - s_j)}} - 1 \geq 0$. Then, $\kappa_j \rightarrow \infty$ since by assumption $\underline{\lambda}_j^2 \ln(p_j - s_j) \rightarrow \infty$. Moreover, condition (15pp.97) implies that $\lim_{n \rightarrow \infty} \sqrt{n\bar{\lambda}\beta_{\min,j}^*} - \sqrt{2\kappa_j} \leq c$. By Proposition ??, we have that $\lim_{n \rightarrow \infty} P(\hat{S}^{ei} = S) < 1$.

15.12 Proof of Corollary 3.6

Since $\hat{S}^{sd}$ is $\hat{S}^{ei}$ with $b = 1$, the assumptions of Corollary ?? for $\hat{S}^{sd}$ are met and $\lim_{n \rightarrow \infty} P(\hat{S}^{sd} = S) < 1$. Since Assumptions A4 and A5 hold, by Theorem 3.3 we have that $\lim_{n \rightarrow \infty} P(\hat{S}^{ei} = S) = 1$.

15.13 Proof of Theorem 3.7

The proof strategy is the same as that of Theorem 3.3, with suitable adjustments. The first step is to use Lemma 13.8 to bound $\mathbb{E}(NC(M))$ for every $M \notin \mathcal{T}(\kappa)$. The main difference

is that in the proof of Theorem 3.3 we took $T = S$ in Lemma 13.8, whereas now we take a model $T = T^M \in \mathcal{T}(\kappa)$ that depends on M . Intuitively, T^M contains large truly non-zero parameters that are missed by M , and hence T^M should be chosen over M asymptotically. More precisely, we choose $T^M \in \mathcal{T}(\kappa)$ such that $T^M \setminus M \subseteq S^L(\kappa)$ and the elements in $M \setminus T^M$ are either inactive or in $S^S(\kappa)$. The latter condition and Assumption A4 ensure that the assumptions of Lemma 13.8 are met. We then get a bound $\mathbb{E}(NC(M)) \leq e^{-\psi A_{T^M}}$ for every large enough n and any $\psi \in (0, 1)$, where $A_{T^M} = \gamma \Delta_{MT^M} + \frac{1-\gamma}{6} \mu_{Q_{T^M}M}$ (cf (39) and (40)), $\gamma \in (1/2, 1)$ is defined in (24) and $Q_{T^M} = M \cup T^M$. The second step is to obtain a lower bound for A_{T^M} , which gives an upper bound for $\mathbb{E}(NC(M))$ (distinct to that obtained in the proof of Theorem 3.3). The final step is to get an upper-bound $\sum_{M \in \mathcal{M} \setminus \mathcal{T}(\kappa)} \mathbb{E}(NC(M))$ that vanishes (as n grows) under the assumptions of Theorem 3.7. Then Lemma 13.7 immediately implies that $P(\hat{S}^{ei} \notin \mathcal{T}(\kappa))$ also vanishes.

For the first step of the proof, recall that the set $\mathcal{T}(\kappa)$ contains all the models that are the union of $S^L(\kappa)$ (large signals) and some subset of $S^I(\kappa)$ (intermediate signals). For any $M \notin \mathcal{T}(\kappa)$, take the unique $T^M \in \mathcal{T}(\kappa)$ such that $T^M = [M \cap S^I(\kappa)] \cup S^L(\kappa)$, which implies $M \cap S^I(\kappa) = T^M \cap S^I(\kappa)$. That is, T^M contains all the large signals plus the intermediate signals in M . To show that $\mathbb{E}(NC(M)) \leq e^{-\psi A_{T^M}}$ we show that A_{T^M} satisfies the conditions of Lemma 13.8, taking $T = T^M$. That is, we wish to show that three conditions hold: $A_{T^M} > 0$, $|M \setminus T^M| = o(A_{T^M})$, and $\mu_{Q_{T^M}T^M} = o(A_{T^M})$.

Observe that Δ_{MT^M} , defined in (39), can be rewritten as $\Delta_{MT^M} = \sum_{j=1}^b (|M_j \setminus T_j^M| - |T_j^M \setminus M_j|) \kappa_j$ and then:

$$A_{T^M} = \sum_{j=1}^b |M_j \setminus T_j^M| \gamma \kappa_j + \frac{1-\gamma}{6} \mu_{Q_{T^M}M} - \sum_{j=1}^b |T_j^M \setminus M_j| \gamma \kappa_j$$

Since $\gamma < 1$, $(1 - \gamma)/6 > 0$, by Lemma 13.2, for every $n \in \mathbb{N}$ we have

$$A_{T^M} \geq \sum_{j=1}^b |M_j \setminus T_j^M| \gamma \kappa_j + \sum_{j=1}^b |T_j^M \setminus M_j| \left(\frac{1-\gamma}{6} n \rho(\mathbf{X}) \min_{i \in T_j^M \setminus M_j} \beta_i^{*2} - \gamma \kappa_j \right). \quad (15\text{pp.98})$$

We have that $T^M \subseteq (S^I(\kappa) \cup S^L(\kappa))$ since $T^M \in \mathcal{T}(\kappa)$ and that $T^M \cap S^I(\kappa) = M \cap S^I(\kappa)$. It follows that $T^M \setminus M \subseteq S^L(\kappa)$. By definition of $S^L(\kappa)$, the rightmost component in

(15pp.98) is nonnegative and, if $|T^M \setminus M| \neq 0$, it is positive, for every n large enough. By Assumption A4, component $\gamma \sum_{j=1}^b |M_j \setminus T_j^M| \kappa_j$ is nonnegative, and, if $|M \setminus T^M| \neq 0$, positive. Now, $M \neq T^M$ implies that necessarily $|M \setminus T^M| \neq 0$ or $|T^M \setminus M| \neq 0$ and then, for every n large enough, $A_{T^M} > 0$, establishing the first condition required by Lemma 13.8. Regarding its second condition, if $|M \setminus T^M| = 0$, we immediately have $|M \setminus T^M| = o(A_{T^M})$. If $|M \setminus T^M| \neq 0$, since the rightmost component in (15pp.98) is nonnegative, we have

$$\frac{|M \setminus T^M|}{A_{T^M}} \leq \left[\gamma \sum_{j=1}^b \frac{|M_j \setminus T_j^M|}{|M \setminus T^M|} \kappa_j \right]^{-1} \leq \left[\gamma \min_{j=1, \dots, b} \kappa_j \right]^{-1}$$

where the last inequality follows from $\sum_{j=1}^b \frac{|M_j \setminus T_j^M|}{|M \setminus T^M|} = 1$. By Assumption A4, $\min_{j=1, \dots, b} \kappa_j \rightarrow \infty$ as $n \rightarrow \infty$ and hence $|M \setminus T^M| = o(A_{T^M})$ when $|M \setminus T^M| \neq 0$ too.

Finally, consider the third condition in Lemma 13.8. In the proof of Theorem 3.3, $\mu_{Q_S S} = o(A_S)$ is immediate because $Q_S \setminus S = M \setminus S \subseteq S^C$, i.e. since all parameters in $M \setminus S$ are truly zero we have that $\mu_{Q_S S} = 0$. Here $M \setminus T^M$ is not necessarily a subset of S^C , hence $\mu_{Q_{T^M} T^M} \geq 0$. Note that, since $T^M \in \mathcal{T}(\kappa)$, we have that $S^L(\kappa) \subseteq T^M$. Moreover we have $M \cap S^I(\kappa) = T^M \cap S^I(\kappa)$. It follows that $M \setminus T^M \subseteq (S^I(\kappa) \cup S^L(\kappa))^C$, that is the elements of $M \setminus T^M$ are either inactive or belong to $S^S(\kappa)$. If $M \cap S^S(\kappa) = \emptyset$ then $M \setminus T^M \subseteq S^C$ and we immediately get $\mu_{Q_{T^M} T^M} = o(A_{T^M})$ because $\beta_{M \setminus T^M}^* = \mu_{Q_{T^M} T^M} = 0$. Assume now that $M \cap S^S(\kappa) \neq \emptyset$. Using that the rightmost component in (15pp.98) is nonnegative and Lemma 13.9, we have

$$\frac{\mu_{Q_{T^M} T^M}}{A_{T^M}} \leq \frac{\mu_{Q_{T^M} T^M}}{\gamma \sum_{j=1}^b |M_j \setminus T_j^M| \kappa_j} \leq \frac{n \bar{\lambda} \sum_{j=1}^b |(S_j \cap M_j) \setminus T_j^M| \max_{i \in (S_j \cap M_j) \setminus T_j^M} \beta_i^{*2}}{\gamma \sum_{j=1}^b |M_j \setminus T_j^M| \kappa_j}.$$

Observe that for all $j = 1, \dots, b$, $M_j \setminus T_j^M \supseteq (S_j \cap M_j) \setminus T_j^M$ and then $|M_j \setminus T_j^M| \geq |(S_j \cap M_j) \setminus T_j^M|$. We then get

$$\frac{\mu_{Q_{T^M} T^M}}{A_{T^M}} \leq \frac{n \bar{\lambda} \sum_{j=1}^b |(S_j \cap M_j) \setminus T_j^M| \max_{i \in (S_j \cap M_j) \setminus T_j^M} \beta_i^{*2}}{\gamma \sum_{j=1}^b |(S_j \cap M_j) \setminus T_j^M| \kappa_j}.$$

Moreover, since $M \setminus T^M \subseteq (S^I(\kappa) \cup S^L(\kappa))^C$ as discussed earlier, we have, for all $j = 1, \dots, b$, $(S_j \cap M_j) \setminus T_j^M \subseteq S_j^S(\kappa)$. It follows $\max_{i \in (S_j \cap M_j) \setminus T_j^M} \beta_i^{*2} \leq \max_{i \in S_j^S(\kappa)} \beta_i^{*2}$ and

$$\frac{\mu_{Q_{T^M} T^M}}{A_{T^M}} \leq \frac{n \bar{\lambda} \sum_{j=1}^b |(S_j \cap M_j) \setminus T_j^M| \max_{i \in S_j^S(\kappa)} \beta_i^{*2}}{\gamma \sum_{j=1}^b |(S_j \cap M_j) \setminus T_j^M| \kappa_j}. \quad (15pp.99)$$

Let $\bar{r} := \sum_{j=1}^b n\bar{\lambda} \max_{i \in S_j^S(\kappa)} \beta_i^{*2} / (\gamma\kappa_j)$. We show next that $\bar{r}$ is an upper bound on $\mu_{Q_{TM}TM} / A_{TM}$. By restricting the sum in $\bar{r}$ to the j such that $|(S_j \cap M_j) \setminus T_j^M| \neq 0$ and multiplying the numerator and denominator of the summand by $|(S_j \cap M_j) \setminus T_j^M|$, we get

$$\bar{r} \geq \sum_{j=1, |(S_j \cap M_j) \setminus T_j^M| \neq 0}^b \frac{|(S_j \cap M_j) \setminus T_j^M| n\bar{\lambda} \max_{i \in S_j^S(\kappa)} \beta_i^{*2}}{|(S_j \cap M_j) \setminus T_j^M| \gamma\kappa_j} \quad (15pp.100)$$

Using the property of (15pp.93) in the right-hand side of (15pp.100), we get

$$\begin{aligned} \bar{r} &\geq \frac{\sum_{j=1, |(S_j \cap M_j) \setminus T_j^M| \neq 0}^b |(S_j \cap M_j) \setminus T_j^M| n\bar{\lambda} \max_{i \in S_j^S(\kappa)} \beta_i^{*2} \left(1 + \sum_{l \neq j} \frac{|(S_l \cap M_l) \setminus T_l^M| \gamma\kappa_l}{|(S_j \cap M_j) \setminus T_j^M| \gamma\kappa_j}\right)}{\sum_{j=1, |(S_j \cap M_j) \setminus T_j^M| \neq 0}^b |(S_j \cap M_j) \setminus T_j^M| \gamma\kappa_j} \\ &\geq \frac{\sum_{j=1}^b |(S_j \cap M_j) \setminus T_j^M| n\bar{\lambda} \max_{i \in S_j^S(\kappa)} \beta_i^{*2}}{\sum_{j=1}^b |(S_j \cap M_j) \setminus T_j^M| \gamma\kappa_j} \end{aligned}$$

where last inequality follows from $\left(1 + \sum_{l \neq j} \frac{|(S_l \cap M_l) \setminus T_l^M| \gamma\kappa_l}{|(S_j \cap M_j) \setminus T_j^M| \gamma\kappa_j}\right) \geq 1$ for all j and from the identity $\sum_{j=1, |(S_j \cap M_j) \setminus T_j^M| \neq 0}^b |(S_j \cap M_j) \setminus T_j^M| \gamma\kappa_j = \sum_{j=1}^b |(S_j \cap M_j) \setminus T_j^M| \gamma\kappa_j$. Then, by (15pp.99),

$$\frac{\mu_{Q_{TM}TM}}{A_{TM}} \leq \bar{r} = \sum_{j=1}^b \frac{n\bar{\lambda} \max_{i \in S_j^S(\kappa)} \beta_i^{*2}}{\gamma\kappa_j}.$$

By definition of $S_j^S(\kappa)$, $n\bar{\lambda} \max_{i \in S_j^S(\kappa)} \beta_i^{*2} = o(\kappa_j)$ for all j and $\mu_{Q_{TM}TM} = o(A_{TM})$ when $M \cap S^S(\kappa) \neq \emptyset$ too. We can now apply Lemma 13.8 and get that for every $M \in \mathcal{M} \setminus \mathcal{T}(\kappa)$, for any $\psi \in (0, 1)$ and every n large enough, $\mathbb{E}(NC(M)) \leq e^{-\psi A_{TM}}$.

The second step of the proof is to lower-bound $A_{TM} = \gamma\Delta_{MTM} + \frac{1-\gamma}{6}\mu_{Q_{TM}M}$. Let

$$A_{TM}^* := \gamma \sum_{j=1}^b |M_j \setminus T_j^M| \kappa_j + \sum_{j=1}^b |T_j^M \setminus M_j| \left(\frac{1-\gamma}{6} n\rho(\mathbf{X}) \min_{i \in S_j^L(\kappa)} \beta_i^{*2} - \gamma\kappa_j \right)$$

Recall that, for every j , $T_j^M \setminus M_j \subseteq S_j^L(\kappa)$. We have then $\min_{i \in T_j^M \setminus M_j} \beta_i^{*2} \geq \min_{i \in S_j^L(\kappa)} \beta_i^{*2}$, and by (15pp.98), $A_{TM} \geq A_{TM}^*$. It follows that for any $\psi \in (0, 1)$ and for all n large enough,

$$\mathbb{E}(NC(M)) \leq e^{-\psi A_{TM}^*}. \quad (15pp.101)$$

To conclude the second part of the proof we lower-bound $\psi A_{TM}^* = \sum_{j=1}^b |M_j \setminus T_j^M| \psi \gamma \kappa_j + \sum_{j=1}^b |T_j^M \setminus M_j| \psi \left(\frac{1-\gamma}{6} n\rho(\mathbf{X}) \min_{i \in S_j^L(\kappa)} \beta_i^{*2} - \gamma\kappa_j \right)$. To do this, we obtain a lower bound for $\psi \gamma \kappa_j$ and for $\psi \left(\frac{1-\gamma}{6} n\rho(\mathbf{X}) \min_{i \in S_j^L(\kappa)} \beta_i^{*2} - \gamma\kappa_j \right)$.

The definition of $S_j^L(\kappa)$ implies there exists some $g'_j \rightarrow \infty$ such that

$$\frac{(1-\gamma)n\rho(\mathbf{X})}{6} \min_{i \in S_j^L(\kappa)} \beta_i^{*2} - \kappa_j = \ln(s_j) + g'_j. \quad (15\text{pp.102})$$

Let $\delta \in (0, 1)$ and denote $\bar{m}_j = \max \left\{ \frac{2\ln(p_j - s_j)}{f_j}, \frac{2\ln(s_j)}{g'_j} \right\}$, where f_j is given in Assumption A4.

Take $\psi = \max_{j=1, \dots, b} \frac{\xi + \delta + \bar{m}_j}{1 + \bar{m}_j}$ for some $\xi \in (0, 1 - \delta)$ then $\psi \in (0, 1)$ and we have, for every $j = 1, \dots, b$,

$$\psi > \frac{\delta + \frac{2\ln(p_j - s_j)}{f_j}}{1 + \frac{2\ln(p_j - s_j)}{f_j}} = \frac{\delta f_j/2 + \ln(p_j - s_j)}{f_j/2 + \ln(p_j - s_j)} \quad (15\text{pp.103})$$

$$\psi > \frac{\delta + \frac{2\ln(s_j)}{g'_j}}{1 + \frac{2\ln(s_j)}{g'_j}} = \frac{\delta g'_j/2 + \ln(s_j)}{g'_j/2 + \ln(s_j)} \geq \frac{\delta g'_j/2 + \ln(s_j)}{g'_j + \ln(s_j)}. \quad (15\text{pp.104})$$

Recall that Assumptions A4 defines $f_j = \kappa_j - \ln(p_j - s_j)$ and $\gamma = \frac{1}{2}(1 + \max_j \frac{\ln(p_j - s_j)}{\kappa_j})$ respectively. Hence,

$$\gamma \kappa_j \geq \frac{1}{2} \left(1 + \frac{\ln(p_j - s_j)}{\kappa_j} \right) \kappa_j = \ln(p_j - s_j) + \frac{1}{2}(\kappa_j - \ln(p_j - s_j)) = \ln(p_j - s_j) + \frac{1}{2}f_j.$$

Hence, by (15pp.103), we have

$$\psi \gamma \kappa_j \geq \psi \left(\ln(p_j - s_j) + \frac{1}{2}f_j \right) \geq \ln(p_j - s_j) + \delta \frac{1}{2}f_j. \quad (15\text{pp.105})$$

Further,

$$\psi \left(\frac{1-\gamma}{6} n\rho(\mathbf{X}) \min_{i \in S_j^L(\kappa)} \beta_i^{*2} - \gamma \kappa_j \right) \geq \psi \left(\ln(s_j) + g'_j \right) \geq \ln(s_j) + \delta \frac{1}{2}g'_j. \quad (15\text{pp.106})$$

where the first inequality follows from (15pp.102) and the second inequality from (15pp.104). In (15pp.101), $\psi A_{TM}^* = \sum_{j=1}^b |M_j \setminus T_j^M| \psi \gamma \kappa_j + \sum_{j=1}^b |T_j^M \setminus M_j| \psi \left(\frac{1-\gamma}{6} n\rho(\mathbf{X}) \min_{i \in S_j^L(\kappa)} \beta_i^{*2} - \gamma \kappa_j \right)$. Then by (15pp.105) and (15pp.106), we get

$$\mathbb{E}(NC(M)) \leq \exp \left\{ - \sum_{j=1}^b |M_j \setminus T_j^M| (\ln(p_j - s_j) + \delta \frac{f_j}{2}) - \sum_{j=1}^b |T_j^M \setminus M_j| (\ln(s_j) + \delta \frac{g'_j}{2}) \right\}. \quad (15\text{pp.107})$$

For the final step of the proof, denote $\mathcal{S} = \sum_{M \in \mathcal{M} \setminus \mathcal{T}(\kappa)} \mathbb{E}(NC(M))$ for convenience. By (15pp.107) we have

$$\mathcal{S} \leq \sum_{M \in \mathcal{M} \setminus \mathcal{T}(\kappa)} e^{-\sum_{j=1}^b |M_j \setminus T_j^M| \left(\ln(p_j - s_j) + \delta \frac{f_j}{2} \right) - \sum_{j=1}^b |T_j^M \setminus M_j| \left(\ln(s_j) + \delta \frac{g'_j}{2} \right)}.$$

We split the sum in the right-hand side above into sums over models M such that $T^M = T$ for some common $T \in \mathcal{T}(\kappa)$. Denote for any $T \in \mathcal{T}$, $\mathcal{M}(T) := \{M \in \mathcal{M} \setminus \mathcal{T}(\kappa) \mid T^M = T\}$, then

$$\mathcal{S} \leq \sum_{T \in \mathcal{T}(\kappa)} \sum_{M \in \mathcal{M}(T)} e^{-\sum_{j=1}^b |M_j \setminus T_j| \left(\ln(p_j - s_j) + \delta \frac{f_j}{2} \right) - \sum_{j=1}^b |T_j \setminus M_j| \left(\ln(s_j) + \delta \frac{g'_j}{2} \right)}. \quad (15\text{pp.108})$$

The right hand-side of (15pp.108) is composed of a double sum over $T \in \mathcal{T}(\kappa)$ and over $M \in \mathcal{M}(T)$. Consider the sum over $M \in \mathcal{M}(T)$, add T to it, and denote it

$$\mathcal{S}(T) = \sum_{M \in \mathcal{M}(T) \cup T} e^{-\sum_{j=1}^b |M_j \setminus T_j| \left(\ln(p_j - s_j) + \delta \frac{f_j}{2} \right) - \sum_{j=1}^b |T_j \setminus M_j| \left(\ln(s_j) + \delta \frac{g'_j}{2} \right)}. \quad (15\text{pp.109})$$

In the summand in the right-hand side of (15pp.109), the case $M = T$ correspond to $|M_j \setminus T_j^M| = |T_j^M \setminus M_j| = 0$ for all j and the summand is then 1. By (15pp.108), we then get that

$$\mathcal{S} \leq \sum_{T \in \mathcal{T}(\kappa)} (\mathcal{S}(T) - 1). \quad (15\text{pp.110})$$

For each $T = T^M$, we further split $\mathcal{S}(T)$ into sums over subsets of models M that have u_j more parameters than T^M in block j , and are missing w_j parameters from T^M in block j . Specifically, consider models M such that, for all j , $|M_j \setminus T_j^M| = u_j$ and $|T_j^M \setminus M_j| = w_j$ with $u_j \in \{0, \dots, p_j - s_j, \dots, p_j - s_j + |S_j^S(\kappa)|\}$ and $w_j \in \{0, \dots, |S_j^L(\kappa)|\}$. Denote by

$$S_{\mathbf{w}}^{\mathbf{u}}(T) = \sum_{M \in \mathcal{M}(T) \cup T : \forall j |M_j \setminus S_j| = u_j, |S_j \setminus M_j| = w_j} e^{-\sum_{j=1}^b u_j \left(\ln(p_j - s_j) + \delta \frac{f_j}{2} \right) - \sum_{j=1}^b w_j \left(\ln(s_j) + \delta \frac{g'_j}{2} \right)}.$$

We get

$$\mathcal{S}(T) = \sum_{w_1=0}^{|S_1^L(\kappa)|} \cdots \sum_{w_b=0}^{|S_b^L(\kappa)|} \sum_{u_1=0}^{p_1 - s_1 + |S_1^S(\kappa)|} \cdots \sum_{u_b=0}^{p_b - s_b + |S_b^S(\kappa)|} S_{\mathbf{w}}^{\mathbf{u}}(T). \quad (15\text{pp.111})$$

The number of models missing, for all j , w_j out of the $|S_j^L(\kappa)|$ large active parameters and having u_j inactive or small active parameters from B_j is $\prod_{j=1}^b \binom{p_j - s_j + |S_j^S(\kappa)|}{u_j} \binom{|S_j^L(\kappa)|}{w_j}$. We thus have

$$\begin{aligned} S_{\mathbf{w}}^{\mathbf{u}}(T) &= \left(\prod_{j=1}^b \binom{p_j - s_j + |S_j^S(\kappa)|}{u_j} \binom{|S_j^L(\kappa)|}{w_j} \right) e^{-\sum_{j=1}^b u_j \left(\ln(p_j - s_j) + \delta \frac{f_j}{2} \right) - \sum_{j=1}^b w_j \left(\ln(s_j) + \delta \frac{g'_j}{2} \right)} \\ &= \prod_{j=1}^b \binom{p_j - s_j + |S_j^S(\kappa)|}{u_j} e^{-u_j \left(\ln(p_j - s_j) + \delta \frac{f_j}{2} \right)} \binom{|S_j^L(\kappa)|}{w_j} e^{-w_j \left(\ln(s_j) + \delta \frac{g'_j}{2} \right)}. \end{aligned}$$

Inputting the expression above in (15pp.111) and factorizing over terms in u_j and w_j gives

$$\mathcal{S}(T) \leq \prod_{j=1}^b \left(\sum_{u_j=0}^{p_j-s_j+|S_j^S(\kappa)|} \binom{p_j-s_j+|S_j^S(\kappa)|}{u_j} e^{-u_j(\ln(p_j-s_j)+\delta\frac{f_j}{2})} \right) \\ \cdot \left(\sum_{w_j=0}^{|S_j^L(\kappa)|} \binom{|S_j^L(\kappa)|}{w_j} e^{-w_j(\ln(s_j)+\delta\frac{g'_j}{2})} \right).$$

By the bound in (15pp.78) and taking the terms in $u_j = 0$ and $w_j = 0$ out of the sums above, we have

$$\mathcal{S}(T) \leq \prod_{j=1}^b \left(1 + \sum_{u_j=1}^{p_j-s_j+|S_j^S(\kappa)|} e^{-u_j \left(\delta\frac{f_j}{2} - \ln \left(1 + \frac{|S_j^S(\kappa)|}{p_j-s_j} \right) - 1 \right)} \right) \quad (15pp.112) \\ \cdot \left(1 + \sum_{w_j=1}^{|S_j^L(\kappa)|} e^{-w_j \left(\delta\frac{g'_j}{2} + \ln \left(\frac{s_j}{|S_j^L(\kappa)|} \right) - 1 \right)} \right).$$

Denote

$$d_j = e^{1+\ln \left(1 + \frac{|S_j^S(\kappa)|}{p_j-s_j} \right) - \delta\frac{f_j}{2}}, \quad h_j = e^{1-\ln \left(\frac{s_j}{|S_j^L(\kappa)|} \right) - \delta\frac{g'_j}{2}}.$$

By assumption $|S_j^S(\kappa)| = O(p_j - s_j)$, and by definition of $|S_j^L(\kappa)|$ we have $s_j \geq |S_j^L(\kappa)|$. By Assumptions A4 and the definition of $|S_j^L(\kappa)|$, we also have $f_j \rightarrow \infty$ and $g'_j \rightarrow \infty$, and then $\lim_{n \rightarrow \infty} d_j = \lim_{n \rightarrow \infty} h_j = 0$. Using the properties of geometric series, for every j we have

$$1 + \sum_{u_j=1}^{p_j-s_j+|S_j^S(\kappa)|} e^{-u_j \left(\delta\frac{f_j}{2} - \ln \left(1 + \frac{|S_j^S(\kappa)|}{p_j-s_j} \right) - 1 \right)} = \frac{1-d_j^{p_j-s_j+|S_j^S(\kappa)|+1}}{1-d_j} \\ 1 + \sum_{w_j=1}^{|S_j^L(\kappa)|} e^{-w_j \left(\delta\frac{g'_j}{2} + \ln \left(\frac{s_j}{|S_j^L(\kappa)|} \right) - 1 \right)} = \frac{1-h_j^{|S_j^L(\kappa)|+1}}{1-h_j},$$

where both expressions converge to 1 as n grows. By (15pp.110) and (15pp.112):

$$\mathcal{S} \leq \sum_{T \in \mathcal{T}} \left(\prod_{j=1}^b \left(\frac{1-d_j^{p_j-s_j+|S_j^S(\kappa)|+1}}{1-d_j} \right) \left(\frac{1-h_j^{|S_j^L(\kappa)|+1}}{1-h_j} \right) - 1 \right).$$

Each of the summand vanishes as $n \rightarrow \infty$. Moreover, by assumption $|S^I(\kappa)| = O(1)$ and then $|\mathcal{T}(\kappa)| = 2^{|S^I(\kappa)|} = O(1)$. We thus have $\lim_{n \rightarrow \infty} \mathcal{S} = \lim_{n \rightarrow \infty} \sum_{M \in \mathcal{M} \setminus \mathcal{T}(\kappa)} \mathbb{E}(NC(M)) = 0$.

Further, by Lemma 13.7, $P(\hat{S}^{ei} \notin \mathcal{T}(\kappa)) \leq (|\mathcal{T}(\kappa)| + 1)\mathcal{S} = (2^{|S^I(\kappa)|} + 1)\mathcal{S}$. Since $\mathcal{S}$ vanishes and $|S^I(\kappa)| = O(1)$, $\lim_{n \rightarrow \infty} P(\hat{S}^{ei} \in \mathcal{T}(\kappa)) = 1$ as claimed.

15.14 Proof of Theorem ??

The proof is essentially the same as the proof of Theorem 3.3 replacing κ_j by $\tilde{\kappa}_j$ for every $j = 1, \dots, b$ and where for any model M δ_{MS} is taken as in (??). We first use Lemma 13.8 with $T = S$ to show that for every $M \neq S$, $\mathbb{E}(NC(M)) \leq e^{-\psi A_S}$ for every large enough n and any $\psi \in (0, 1)$, where $A_S = \gamma \Delta_{MS} + \frac{1-\gamma}{6} \mu_{Q_S M}$ (cf (??) and (40)), $\gamma \in (1/2, 1)$ is defined in Assumption A5bis and $Q_S = M \cup S$. The second step is to obtain a lower bound for A_S , which gives a new upper bound for $\mathbb{E}(NC(M))$. The final step is to use these bounds to get an upper-bound on $\sum_{M \in \mathcal{M} \setminus \{S\}} \mathbb{E}(NC(M))$ that vanishes under Assumptions A4bis and A5bis. We then use Lemma 3.2 to conclude on the vanishing of $P(\hat{S}^{ei} \neq S)$.

First, to show that $\mathbb{E}(NC(M)) \leq e^{-\psi A_S}$ for any $M \in \mathcal{M} \setminus \{S\}$, we show that A_S satisfies the conditions of Lemma 13.8, taking $T = S$. That is, we wish to show that, $A_S > 0$, $|M \setminus S| = o(A_S)$, and $\mu_{Q_S S} = o(A_S)$. Observe that Δ_{MS} , defined in (??), can be rewritten as $\Delta_{MS} = \sum_{j=1}^b (|M_j \setminus S_j| - |S_j \setminus M_j|) \tilde{\kappa}_j(M)$. By Lemma 10.2, for every $n \in \mathbb{N}$ we have

$$\begin{aligned} A_S &= \gamma \Delta_{MS} + \frac{1-\gamma}{6} \mu_{Q_S M} \\ &\geq \gamma \sum_{j=1}^b |M_j \setminus S_j| \tilde{\kappa}_j(M) + \sum_{j=1}^b |S_j \setminus M_j| \left(\frac{1-\gamma}{6} n \rho(\mathbf{X}) \beta_{\min, j}^*{}^2 - \gamma \tilde{\kappa}_j(M) \right) \end{aligned} \quad (15pp.113)$$

Since $M \neq S$, $|M \setminus S| \neq 0$ or $|S \setminus M| \neq 0$, then by Assumptions A4bis and A5bis, for every n large enough, $A_S > 0$. We immediately have $\mu_{Q_S S} = o(A_S)$ because $\beta_{Q_S \setminus S}^* = \beta_{M \setminus S}^* = 0$ (any parameter outside the true support S is by definition 0) and hence $\mu_{Q_S S} = 0$. If $|M \setminus S| = 0$, $|M \setminus S| = o(A_S)$ also immediately. Consider now the case $|M \setminus S| \neq 0$. By Assumption A5bis the last term in (15pp.113) is nonnegative, and hence

$$\frac{|M \setminus S|}{A_S} = \frac{|M \setminus S|}{\gamma \Delta_{MS} + \frac{1-\gamma}{6} \mu_{Q_S M}} \leq \left[\gamma \sum_{j=1}^b \frac{|M_j \setminus S_j|}{|M \setminus S|} \tilde{\kappa}_j(M) \right]^{-1} \leq \left[\gamma \min_{j=1, \dots, b | \tilde{\kappa}_j(M) \neq 0} \tilde{\kappa}_j(M) \right]^{-1}$$

where the last inequality follows from $\sum_{j=1}^b \frac{|M_j \setminus S_j|}{|M \setminus S|} = 1$. By Assumption A4bis, $\min_{j | \tilde{\kappa}_j(M) \neq 0} \tilde{\kappa}_j(M) \rightarrow \infty$ as $n \rightarrow \infty$, and hence $|M \setminus S| = o(A_S)$. Thus, by Lemma 13.8, for any $\psi \in (0, 1)$ and all n large enough, $\mathbb{E}(NC(M)) \leq e^{-\psi A_S}$.

For the second step of the proof, let A_S^* be the lower bound for A_S given in (15pp.113).

That is

$$A_S^* := \gamma \sum_{j=1}^b |M_j \setminus S_j| \tilde{\kappa}_j(M) + \sum_{j=1}^b |S_j \setminus M_j| \left(\frac{1-\gamma}{6} n \rho(\mathbf{X}) \beta_{\min,j}^*{}^2 - \gamma \tilde{\kappa}_j(M) \right).$$

By (15pp.113), we have for all n large enough,

$$\mathbb{E}(NC(M)) \leq e^{-\psi A_S^*}. \quad (15pp.114)$$

Assumption A5bis implies that there exists $g'_j \rightarrow \infty$ such that

$$\frac{(1-\gamma)n\rho(\mathbf{X})}{6} \beta_{\min,j}^*{}^2 - \tilde{\kappa}_j(M) = \ln(s_j) + g'_j. \quad (15pp.115)$$

Let $\delta \in (0, 1)$ and denote $\bar{m}_j = \max \left\{ \frac{2\ln(p_j - s_j)}{f_j(M)}, \frac{2\ln(s_j)}{g'_j} \right\}$ where f_j is given in Assumption A4bis. Take $\psi = \max_{j=1, \dots, b} \frac{\xi + \delta + \bar{m}_j}{1 + \bar{m}_j}$ for some $\xi \in (0, 1 - \delta)$ then $\psi \in (0, 1)$ and we have, for every $j = 1, \dots, b$,

$$\psi > \frac{\delta + \frac{2\ln(p_j - s_j)}{f_j(M)}}{1 + \frac{2\ln(p_j - s_j)}{f_j(M)}} = \frac{\delta f_j(M)/2 + \ln(p_j - s_j)}{f_j(M)/2 + \ln(p_j - s_j)} \geq \frac{\delta f_j(M)/2 + \ln((p_j - s_j)/|M_j \setminus S_j| \vee 1)}{f_j(M)/2 + \ln((p_j - s_j)/|M_j \setminus S_j| \vee 1)} \quad (15pp.116)$$

$$\psi > \frac{\delta + \frac{2\ln(s_j)}{g'_j}}{1 + \frac{2\ln(s_j)}{g'_j}} = \frac{\delta g'_j/2 + \ln(s_j)}{g'_j/2 + \ln(s_j)} \geq \frac{\delta g'_j/2 + \ln(s_j)}{g'_j + \ln(s_j)}. \quad (15pp.117)$$

By definition of γ in Assumption A5bis, for all M such that $|M_j \setminus S_j| > 0$,

$$\gamma \geq \frac{1}{2} \left(1 + \frac{\ln[(p_j - s_j)/|M_j \setminus S_j|]}{\ln[(p_j - s_j)/|M_j \setminus S_j|] + f_j(M)} \right),$$

and it follows that

$$\begin{aligned} \gamma \tilde{\kappa}_j(M) &\geq \frac{1}{2} \left(1 + \frac{\ln[(p_j - s_j)/|M_j \setminus S_j|]}{\tilde{\kappa}_j(M)} \right) \tilde{\kappa}_j(M) \\ &= \ln \left(\frac{p_j - s_j}{|M_j \setminus S_j|} \right) + \frac{1}{2} \left(\tilde{\kappa}_j(M) - \ln \left(\frac{p_j - s_j}{|M_j \setminus S_j|} \right) \right) \\ &= \ln \left(\frac{p_j - s_j}{|M_j \setminus S_j|} \right) + \frac{1}{2} f_j(M). \end{aligned}$$

Hence, by (15pp.116), when $|M_j \setminus S_j| > 0$ we have

$$\psi \gamma \tilde{\kappa}_j(M) \geq \psi \left(\ln \left(\frac{p_j - s_j}{|M_j \setminus S_j|} \right) + \frac{1}{2} f_j(M) \right) \geq \ln \left(\frac{p_j - s_j}{|M_j \setminus S_j|} \right) + \delta \frac{1}{2} f_j(M).$$

Taking the minimum of $f_j(M)$ over $M \in \mathcal{M}$ such that $|M_j \setminus S_j| > 0$, we get

$$\psi \gamma \tilde{\kappa}_j(M) \geq \ln \left(\frac{p_j - s_j}{|M_j \setminus S_j| \vee 1} \right) + \delta \frac{1}{2} \min_{M: |M_j \setminus S_j| > 0} f_j(M),$$

and then, for any j such that $|M_j \setminus S_j| \geq 0$,

$$|M_j \setminus S_j| \psi \gamma \tilde{\kappa}_j(M) \geq |M_j \setminus S_j| \left(\ln \left(\frac{p_j - s_j}{|M_j \setminus S_j| \vee 1} \right) + \delta \frac{1}{2} \min_{M: |M_j \setminus S_j| > 0} f_j(M) \right). \quad (15\text{pp.118})$$

Further using that $\gamma \in (0, 1)$,

$$\psi \left(\frac{1-\gamma}{6} n \rho(\mathbf{X}) \beta_{\min, j}^*{}^2 - \gamma \tilde{\kappa}_j(M) \right) \geq \psi \left(\ln(s_j) + g'_j \right) \geq \ln(s_j) + \delta \frac{1}{2} g'_j. \quad (15\text{pp.119})$$

where the first inequality follows from (15pp.115) and the second inequality from (15pp.117). In (15pp.114), $\psi A_S^* = \sum_{j=1}^b |M_j \setminus S_j| \psi \gamma \tilde{\kappa}_j(M) + \sum_{j=1}^b |S_j \setminus M_j| \psi \left(\frac{1-\gamma}{6} n \rho(\mathbf{X}) \beta_{\min, j}^*{}^2 - \gamma \tilde{\kappa}_j(M) \right)$. Then by (15pp.118) and (15pp.119), we get

$$\begin{aligned} \mathbb{E}(NC(M)) \leq \exp \left\{ - \sum_{j=1}^b |M_j \setminus S_j| \left(\ln \left(\frac{p_j - s_j}{|M_j \setminus S_j| \vee 1} \right) + \delta \frac{\min_{M: |M_j \setminus S_j| > 0} f_j(M)}{2} \right) \right. \\ \left. - \sum_{j=1}^b |S_j \setminus M_j| \left(\ln(s_j) + \delta \frac{g'_j}{2} \right) \right\}. \end{aligned} \quad (15\text{pp.120})$$

For the final step of the proof, denote $\mathcal{S} = \sum_{M \in \mathcal{M} \setminus \{S\}} \mathbb{E}(NC(M))$ for convenience. By (15pp.120) we have that

$$\mathcal{S} \leq \sum_{M \in \mathcal{M} \setminus \{S\}} e^{-\sum_{j=1}^b |M_j \setminus S_j| \left(\ln \left(\frac{p_j - s_j}{|M_j \setminus S_j| \vee 1} \right) + \delta \frac{\min_{M: |M_j \setminus S_j| > 0} f_j(M)}{2} \right) - \sum_{j=1}^b |S_j \setminus M_j| \left(\ln(s_j) + \delta \frac{g'_j}{2} \right)}.$$

Observe that if $|M_j \setminus S_j| = 0$ and $|S_j \setminus M_j| = 0$ for all j , then $M = S$ and the summand in the right-hand side above is 1. Then by adding and subtracting 1 we get that

$$\mathcal{S} \leq -1 + \sum_{M \in \mathcal{M}} e^{-\sum_{j=1}^b |M_j \setminus S_j| \left(\ln \left(\frac{p_j - s_j}{|M_j \setminus S_j| \vee 1} \right) + \delta \frac{\min_{M: |M_j \setminus S_j| > 0} f_j(M)}{2} \right) - \sum_{j=1}^b |S_j \setminus M_j| \left(\ln(s_j) + \delta \frac{g'_j}{2} \right)}.$$

We can split the sum in the right-hand side above into sums over the models that have the same number of inactive variables and missing the same number of truly active variables in every block. That is, the models M such that for all j , $|M_j \setminus S_j| = u_j$ and $|S_j \setminus M_j| = w_j$ with $u_j \in \{0, \dots, p_j - s_j\}$ and $w_j \in \{0, \dots, s_j\}$. Denote

$$S_{\mathbf{w}}^{\mathbf{u}} = \sum_{\substack{M \in \mathcal{M}: \forall j \ |M_j \setminus S_j| = u_j, \\ |S_j \setminus M_j| = w_j}} e^{-\sum_{j=1}^b u_j \left(\ln \left(\frac{p_j - s_j}{u_j \vee 1} \right) + \delta \frac{\min_{M: |M_j \setminus S_j| > 0} f_j(M)}{2} \right) - \sum_{j=1}^b w_j \left(\ln(s_j) + \delta \frac{g'_j}{2} \right)}.$$

We get that

$$\mathcal{S} \leq -1 + \sum_{w_1=0}^{s_1} \cdots \sum_{w_b=0}^{s_b} \sum_{u_1=0}^{p_1-s_1} \cdots \sum_{u_b=0}^{p_b-s_b} S_{\mathbf{w}}^{\mathbf{u}}. \quad (15\text{pp.121})$$

The number of models having, for all j , u_j inactive parameters and missing w_j out of the s_j active parameters is $\prod_{j=1}^b \binom{p_j-s_j}{u_j} \binom{s_j}{w_j}$. We thus have that

$$\begin{aligned} S_{\mathbf{w}}^{\mathbf{u}} &= \left(\prod_{j=1}^b \binom{p_j-s_j}{u_j} \binom{s_j}{w_j} \right) e^{-\sum_{j=1}^b u_j \left(\ln \left(\frac{p_j-s_j}{u_j \vee 1} \right) + \delta \frac{\min_{M: |M_j \setminus S_j| > 0} f_j(M)}{2} \right) - \sum_{j=1}^b w_j \left(\ln(s_j) + \delta \frac{g'_j}{2} \right)} \\ &= \prod_{j=1}^b \binom{p_j-s_j}{u_j} e^{-u_j \left(\ln \left(\frac{p_j-s_j}{u_j \vee 1} \right) + \delta \frac{\min_{M: |M_j \setminus S_j| > 0} f_j(M)}{2} \right)} \binom{s_j}{w_j} e^{-w_j \left(\ln(s_j) + \delta \frac{g'_j}{2} \right)}. \end{aligned}$$

Plugging the expression above into (15pp.121) gives that

$$\begin{aligned} \mathcal{S} \leq -1 + \sum_{w_1=0}^{s_1} \cdots \sum_{w_b=0}^{s_b} \sum_{u_1=0}^{p_1-s_1} \cdots \sum_{u_b=0}^{p_b-s_b} \prod_{j=1}^b \binom{p_j-s_j}{u_j} e^{-u_j \left(\ln \left(\frac{p_j-s_j}{u_j \vee 1} \right) + \delta \frac{\min_{M: |M_j \setminus S_j| > 0} f_j(M)}{2} \right)} \\ \cdot \binom{s_j}{w_j} e^{-w_j \left(\ln(s_j) + \delta \frac{g'_j}{2} \right)} \end{aligned}$$

and by factorizing,

$$\begin{aligned} \mathcal{S} \leq -1 + \prod_{j=1}^b \left(1 + \sum_{u_j=1}^{p_j-s_j} \binom{p_j-s_j}{u_j} e^{-u_j \left(\ln \left(\frac{p_j-s_j}{u_j} \right) + \delta \frac{\min_{M: |M_j \setminus S_j| > 0} f_j(M)}{2} \right)} \right) \\ \cdot \left(1 + \sum_{w_j=1}^{s_j} \binom{s_j}{w_j} e^{-w_j \left(\ln(s_j) + \delta \frac{g'_j}{2} \right)} \right). \end{aligned}$$

where the second inequality follows from first factorizing over terms in u_j and w_j and then taking the term in 0 out of every sum. By the bound on the binomial coefficient in (15pp.78), we have that

$$\mathcal{S} \leq -1 + \prod_{j=1}^b \left(1 + \sum_{u_j=1}^{p_j-s_j} e^{-u_j \left(\delta \frac{\min_{M: |M_j \setminus S_j| > 0} f_j(M)}{2} - 1 \right)} \right) \left(1 + \sum_{w_j=1}^{s_j} e^{-w_j \left(\delta \frac{g'_j}{2} - 1 \right)} \right). \quad (15\text{pp.122})$$

Denote

$$d_j = e^{1-\delta \frac{\min_{M: |M_j \setminus S_j| > 0} f_j(M)}{2}}, \quad h_j = e^{1-\delta \frac{g'_j}{2}}.$$

where both expressions go to zero as n increases since $\min_{M: |M_j \setminus S_j| > 0} f_j(M) \rightarrow \infty$ and

$g'_j \rightarrow \infty$. For every j , by the properties of geometric sums, we have

$$1 + \sum_{u_j=1}^{p_j-s_j} e^{-u_j \left(\delta^{\frac{\min_{M: |M_j \setminus S_j| > 0} f_j(M)}{2}} - 1 \right)} = \frac{1 - d_j^{p_j-s_j+1}}{1 - d_j},$$

$$1 + \sum_{w_j=1}^{s_j} e^{-w_j \left(\delta^{\frac{g'_j}{2}} - 1 \right)} = \frac{1 - h_j^{s_j+1}}{1 - h_j}.$$

Since both expressions converge to 1 as n grows, we get

$$\lim_{n \rightarrow \infty} \mathcal{S} = \lim_{n \rightarrow \infty} \sum_{M \in \mathcal{M} \setminus \{S\}} \mathbb{E}(NC(M)) = 0.$$

By Lemma 4.2, $P(\hat{S}^{ei} \neq S) \leq 2\mathcal{S}$ and then $\lim_{n \rightarrow \infty} P(\hat{S}^{ei} = S) = 1$.

15.15 Proof of Corollary ??

The proof strategy is to first show that η satisfies Assumption [A4bis](#), then that (??) implies Assumption [A5bis](#). The consistency results then follow from Theorem ??.

To show that η satisfies Assumption [A4bis](#), we show that, for all $M \neq S$ such that $|M \setminus S| > 0$ and $\tilde{\kappa}(M)$ defined in (??), the function $f(M) := \tilde{\kappa}(M) - \ln(p - s/|M \setminus S|)$ is lower bounded by a diverging sequence.

Let $M \in \mathcal{M}$ such that $M \neq S$ and $|M \setminus S| > 0$. Denote $|M \setminus S| = u$ and $|S \setminus M| = w$. We have $|M| = u + s - w$ and $|M| - |S| = u - w$. Since $M \neq S$ and $|M \setminus S| > 0$, we have $u - w \neq 0$ and $u > 0$. We consider first the case where $u - w > 0$, we have

$$\tilde{\kappa}(M) = \frac{\zeta}{u - w} \ln \left[\frac{\binom{p}{|M|}}{\binom{p}{|S|}} \right] + \frac{1}{2} \ln(n).$$

A well-known property of binomial coefficients is that for any positive integers n, h, k we have

$$\binom{n}{h} \binom{n-h}{k} = \binom{n}{h+k} \binom{h+k}{h}. \quad (15\text{pp.123})$$

Taking $n = p$, $k = u - w$ and $h = s$ in [\(15pp.123\)](#), we get

$$\tilde{\kappa}(M) = \frac{\zeta}{u - w} \ln \left[\frac{\binom{p-s}{u-w}}{\binom{p}{s}} \right] + \frac{1}{2} \ln(n). \quad (15\text{pp.124})$$

Standard bounds on binomial coefficient for $1 \leq k \leq n$ are

$$\left(\frac{n}{k}\right)^k \leq \binom{n}{k} \leq \frac{n^n}{k^k(n-k)^{n-k}}. \quad (15\text{pp.125})$$

Using the bounds in (15pp.125) in (15pp.124), we get

$$\frac{\zeta}{u-w} \ln \left[\frac{\binom{p}{|M|}}{\binom{p}{s}} \right] \geq \zeta \ln \left(\frac{p-s}{u-w} \right) - \zeta \left[\frac{s}{u-w} \ln \left(\frac{u-w}{s} + 1 \right) + \ln \left(1 + \frac{s}{u-w} \right) \right]. \quad (15\text{pp.126})$$

We have

$$\zeta \ln \left(\frac{p-s}{u-w} \right) \geq \zeta \ln \left(\frac{p-s}{u} \right) \geq \ln \left(\frac{p-s}{u} \right). \quad (15\text{pp.127})$$

where the first inequality follows from $u-w \leq u$ and the second from $\zeta \geq 1$. Observe also that $h(x) = \frac{\ln(x+1)}{x} + \ln(1+x^{-1})$ is decreasing for $x > 0$ and that $\frac{u-w}{s} \geq s^{-1}$. By (15pp.126) and (15pp.127), we then have

$$\tilde{\kappa}(M) \geq \ln \left(\frac{p-s}{u} \right) - \zeta s \ln(s^{-1} + 1) + \ln \left(\frac{\sqrt{n}}{(1+s)^\zeta} \right),$$

and for every M such that $u-w > 0$,

$$f(M) \geq -\zeta s \ln(s^{-1} + 1) + \ln \left(\frac{\sqrt{n}}{(1+s)^\zeta} \right). \quad (15\text{pp.128})$$

Since $\lim_{n \rightarrow \infty} s \ln(s^{-1} + 1) = 1$ and by assumption $s^{\zeta+1} = o(\sqrt{n})$, η satisfies Assumption A4bis in the case of M such that $u-w > 0$.

Consider now the case where M is such that $u-w < 0$. We have

$$\tilde{\kappa}(M) = \frac{\zeta}{w-u} \ln \left[\frac{\binom{p}{s}}{\binom{p}{|M|}} \right] + \frac{1}{2} \ln n.$$

Taking $n = p$, $k = w-u$ and $h = |M|$ in (15pp.123) gives

$$\tilde{\kappa}(M) = \frac{\zeta}{w-u} \ln \left[\frac{\binom{p-|M|}{w-u}}{\binom{s}{|M|}} \right] + \frac{1}{2} \ln n. \quad (15\text{pp.129})$$

Using the bounds in (15pp.125), we get

$$\frac{\zeta}{w-u} \ln \left[\frac{\binom{p}{s}}{\binom{p}{|M|}} \right] \geq \zeta \ln \left(\frac{p-|M|}{w-u} \right) + \zeta \left[\frac{s}{w-u} \ln \left(1 - \frac{w-u}{s} \right) - \ln \left(\frac{s}{w-u} - 1 \right) \right]. \quad (15\text{pp.130})$$

We have

$$\ln \left(\frac{p - |M|}{w - u} \right) = \ln \left(\frac{p - s}{u} \right) + \ln \left(\frac{p - |M|}{p - s} \right) + \ln \left(\frac{u}{w - u} \right).$$

Since $u - w < 0$, we have $|M| < s$ and $\ln((p - |M|)/(p - s)) \geq 0$. Since $w \leq s$ and $u \geq 1$, $\ln(u/(w - u)) \geq \ln(1/(s - 1))$. Using also that $\zeta \geq 1$, we have

$$\zeta \ln \left(\frac{p - |M|}{w - u} \right) \geq \ln \left(\frac{p - s}{u} \right) - \ln(s - 1) \quad (15\text{pp.131})$$

where the right-hand side is well defined because since $u > 0$ and $u - w < 0$, we have $2 \leq w \leq s$. Observe that $g : x \mapsto \frac{\ln(1-x)}{x} - \ln(x^{-1} - 1)$ for $x \in (0, 1)$ is increasing and that $1 > \frac{w-u}{s} \geq s^{-1}$. By (15pp.130) and (15pp.131), we then get for all $M \in \mathcal{M}$ such that $u - w < 0$,

$$\tilde{\kappa}(M) \geq \ln \left(\frac{p - s}{u} \right) + \zeta s \ln(1 - s^{-1}) + \ln \left(\frac{\sqrt{n}}{(s-1)^{\zeta+1}} \right).$$

and for every $M \in \mathcal{M}$ such that $w - u > 0$,

$$f(M) \geq \zeta s \ln(1 - s^{-1}) + \ln \left(\frac{\sqrt{n}}{(s-1)^{\zeta+1}} \right). \quad (15\text{pp.132})$$

Since $\lim_{n \rightarrow \infty} s \ln(1 - s^{-1}) = -1$ and by assumption $s^{\zeta+1} = o(\sqrt{n})$, η satisfies Assumption A4bis in the case $u - w < 0$ too.

We now show that if (??) holds then Assumption A5bis holds. Assumption A5bis for η as in (??) states that there exists $l_j \rightarrow \infty$ such that for large enough n ,

$$\sqrt{\frac{(1 - \gamma)n\rho(\mathbf{X})}{6}} \beta_{\min}^* - \sqrt{\max_{M \in \mathcal{M}} \tilde{\kappa}(M)} = \sqrt{\ln(s)} + l_j.$$

where γ takes value

$$\gamma = \frac{1}{2} \left(1 + \frac{\ln(p - s)}{\ln(p - s) + \min_{M: |M \setminus S| > 0} f(M)} \right). \quad (15\text{pp.133})$$

To show that if (??) holds then Assumption A5bis holds for η , it suffices to show that the following two inequalities

$$-\sqrt{\max_{M \in \mathcal{M}} \tilde{\kappa}(M)} \geq -\sqrt{\zeta \ln(p - s + 1) + \zeta - 1 + \frac{1}{2} \ln(n)}, \quad \text{and} \quad (15\text{pp.134})$$

$$\sqrt{\frac{(1 - \gamma)n\rho(\mathbf{X})}{6}} \beta_{\min, j}^* \geq \sqrt{\frac{\ln \left(\frac{\sqrt{n}}{(1+s)^\zeta} \right) + \zeta s \ln(1 - s^{-1})}{\ln(p - s) + \ln \left(\frac{\sqrt{n}}{(1+s)^\zeta} \right) + \zeta s \ln(1 - s^{-1})} \frac{n\rho(\mathbf{X})}{12}} \beta_{\min, j}^* \quad (15\text{pp.135})$$

hold. We start with (15pp.134). If $u - w > 0$, by (15pp.124), the upper bound in (15pp.78), and the lower bound in (15pp.125), then

$$\tilde{\kappa}(M) \leq \zeta \ln \left(\frac{(p-s)e}{u-w} \right) - \zeta \left(1 + \frac{s}{u-w} \right) \ln \left(1 + \frac{u-w}{s} \right) + \frac{1}{2} \ln(n).$$

Using that, for $x \geq 1$, $(1 + \frac{1}{x}) \ln(1+x) \geq 1$ and $\ln \left(\frac{(p-s)e}{x} \right) \leq \ln(p-s) + 1$. We get that, for all M such that $u - w > 0$,

$$\tilde{\kappa}(M) \leq \zeta \ln(p-s) + \zeta - 1 + \frac{1}{2} \ln(n). \quad (15pp.136)$$

If $w - u > 0$, by (15pp.129), the upper bound in (15pp.78), and the lower bound in (15pp.125), then

$$\tilde{\kappa}(M) \leq \zeta \ln \left(\frac{(p-s+(w-u))e}{w-u} \right) + \zeta \left(\frac{s}{w-u} - 1 \right) \ln \left(1 - \frac{w-u}{s} \right) + \frac{1}{2} \ln(n).$$

Using that, for $x \geq 1$, $(1 + \frac{1}{x}) \ln(1+x) \geq 1$ and $\ln \left(\frac{(p-s+x)e}{x} \right) \leq \ln(p-s+1) + 1$, we get that for all M such that $w - u > 0$,

$$\tilde{\kappa}(M) \leq \zeta \ln(p-s+1) + \zeta - 1 + \frac{1}{2} \ln(n). \quad (15pp.137)$$

By (15pp.136) and (15pp.137),

$$\max_{M \in \mathcal{M}} \tilde{\kappa}(M) \leq \zeta \ln(p-s+1) + \zeta - 1 + \frac{1}{2} \ln(n).$$

which shows (15pp.134). We now show (15pp.135). By (15pp.128) and (15pp.132), we have

$$\min_{M: |M \setminus S| > 0} f(M) \geq \zeta s \ln(1-s^{-1}) + \ln \left(\frac{\sqrt{n}}{(1+s)^\zeta} \right).$$

It follows that

$$\begin{aligned} \gamma &= \frac{1}{2} \left(1 + \frac{\ln(p-s)}{\ln(p-s) + \min_{M: |M \setminus S| > 0} f(M)} \right) \\ &\leq \frac{1}{2} \left(1 + \frac{\ln(p-s)}{\ln(p-s) + \ln \left(\frac{\sqrt{n}}{(1+s)^\zeta} \right) + \zeta s \ln(s^{-1}-1)} \right). \end{aligned}$$

Simple algebra gives

$$1 - \gamma \geq \frac{1}{2} \frac{\ln \left(\frac{\sqrt{n}}{(1+s)^\zeta} \right) + \zeta s \ln(1-s^{-1})}{\ln(p-s) + \ln \left(\frac{\sqrt{n}}{(1+s)^\zeta} \right) + \zeta s \ln(1-s^{-1})}$$

which shows (15pp.135) and that (??) is sufficient for Assumption A5bis to hold.

Since Assumptions A4bis and A5bis hold for η as in (??), by Theorem ??, $\lim_{n \rightarrow \infty} P(\hat{S}^{ei} \neq S) = 1$, as claimed.

15.16 Proof of Proposition ??

15.16.1 Part (i)

Consider $\mathcal{O}_j$ the set of models overfitting S by one variable from block B_j , defined in (16). The event $\hat{S}^{ei} = S$ (correct recovery of S) requires the event $\max_{M \in \mathcal{O}_j} \frac{NC(M)}{NC(S)} < 1$ where NC is the normalized criterion defined in (38). That is S is preferred to any model in $\mathcal{O}_j$, i.e. over-fitting S by 1 variable in block j .

$$P(\hat{S}^{ei} = S) \leq P\left(\max_{M \in \mathcal{O}_j} \frac{NC(M)}{NC(S)} < 1\right) = P\left(\max_{M \in \mathcal{O}_j} e^{C(M) - C(S)} < 1\right)$$

Using that $C(S) - C(M) = L_{SM}/2 + \Delta_{MS}$ for L_{SM} and Δ_{MS} defined in (??), and that $\Delta_{MS} = \eta_j(s_j + 1) - \eta_j(s_j)$, we obtain

$$P(\hat{S}^{ei} = S) \leq P\left(\max_{M \in \mathcal{O}_j} L_{MS} < 2(\eta_j(s_j + 1) - \eta_j(s_j))\right). \quad (15pp.138)$$

By Lemma 10.1, for every $M \in \mathcal{O}_j$, L_{MS} is distributed χ_1^2 which has positive support. It follows that if $\eta_j(s_j + 1) < \eta_j(s_j)$, then $P(\hat{S}^{ei} = S) = 0$. Consider now the case $\eta_j(s_j + 1) \geq \eta_j(s_j)$. Taking square root on both sides of the inequality in the right-most probability in (15pp.138), we have

$$P(\hat{S}^{ei} = S) \leq P\left(\max_{M \in \mathcal{O}_j} \sqrt{L_{MS}} < \sqrt{2(\eta_j(s_j + 1) - \eta_j(s_j))}\right). \quad (15pp.139)$$

The inequality in (15pp.139) is the same as (15pp.88) in the proof of Proposition ?? with κ_j replaced by $\eta_j(s_j + 1) - \eta_j(s_j)$. Proceeding as in that proof, that is, using that the L_{MS} are squared standard Gaussians and the lower bound on the maximum of standard Gaussian in Theorem 3.4 in ?, we get that if

$$\lim_{n \rightarrow \infty} \frac{\eta_j(s_j + 1) - \eta_j(s_j)}{\underline{\lambda}_j^2 \ln(p_j - s_j)} < 1$$

where $\underline{\lambda}_j$ is defined in (18), then $\lim_{n \rightarrow \infty} P(\hat{S}^{ei} = S) = 0$.

15.16.2 Part (ii)

Consider the model M under-fitting S by the variable in block j associated to $\beta_{\min,j}^*$. The event $\hat{S}^{ei} = S$ (correct recovery of S) requires the event $\frac{NC(M)}{NC(S)} < 1$ where NC is the normalized criterion in (38). That is S is preferred to model M .

$$P(\hat{S}^{ei} = S) \leq P\left(\frac{NC(M)}{NC(S)} < 1\right) = P\left(e^{C(M)-C(S)} < 1\right)$$

Using that $C(S) - C(M) = L_{SM}/2 + \Delta_{MS}$ for L_{SM} and Δ_{MS} defined in (??), and that $\Delta_{MS} = \eta_j(|S_j - 1|) - \eta_j(s_j)$ we obtain

$$P(\hat{S}^{ei} = S) \leq P\left(L_{SM} > 2\left(\eta_j(s_j) - \eta_j(s_j - 1)\right)\right). \quad (15pp.140)$$

By assumption, $\eta_j(s_j) - \eta_j(s_j - 1)$ is nonnegative and taking square root on both sides of the inequality in the right-most probability in (15pp.140), we get

$$P(\hat{S}^{ei} = S) \leq P\left(\sqrt{L_{SM}} > \sqrt{2\left(\eta_j(s_j) - \eta_j(s_j - 1)\right)}\right). \quad (15pp.141)$$

The inequality in (15pp.141) is the same as (15pp.95) in the proof of Proposition ?? with κ_j replaced by $\eta_j(s_j) - \eta_j(s_j - 1)$. Proceeding as in that proof, that is, using that L_{SM} is a squared Gaussian, the symmetry of the standard Gaussian for the case $2\left(\eta_j(s_j) - \eta_j(s_j - 1)\right) > \mu_{SM}$ and Lemma 13.1 (ii) and Lemma ?? for the case $2\left(\eta_j(s_j) - \eta_j(s_j - 1)\right) \leq \mu_{SM}$, we have that if

$$\lim_{n \rightarrow \infty} \sqrt{n\bar{\lambda}\beta_{\min,j}^*} - \sqrt{2\left(\eta_j(s_j) - \eta_j(s_j - 1)\right)} < \infty$$

where $\bar{\lambda}$ is defined in Lemma ??, then $P(\hat{S}^{ei} = S) < 1$.

15.17 Proof of Corollary ??

First, observe that if $\eta_j(s_j + 1) - \eta_j(s_j) < 0$ then $\lim_{n \rightarrow \infty} (\eta_j(s_j + 1) - \eta_j(s_j)) / (\lambda_j^2 \ln(p_j - s_j)) < 1$, and, by Proposition ?? (i), $\lim_{n \rightarrow \infty} P(\hat{S}^{ei} = S) < 1$. In the rest of the proof, we then assume $\eta_j(s_j + 1) - \eta_j(s_j) \geq 0$ which implies $\eta_j(s_j) - \eta_j(s_j - 1) \geq 0$ by the concavity

assumption. Next, we re-write

$$\begin{aligned} \sqrt{n\bar{\lambda}\beta_{\min,j}^*} - \underline{\lambda}_j \sqrt{2\ln(p_j - s_j)} &= \sqrt{n\bar{\lambda}\beta_{\min,j}^*} - \sqrt{2(\eta_j(s_j) - \eta_j(s_j - 1))} + \\ &\quad \sqrt{2(\eta_j(s_j) - \eta_j(s_j - 1))} - \sqrt{2(\eta_j(s_j + 1) - \eta_j(s_j))} + \\ &\quad \underline{\lambda}_j \sqrt{2\ln(p_j - s_j)} \left(\sqrt{\frac{\eta_j(s_j + 1) - \eta_j(s_j)}{\underline{\lambda}_j^2 \ln(p_j - s_j)}} - 1 \right). \end{aligned}$$

By the concavity assumption, $\sqrt{2(\eta_j(s_j) - \eta_j(s_j - 1))} - \sqrt{2(\eta_j(s_j + 1) - \eta_j(s_j))} \geq 0$ and then

$$\begin{aligned} \sqrt{n\bar{\lambda}\beta_{\min,j}^*} - \underline{\lambda}_j \sqrt{2\ln(p_j - s_j)} &\geq \sqrt{n\bar{\lambda}\beta_{\min,j}^*} - \sqrt{2(\eta_j(s_j) - \eta_j(s_j - 1))} + \\ &\quad \underline{\lambda}_j \sqrt{2\ln(p_j - s_j)} \left(\sqrt{\frac{\eta_j(s_j + 1) - \eta_j(s_j)}{\underline{\lambda}_j^2 \ln(p_j - s_j)}} - 1 \right). \end{aligned}$$

Condition (??) then implies that there exists $c \in \mathbb{R}^+$ such that

$$\lim_{n \rightarrow \infty} \sqrt{n\bar{\lambda}\beta_{\min,j}^*} - \sqrt{2(\eta_j(s_j) - \eta_j(s_j - 1))} + \underline{\lambda}_j \sqrt{2\ln(p_j - s_j)} \left(\sqrt{\frac{\eta_j(s_j + 1) - \eta_j(s_j)}{\underline{\lambda}_j^2 \ln(p_j - s_j)}} - 1 \right) \leq c \quad (15pp.142)$$

Consider the case $\lim_{n \rightarrow \infty} \sqrt{\frac{\eta_j(s_j + 1) - \eta_j(s_j)}{\underline{\lambda}_j^2 \ln(p_j - s_j)}} - 1 < 0$. Then by Proposition ?? (i), we have $\lim_{n \rightarrow \infty} P(\hat{S}^{ei} = S) < 1$. Now consider the case $\lim_{n \rightarrow \infty} \sqrt{\frac{\eta_j(s_j + 1) - \eta_j(s_j)}{\underline{\lambda}_j^2 \ln(p_j - s_j)}} - 1 \geq 0$. Since by assumption $\underline{\lambda}_j^2 \ln(p_j - s_j) \rightarrow \infty$, $\eta_j(s_j + 1) - \eta_j(s_j) \rightarrow \infty$ and by concavity $\eta_j(s_j) - \eta_j(s_j - 1) \rightarrow \infty$ too. Moreover, condition (15pp.142) implies that $\lim_{n \rightarrow \infty} \sqrt{n\bar{\lambda}\beta_{\min,j}^*} - \sqrt{2(\eta_j(s_j) - \eta_j(s_j - 1))} \leq c$. By Proposition ?? (ii), we have that $\lim_{n \rightarrow \infty} P(\hat{S}^{ei} = S) < 1$.

15.18 Proof of Corollary ??

Since $\hat{S}^{sd}$ is $\hat{S}^{ei}$ with $b = 1$, the assumptions of Corollary ?? for $\hat{S}^{sd}$ are met and $\lim_{n \rightarrow \infty} P(\hat{S}^{sd} = S) < 1$. Since Assumptions A4bis and A5bis hold, by Theorem ?? we have that $\lim_{n \rightarrow \infty} P(\hat{S}^{ei} = S) = 1$.

16 Proofs of Section 3.4

16.1 Proof of Proposition 3.9

Denote

$$A_j := \frac{1}{p_j} \sum_{i \in B_j} \sum_{M \in \mathcal{T}(\kappa) | i \in M} NC(M) \quad \text{and} \quad C_j := \frac{1}{p_j} \sum_{i \in B_j} \sum_{M \in \mathcal{M} \setminus \mathcal{T}(\kappa) | i \in M} NC(M).$$

For every $j = 1, \dots, b$, we have the decomposition

$$\frac{\hat{s}_j}{p_j} = \frac{\sum_{i \in B_j} \sum_{M \in \mathcal{M} | i \in M} NC(M)}{p_j} = A_j + C_j. \quad (16\text{pp.143})$$

To show the lower bound on $\hat{s}_j/p_j$, we decompose A_j

$$\begin{aligned} A_j &= \sum_{M \in \mathcal{T}(\kappa)} NC(M) \sum_{i \in B_j} \frac{I(i \in M_j)}{p_j} \\ &= \sum_{M \in \mathcal{T}(\kappa)} NC(M) \sum_{i \in S_j^L(\kappa)} \frac{I(i \in M_j)}{p_j} + \sum_{M \in \mathcal{T}(\kappa)} NC(M) \sum_{i \in B_j \setminus S_j^L(\kappa)} \frac{I(i \in M_j)}{p_j} \\ &= \frac{|S_j^L(\kappa)|}{p_j} \sum_{M \in \mathcal{T}(\kappa)} NC(M) + \sum_{M \in \mathcal{T}(\kappa)} NC(M) \sum_{i \in B_j \setminus S_j^L(\kappa)} \frac{I(i \in M_j)}{p_j}. \end{aligned} \quad (16\text{pp.144})$$

where the last equality follows from $I(i \in M_j) = 1$ for all $i \in S^L(\kappa)$ when $M \in \mathcal{T}(\kappa)$. The rightmost term above and C_j are nonnegative, then by the linearity of the expectation

$$\mathbb{E}\left(\frac{\hat{s}_j}{p_j}\right) \geq \frac{|S_j^L(\kappa)|}{p_j} \sum_{M \in \mathcal{T}(\kappa)} \mathbb{E}(NC(M)).$$

By Theorem 3.7, $\lim_{n \rightarrow \infty} \sum_{M \in \mathcal{T}(\kappa)} \mathbb{E}(NC(M)) = 1$. It follows that $\lim_{n \rightarrow \infty} \mathbb{E}\left(\frac{\hat{s}_j}{p_j}\right) \geq \frac{|S_j^L(\kappa)|}{p_j}$ for every $j = 1, \dots, b$.

We now prove the upper bound. Recall that $\mathcal{T}(\kappa)$ by definition includes models that have no small signals, i.e. all parameters are in $S^L(\kappa) \cup S^I(\kappa)$. That is, for all $M \in \mathcal{T}(\kappa)$, we have that $I(i \in M_j) = 0$ for all $i \in B_j \setminus (S_j^L(\kappa) \cup S_j^I(\kappa))$. Hence, A_j in (16pp.144) satisfies

$$\begin{aligned} A_j &= \frac{|S_j^L(\kappa)|}{p_j} \sum_{M \in \mathcal{T}(\kappa)} NC(M) + \sum_{M \in \mathcal{T}(\kappa)} NC(M) \sum_{i \in S_j^I(\kappa)} \frac{I(i \in M_j)}{p_j} \\ &\leq \frac{|S_j^L(\kappa)|}{p_j} \sum_{M \in \mathcal{T}(\kappa)} NC(M) + \frac{|S_j^I(\kappa)|}{p_j} \sum_{M \in \mathcal{T}(\kappa)} NC(M) \end{aligned}$$

where the inequality follows from $\sum_{i \in S_j^I(\kappa)} I(i \in M_j) \leq |S_j^I(\kappa)|$ for all M . By (16pp.143), we then have

$$\frac{\hat{s}_j}{p_j} \leq \frac{|S_j^L(\kappa)| + |S_j^I(\kappa)|}{p_j} \sum_{M \in \mathcal{T}(\kappa)} NC(M) + C_j. \quad (16pp.145)$$

Moreover, for every $j = 1, \dots, b$, C_j satisfies

$$C_j = \sum_{M \in \mathcal{M} \setminus \mathcal{T}(\kappa)} NC(M) \sum_{i \in B_j} \frac{I(i \in M_j)}{p_j} \leq \sum_{M \in \mathcal{M} \setminus \mathcal{T}(\kappa)} NC(M) \quad (16pp.146)$$

where the inequality follows from $\sum_{i \in B_j} \frac{I(i \in M_j)}{p_j} \leq 1$ for all M . Taking expectations in (16pp.145) and (16pp.146) gives

$$\mathbb{E}\left(\frac{\hat{s}_j}{p_j}\right) \leq \frac{|S_j^L(\kappa)| + |S_j^I(\kappa)|}{p_j} \sum_{M \in \mathcal{T}(\kappa)} \mathbb{E}(NC(M)) + \sum_{M \in \mathcal{M} \setminus \mathcal{T}(\kappa)} \mathbb{E}(NC(M)).$$

By Theorem 3.7, we have on one hand $\lim_{n \rightarrow \infty} \sum_{M \in \mathcal{T}(\kappa)} \mathbb{E}(NC(M)) = 1$ and, on the other hand, $\lim_{n \rightarrow \infty} \sum_{M \in \mathcal{M} \setminus \mathcal{T}(\kappa)} \mathbb{E}(NC(M)) = 0$. It follows that $\lim_{n \rightarrow \infty} \mathbb{E}\left(\frac{\hat{s}_j}{p_j}\right) \leq \frac{|S_j^L(\kappa)| + |S_j^I(\kappa)|}{p_j} = \frac{s_j - |S_j^S(\kappa)|}{p_j}$ for every $j = 1, \dots, b$, which proves the upper bound.

16.2 Proof of Lemma 3.8

Denote the marginal likelihood of $\mathbf{y}$ given $\boldsymbol{\theta}$ by $H(\boldsymbol{\theta}) = p(\mathbf{y} \mid \boldsymbol{\theta})$. For every $j = 1, \dots, b$, the partial derivative of its logarithm with respect to θ_j is

$$\frac{\partial \ln H(\boldsymbol{\theta})}{\partial \theta_j} = \frac{\partial H(\boldsymbol{\theta})}{\partial \theta_j} H(\boldsymbol{\theta})^{-1}. \quad (16pp.147)$$

Observe that:

$$H(\boldsymbol{\theta}) = \sum_{M \in \mathcal{M}} p(\mathbf{y} \mid M, \boldsymbol{\theta}) p(M \mid \boldsymbol{\theta}) \quad \text{and} \quad \frac{\partial H(\boldsymbol{\theta})}{\partial \theta_j} = \sum_{M \in \mathcal{M}} p(\mathbf{y} \mid M, \boldsymbol{\theta}) \frac{\partial p(M \mid \boldsymbol{\theta})}{\partial \theta_j} \quad (16pp.148)$$

Recall that each model is defined as $M = (m_1, \dots, m_p)$ where $m_i = I(\beta_i \neq 0)$ indicates whether variable j is included under M , and that our choice of model prior in (8) is

$$p(M \mid \boldsymbol{\theta}) = \prod_{j=1}^b (\theta_j)^{\sum_{i \in B_j} m_i} (1 - \theta_j)^{p_j - \sum_{i \in B_j} m_i}.$$

Hence, simple algebra shows that for every $M \in \mathcal{M}$

$$\frac{\partial p(M \mid \boldsymbol{\theta})}{\partial \theta_j} = p(M \mid \boldsymbol{\theta}) \left(\frac{\sum_{i \in B_j} m_i}{\theta_j} - \frac{p_j - \sum_{i \in B_j} m_i}{1 - \theta_j} \right). \quad (16pp.149)$$

Replacing (16pp.149) into (16pp.148), and using that for any function f

$$\sum_{M \in \mathcal{M}} \sum_{i \in B_j} m_i f(M) = \sum_{i \in B_j} \sum_{M \in \mathcal{M}: m_i=1} f(M),$$

we get

$$\begin{aligned} \frac{\partial H(\boldsymbol{\theta})}{\partial \theta_j} &= \frac{1}{\theta_j} \sum_{i \in B_j} \sum_{M \in \mathcal{M}: m_i=1} p(\mathbf{y} \mid M, \boldsymbol{\theta}) p(M \mid \boldsymbol{\theta}) \\ &\quad - \frac{1}{1 - \theta_j} \left[p_j H(\boldsymbol{\theta}) - \sum_{i \in B_j} \sum_{M \in \mathcal{M}: m_i=1} p(\mathbf{y} \mid M, \boldsymbol{\theta}) p(M \mid \boldsymbol{\theta}) \right] \end{aligned} \quad (16pp.150)$$

Note that

$$\frac{\sum_{M \in \mathcal{M}: m_i=1} p(\mathbf{y} \mid M, \boldsymbol{\theta}) p(M \mid \boldsymbol{\theta})}{H(\boldsymbol{\theta})} = \frac{\sum_{M \in \mathcal{M}: m_i=1} p(\mathbf{y}, M \mid \boldsymbol{\theta})}{p(\mathbf{y} \mid \boldsymbol{\theta})} = P(m_i = 1 \mid \mathbf{y}, \boldsymbol{\theta})$$

By (16pp.147), we then get the following expression of the partial derivative, for every $j = 1, \dots, b$,

$$\frac{\partial \ln H(\boldsymbol{\theta})}{\partial \theta_j} = \frac{1}{\theta_j} \sum_{i \in B_j} P(m_i = 1 \mid \mathbf{y}, \boldsymbol{\theta}) - \frac{1}{1 - \theta_j} \left[p_j - \sum_{i \in B_j} P(m_i = 1 \mid \mathbf{y}, \boldsymbol{\theta}) \right]$$

Setting the partial derivatives to 0 and solving for θ_j gives the desired result.

16.3 Proof of Theorem ??

The proof strategy is to show that Assumptions A1, A3, A4 and A5 hold to apply Theorem 3.3. Recall that Theorem ?? makes Assumptions A1, A3, A7 and A8, hence it suffices to show that A4-A5 hold. We first derive a convenient decomposition of the empirical Bayes penalties κ_j^{EB} . The second step of the proof is to show that, with probability going to 1, these κ_j^{EB} satisfy Assumption A4. The third step consists in showing that Assumption A8 implies Assumption A5 for κ_j^{EB} . The consistency of $\hat{S}^{EB,ei}$ then follows from Theorem 3.3.

Denote for any $M \in \mathcal{M}$, $NC^\circ(M)$, the normalized criterion value for model M under Step

1 penalty κ° . For this choice of penalty and every $j = 1, \dots, b$, we have

$$\begin{aligned} \frac{\hat{s}_j}{p_j} &= \frac{1}{p_j} \sum_{i \in B_j} \sum_{M \in \mathcal{M}} NC^\circ(M) I(i \in M) \\ &= \frac{1}{p_j} \sum_{M \in \mathcal{M}} NC^\circ(M) \sum_{i \in B_j} I(i \in M) \\ &= \frac{s_j}{p_j} NC^\circ(S) + \sum_{M \in \mathcal{M} | M \neq S} \frac{|M_j|}{p_j} NC^\circ(M). \end{aligned}$$

Using that $NC^\circ(S) = 1 - \sum_{M \in \mathcal{M} | M \neq S} NC^\circ(M)$, we get

$$\frac{\hat{s}_j}{p_j} = \frac{s_j}{p_j} + \sum_{M \in \mathcal{M} | M \neq S} \frac{|M_j| - s_j}{p_j} NC^\circ(M).$$

Consider the decomposition of the sum in the right-hand side above between the sum over models M that contain more parameters than S in block j and the sum over those that contain less parameters than S in block b . Denote

$$\mathcal{O}_j^\circ := \sum_{M \in \mathcal{M} | |M_j| > s_j} \frac{|M_j| - s_j}{p_j} NC^\circ(M) \quad \text{and} \quad U_j^\circ := \sum_{M \in \mathcal{M} | |M_j| < s_j} \frac{s_j - |M_j|}{p_j} NC^\circ(M).$$

We have

$$\frac{\hat{s}_j}{p_j} = \frac{s_j}{p_j} + \mathcal{O}_j^\circ - U_j^\circ. \tag{16pp.151}$$

Observe that we have the following decomposition of Step 2 penalties

$$\kappa_j^{EB} = \ln(p_j - s_j) + \ln\left(\frac{\sqrt{n}}{s_j}\right) + \ln\left(\frac{p_j - \hat{s}_j}{p_j - s_j}\right) + \ln\left(\frac{s_j}{\hat{s}_j}\right).$$

By (16pp.151), it follows that

$$\kappa_j^{EB} = \ln(p_j - s_j) + \ln\left(\frac{\sqrt{n}}{s_j}\right) + \ln\left(1 - \frac{p_j(\mathcal{O}_j^\circ - U_j^\circ)}{p_j - s_j}\right) + \ln\left(\frac{s_j}{\hat{s}_j}\right), \tag{16pp.152}$$

completing the first step of the proof.

We continue with the second step of the proof: showing that the κ_j^{EB} 's satisfy Assumption A4 with probability going to 1. Recall that Assumption A4 states that there exists $f_j \rightarrow \infty$ (as $n \rightarrow \infty$) such that for every sufficiently large n ,

$$\kappa_j = \ln(p_j - s_j) + f_j.$$

Since U_j° is nonnegative, a lower bound on κ_j^{EB} is

$$\kappa_j^{EB} \geq \ln(p_j - s_j) + \ln\left(\frac{\sqrt{n}}{s_j}\right) + \ln\left(1 - \frac{p_j O_j^\circ}{p_j - s_j}\right) + \ln\left(\frac{s_j}{\hat{s}_j}\right). \quad (16\text{pp.153})$$

Plugging in the definition of O_j° , we have that

$$\frac{p_j O_j^\circ}{p_j - s_j} = \sum_{M \in \mathcal{M} \mid |M_j| > s_j} \frac{|M_j| - s_j}{p_j - s_j} NC^\circ(M) \leq \sum_{M \in \mathcal{M} \mid |M_j| > s_j} NC^\circ(M)$$

where the inequality follows from $(|M_j| - s_j)/(p_j - s_j) \leq 1$ for all M . Note that if M is such that $|M_j| > s_j$, then $M \notin \mathcal{T}(\kappa^\circ)$ (this follows immediately from the definition of $\mathcal{T}(\kappa)$ in (25)) and therefore $\sum_{M \in \mathcal{M} \mid |M_j| > s_j} NC^\circ(M) \leq \sum_{M \in \mathcal{M} \setminus \mathcal{T}(\kappa^\circ)} NC^\circ(M)$. Moreover, κ° satisfies Assumption A4 and the assumptions of Theorem 3.7 are met for κ° . Then, by Theorem 3.7, $\lim_{n \rightarrow \infty} \sum_{M \in \mathcal{M} \setminus \mathcal{T}(\kappa^\circ)} \mathbb{E}(NC^\circ(M)) = \lim_{n \rightarrow \infty} \sum_{M \in \mathcal{M} \mid |M_j| > s_j} \mathbb{E}(NC^\circ(M)) = 0$. It follows that $\frac{p_j O_j^\circ}{p_j - s_j}$ vanishes in probability and so does $\ln\left(1 - \frac{p_j O_j^\circ}{p_j - s_j}\right)$. By Assumption A7, we also have that $\ln(\sqrt{n} s_j^{-1}) \rightarrow \infty$. Then, to show that Assumption A4 holds with probability going to 1, it is enough to show that with probability going to 1, $\ln(s_j/\hat{s}_j)$ is nonnegative. Observe that all assumptions in Proposition 3.9 are also met for κ° . By (16pp.145) and (16pp.146) in the proof of Proposition 3.9, we have

$$\frac{\hat{s}_j}{p_j} \leq \frac{|S_j^L(\kappa^\circ)| + |S^I(\kappa^\circ)_j|}{p_j} \sum_{M \in \mathcal{T}(\kappa^\circ)} NC^\circ(M) + \frac{1}{p_j} \sum_{M \in \mathcal{M} \setminus \mathcal{T}(\kappa^\circ)} NC^\circ(M).$$

By Theorem 3.7, $\sum_{M \in \mathcal{T}(\kappa^\circ)} NC(M)$ and $\sum_{M \in \mathcal{M} \setminus \mathcal{T}(\kappa^\circ)} NC(M)$ converge in probability to 1 and 0 respectively. We then have that, with probability going to 1,

$$\frac{\hat{s}_j}{p_j} \leq \frac{|S_j^L(\kappa^\circ)| + |S^I(\kappa^\circ)_j|}{p_j} \implies \ln\left(\frac{s_j}{\hat{s}_j}\right) \geq \ln\left(\frac{s_j}{|S_j^L(\kappa^\circ)| + |S^I(\kappa^\circ)_j|}\right) \geq 0.$$

We then obtain that, with probability going to 1,

$$\kappa_j^{EB} \geq \ln(p_j - s_j) + \ln\left(\frac{\sqrt{n}}{s_j}\right) \quad (16\text{pp.154})$$

and that the κ_j^{EB} satisfy Assumption A4, completing the second part of the proof.

For the third and final part of the proof, we now show that Assumption A8 implies Assumption A5 for the κ_j^{EB} with probability going to 1. Recall that Assumption A5 for the

κ_j^{EB} states that for each block j there exists $g_j \rightarrow \infty$ such that for large enough n ,

$$\sqrt{\frac{(1-\gamma)n\rho(\mathbf{X})}{6}}\beta_{\min,j}^* - \sqrt{\kappa_j^{EB}} = \sqrt{\ln(s_j)} + g_j.$$

where γ takes value

$$\gamma = \frac{1}{2} \left(1 + \max_j \frac{\ln(p_j - s_j)}{\kappa_j^{EB}} \right). \quad (16\text{pp.155})$$

Observe that Assumption A8 and Assumption A5 take the same form. To show that Assumption A8 implies Assumption A5 for the κ_j^{EB} with probability going to 1, it suffices to show that the following two inequalities

$$\sqrt{\frac{(1-\gamma)n\rho(\mathbf{X})}{6}}\beta_{\min,j}^* \geq \sqrt{\frac{(1-\psi)n\rho(\mathbf{X})}{6}}\beta_{\min,j}^*, \quad (16\text{pp.156})$$

$$-\sqrt{\kappa_j^{EB}} \geq -\sqrt{\ln\left(\frac{p}{|S^L(\kappa^\circ)|} - 1\right) + \frac{1}{2}\ln(n)} \quad (16\text{pp.157})$$

hold with probability going to 1 for γ as in (16pp.155) and $\psi = \frac{1}{2} \left(1 + \max_j \frac{\ln(p_j - s_j)}{\ln(p_j/s_j - 1) + 0.5\ln(n)} \right)$ (defined in Assumption A8). We first show (16pp.156) holds with probability going to 1 and then that (16pp.157) does too.

By (16pp.154), with probability going to 1,

$$\frac{\ln(p_j - s_j)}{\kappa_j^{EB}} \leq \frac{\ln(p_j - s_j)}{\ln(p_j - s_j) + \ln\left(\frac{\sqrt{n}}{s_j}\right)} = \frac{\ln(p_j - s_j)}{\ln(p_j/s_j - 1) + 0.5\ln(n)}.$$

It follows that:

$$\gamma = \frac{1}{2} \left(1 + \max_{j=1,\dots,b} \frac{\ln(p_j - s_j)}{\kappa_j^{EB}} \right) \leq \frac{1}{2} \left(1 + \max_j \frac{\ln(p_j - s_j)}{\ln(p_j/s_j - 1) + 0.5\ln(n)} \right) = \psi$$

and (16pp.156) holds with probability going to 1.

We now upper bound κ_j^{EB} to show (16pp.157) holds with probability going to 1. Observe that

$$\ln\left(\frac{\hat{s}_j}{s_j}\right) = \ln\left(1 + \frac{\hat{s}_j - s_j}{s_j}\right) = \ln\left(1 + \frac{p_j(\mathcal{O}_j^\circ - U_j^\circ)}{s_j}\right).$$

where the second equality follows from (16pp.151). Plugging this expression into (16pp.152), and using that $\mathcal{O}_j^\circ \geq 0$, we have that

$$\kappa_j^{EB} \leq \ln(p_j - s_j) + \ln\left(\frac{\sqrt{n}}{s_j}\right) + \ln\left(\frac{1 + \frac{p_j}{p_j - s_j} U_j^\circ}{1 - \frac{p_j}{s_j} U_j^\circ}\right). \quad (16\text{pp.158})$$

We split the sum in U_j° between models in $\mathcal{T}(\kappa^\circ)$ and those not in $\mathcal{T}(\kappa^\circ)$.

$$U_j^\circ = \sum_{M \in \mathcal{T}(\kappa^\circ) \mid |M_j| < s_j} \frac{s_j - |M_j|}{p_j} NC^\circ(M) + \sum_{M \in \mathcal{M} \setminus \mathcal{T}(\kappa^\circ) \mid |M_j| < s_j} \frac{s_j - |M_j|}{p_j} NC^\circ(M).$$

If $M \in \mathcal{T}(\kappa^\circ)$, then by definition $|M_j| \geq |S_j^L(\kappa^\circ)|$ and thus $s_j - |M_j| \leq s_j - |S_j^L(\kappa^\circ)|$. A bound on $s_j - |M_j|$ for $M \notin \mathcal{T}(\kappa^\circ)$ is simply $s_j - |M_j| \leq s_j$. It follows that

$$U_j^\circ \leq \frac{s_j - |S_j^L(\kappa^\circ)|}{p_j} \sum_{M \in \mathcal{T}(\kappa^\circ) \mid |M_j| < s_j} NC^\circ(M) + \frac{s_j}{p_j} \sum_{M \in \mathcal{M} \setminus \mathcal{T}(\kappa^\circ) \mid |M_j| < s_j} NC^\circ(M)$$

By Theorem 3.7, $\sum_{M \in \mathcal{T}(\kappa^\circ) \mid |M_j| < s_j} NC^\circ(M)$ and $\sum_{M \in \mathcal{M} \setminus \mathcal{T}(\kappa^\circ) \mid |M_j| < s_j} NC^\circ(M)$ converge in probability to 1 and 0 respectively. We then get that, with probability going to 1,

$$\begin{aligned} \frac{p_j}{p_j - s_j} U_j^\circ &\leq \frac{s_j - |S_j^L(\kappa^\circ)|}{p_j - s_j} \\ \frac{p_j}{s_j} U_j^\circ &\leq \frac{s_j - |S_j^L(\kappa^\circ)|}{s_j}. \end{aligned} \tag{16pp.159}$$

By the bounds above and (16pp.158), we have that with probability going to 1,

$$\begin{aligned} \kappa_j^{EB} &\leq \ln(p_j - s_j) + \ln\left(\frac{\sqrt{n}}{s_j}\right) + \ln\left(\frac{1 + \frac{s_j - |S_j^L(\kappa^\circ)|}{p_j - s_j}}{1 - \frac{s_j - |S_j^L(\kappa^\circ)|}{s_j}}\right) \\ &= \ln(p_j - s_j) + \ln\left(\frac{\sqrt{n}}{s_j}\right) + \ln\left(\frac{\frac{p_j - |S_j^L(\kappa^\circ)|}{p_j - s_j}}{\frac{|S_j^L(\kappa^\circ)|}{s_j}}\right) \\ &= \ln(p_j / |S_j^L(\kappa^\circ)| - 1) + \frac{1}{2} \ln(n) \end{aligned}$$

which shows (16pp.157) holds with probability going to 1 and that Assumption A8 implies Assumption A5 holds for the κ_j^{EB} with probability going to 1.

Since Assumptions A4 and A5 hold with probability going to 1, by Theorem 3.3, $\lim_{n \rightarrow \infty} P(\hat{S}^{EB,ei} = S) = 1$, as claimed.

16.4 Proof of Theorem 3.10

The proof strategy is similar to that of Theorem ?? and relies on several results therein.

The first step is to show that κ_j^A satisfies Assumption A4 with probability going to 1 as n

grows. The second step is to show that Assumption A9 implies Assumption A5 for the κ_j^A with probability going to 1. The consistency of $\hat{S}^{A,ei}$ then follows from Theorem 3.3.

Observe that $\kappa_j^A = \kappa_j^{EB} + \ln(\hat{s}_j)$, hence by (16pp.153) we have that

$$\kappa_j^A \geq \ln(p_j - s_j) + \ln(\sqrt{n}) + \ln\left(1 - \frac{p_j O_j^\circ}{p_j - s_j}\right).$$

In the proof of Theorem ?? we showed that, since κ° satisfies Assumption A4, by Theorem 3.7 $\ln\left(1 - \frac{p_j O_j^\circ}{p_j - s_j}\right)$ vanishes in probability as n grows. With probability going to 1, we then have that

$$\kappa_j^A \geq \ln(p_j - s_j) + \ln(\sqrt{n}) \quad (16pp.160)$$

and hence that the κ_j^A 's satisfy Assumption A4.

For the second part of the proof, we now show that Assumption A9 implies Assumption A5 for the κ_j^A . Assumption A5 for the κ_j^A states that for each block j there exists $g_j \rightarrow \infty$ such that for large enough n ,

$$\sqrt{\frac{(1-\gamma)n\rho(\mathbf{X})}{6}}\beta_{\min,j}^* - \sqrt{\kappa_j^{EB}} = \sqrt{\ln(s_j)} + g_j.$$

where γ takes value

$$\gamma = \frac{1}{2}\left(1 + \max_j \frac{\ln(p_j - s_j)}{\kappa_j^A}\right). \quad (16pp.161)$$

To show that Assumption A9 implies Assumption A5 for the κ_j^A with probability going to 1, it suffices to show that the following two inequalities

$$\sqrt{\frac{(1-\gamma)n\rho(\mathbf{X})}{6}}\beta_{\min,j}^* \geq \sqrt{\frac{(1-\xi)n\rho(\mathbf{X})}{6}}\beta_{\min,j}^*, \quad \text{and} \quad (16pp.162)$$

$$-\sqrt{\kappa_j^A} \geq -\sqrt{\ln\left(p - |S^L(\kappa^\circ)|\right) + \frac{1}{2}\ln(n)} \quad (16pp.163)$$

hold with probability going to 1 for γ as in (16pp.161) and $\xi = \frac{1}{2}\left(1 + \max_j \frac{\ln(p_j - s_j)}{\ln(p_j - s_j) + 0.5\ln(n)}\right)$ (defined in Assumption A9). We first show (16pp.162) holds with probability going to 1 and then that (16pp.163) does too.

By (16pp.160) we have that with probability going to 1, for any j

$$\frac{\ln(p_j - s_j)}{\kappa_j^A} \leq \frac{\ln(p_j - s_j)}{\ln(p_j - s_j) + \ln(n)/2}.$$

It follows that

$$\gamma = \frac{1}{2} \left(1 + \max_{j=1, \dots, b} \frac{\ln(p_j - s_j)}{\kappa_j^A} \right) \leq \frac{1}{2} \left(1 + \max_j \frac{\ln(p_j - s_j)}{\ln(p_j - s_j) + \ln(n)/2} \right) = \xi$$

and (16pp.162) holds with probability going to 1.

We now upper bound κ_j^A to show (16pp.163) holds with probability going to 1. By (16pp.152), we can write

$$\kappa_j^A = \kappa_j^{EB} + \ln(\hat{s}_j) = \ln(p_j - s_j) + \ln(\sqrt{n}) + \ln \left(1 - \frac{p_j(O_j^\circ - U_j^\circ)}{p_j - s_j} \right).$$

Since $\mathcal{O}_j^\circ \geq 0$, we obtain that

$$\kappa_j^A \leq \ln(p_j - s_j) + \ln(\sqrt{n}) + \ln \left(1 + \frac{p_j}{p_j - s_j} U_j^\circ \right).$$

By (16pp.159), with probability going to 1:

$$\begin{aligned} \kappa_j^A &\leq \ln(p_j - s_j) + \ln(\sqrt{n}) + \ln \left(1 + \frac{s_j - |S_j^L(\kappa^\circ)|}{p_j - s_j} \right) \\ &= \ln(p_j - |S_j^L(\kappa^\circ)|) + \frac{1}{2} \ln(n). \end{aligned} \tag{16pp.164}$$

which shows (16pp.163) holds with probability going to 1 and that Assumption A9 implies Assumption A5 holds for the κ_j^A with probability going to 1.

Since Assumptions A4 and A5 hold with probability going to 1, by Theorem 3.3, $\lim_{n \rightarrow \infty} P(\hat{S}^{A,ei} = S) = 1$, as claimed.

17 Proofs of Section ??

17.1 Auxiliary results

17.1.1 Bounds related to posterior model probabilities

In Lemma 17.1, I use a strategy similar to that used in Lemma 13.8 to bound the expected posterior probability of any $M \in \mathcal{M}$ based on the pairwise comparison $C(T) - C(M)$ (*cf*

(38)) for some $T \subseteq S$. It essentially shows that, if the block prior inclusion probabilities and the signals in $M \setminus T$ are small, $p(M \mid \mathbf{y}, \boldsymbol{\theta})$ is small.

Recall the notation in (39) and (40), for any model M and M' ,

$$\begin{aligned} L_{M'M} &= \|\mathbf{X}_{M'} \tilde{\boldsymbol{\beta}}^{(M')}\|^2 - \|\mathbf{X}_M \tilde{\boldsymbol{\beta}}^{(M)}\|^2 \\ \mu_{M'M} &= \|(I_n - P_M) \mathbf{X}_{M' \setminus M} \boldsymbol{\beta}_{M' \setminus M}^*\|^2 \end{aligned}$$

where $P_M = \mathbf{X}_M (\mathbf{X}_M^\top \mathbf{X}_M)^{-1} \mathbf{X}_M^\top$. In (??), I also defined for all $j = 1, \dots, b$

$$\bar{\kappa}_j = \frac{1}{2} \log(1 + gn) + \log(1/\theta_j - 1).$$

I redefine here, for any model M and M' ,

$$\Delta_{MM'} := \sum_{j=1}^b \bar{\kappa}_j (|M_j| - |M'_j|) \quad (17\text{pp.165})$$

Lemma 17.1. *For any $T \subseteq S$ and $M \in \mathcal{M} \setminus \{T\}$, denote $Q_T = M \cup T$ and $A_T := \omega \Delta_{MT} + \frac{1-\omega}{6} \mu_{Q_TM}$ (cf (17pp.165) and (40)). Suppose that, for some $\omega \in (1/2, 1]$, it holds that $A_T > 0$, $|M \setminus T| = o(A_T)$, and $\mu_{Q_TM} = o(A_T)$. For any $\psi \in (0, 1)$ and every n large enough,*

$$\mathbb{E}(p(M \mid \mathbf{y}, \boldsymbol{\theta})) \leq e^{-\psi A_T}.$$

Proof. For any $M \neq T$, since $\sum_M p(M \mid \mathbf{y}, \boldsymbol{\theta}) = 1$, we have the following bound on the posterior probability

$$p(M \mid \mathbf{y}, \boldsymbol{\theta}) = \frac{p(M \mid \mathbf{y}, \boldsymbol{\theta})}{\sum_{M'} p(M' \mid \mathbf{y}, \boldsymbol{\theta})} = \left[1 + \sum_{M': M' \neq M} \frac{p(M' \mid \mathbf{y}, \boldsymbol{\theta})}{p(M \mid \mathbf{y}, \boldsymbol{\theta})} \right]^{-1} \leq \left[1 + \frac{p(T \mid \mathbf{y}, \boldsymbol{\theta})}{p(M \mid \mathbf{y}, \boldsymbol{\theta})} \right]^{-1} \quad (17\text{pp.166})$$

where the last inequality follows from $\sum_{M': M' \notin \{M, T\}} p(M' \mid \mathbf{y}, \boldsymbol{\theta}) / p(M \mid \mathbf{y}, \boldsymbol{\theta}) \geq 0$. By (17pp.166) and using that for any random variable $Z \geq 0$, $E(Z) = \int_0^\infty P(Z > u) du$, we have that

$$E[p(M \mid \mathbf{y}, \boldsymbol{\theta})] \leq \int_0^1 P \left(\left[1 + \frac{p(T \mid \mathbf{y}, \boldsymbol{\theta})}{p(M \mid \mathbf{y}, \boldsymbol{\theta})} \right]^{-1} \geq u \right) du.$$

Inverting and taking logarithm on both sides of the inequality within the probability in the right-hand side above, we get

$$E [p(M | \mathbf{y}, \boldsymbol{\theta})] \leq \int_0^1 P \left(\log \frac{p(T | \mathbf{y}, \boldsymbol{\theta})}{p(M | \mathbf{y}, \boldsymbol{\theta})} \leq \log \left(\frac{1}{u} - 1 \right) \right) du. \quad (17\text{pp.167})$$

Simple algebra shows that

$$\frac{p(T | \mathbf{y}, \boldsymbol{\theta})}{p(M | \mathbf{y}, \boldsymbol{\theta})} = \exp \left\{ -\frac{gnL_{MT}}{2(1+gn)} \right\} \prod_{j=1}^b \left(\frac{\theta_j}{(1+gn)^{1/2}(1-\theta_j)} \right)^{|T_j|-|M_j|} \quad (17\text{pp.168})$$

Taking logarithm in (17pp.168) and using the notation recalled above gives

$$\log \frac{p(T | \mathbf{y}, \boldsymbol{\theta})}{p(M | \mathbf{y}, \boldsymbol{\theta})} = \frac{gnL_{TM}}{2(1+gn)} + \sum_{j=1}^b |M_j| - |T_j| \bar{\kappa}_j = \frac{gnL_{TM}}{2(1+gn)} + \Delta_{MT} \quad (17\text{pp.169})$$

Inputting (17pp.169) in (17pp.167), we have

$$\begin{aligned} E [p(M | \mathbf{y}, \boldsymbol{\theta})] &\leq \int_0^1 P \left(\frac{gnL_{TM}}{2(1+gn)} + \Delta_{MT} \leq \log \left(\frac{1}{u} - 1 \right) \right) du \\ &= \int_0^1 P \left(-\frac{gnL_{TM}}{2(1+gn)} - \left(\Delta_{MT} - \log \left(\frac{1}{u} - 1 \right) \right) \geq 0 \right) du. \end{aligned}$$

The second step of the proof is to use the union bound to upper bound the probability in the integrand above. Let $Q_T = M \cup T$ and recall that $L_{TM} = L_{Q_TM} - L_{Q_TT}$. For any ω

$$\begin{aligned} -\frac{gnL_{TM}}{2(1+gn)} - \left(\Delta_{MT} - \ln \left(\frac{1}{u} - 1 \right) \right) &= \left(\frac{gnL_{Q_TT}}{2(1+gn)} - \ln \left(\frac{e^{\omega\Delta_{MT}}}{\frac{1}{u} - 1} \right) \right) \\ &\quad - \left(\frac{gnL_{Q_TM}}{2(1+gn)} + (1-\omega)\Delta_{MT} \right). \end{aligned}$$

Observe that for any random variables U, V , and any $\epsilon, \omega' \geq 0$, the event $\{U - V \geq 0\}$ implies $\{U \geq \omega'\epsilon\} \cup \{V < \omega'\epsilon\}$. Let $U = \frac{gnL_{Q_TT}}{2(1+gn)} - \ln \left(\frac{e^{\omega\Delta_{MT}}}{\frac{1}{u} - 1} \right)$ and $V = \frac{gnL_{Q_TM}}{2(1+gn)} + (1-\omega)\Delta_{MT}$. Take $\epsilon = \mu_{Q_TM}$ and $\omega' = \frac{1}{6}(1-\omega)$, and observe that $A_T := \omega\Delta_{MT} + \frac{1-\omega}{6}\mu_{Q_TM} = \omega\Delta_{MT} + \omega'\mu_{Q_TM}$.

We then have

$$\{U \geq \omega'\epsilon\} = \left\{ \frac{gnL_{Q_TT}}{2(1+gn)} \geq \ln \left(\frac{e^{\omega\Delta_{MT}}}{\frac{1}{u} - 1} \right) + \omega'\mu_{Q_TM} \right\} = \left\{ \frac{gnL_{Q_TT}}{2(1+gn)} \geq \ln \left(\frac{e^{A_T}}{\frac{1}{u} - 1} \right) \right\}$$

and,

$$\begin{aligned} \{V < \omega'\epsilon\} &= \left\{ \frac{gnL_{Q_TM}}{2(1+gn)} < -(1-\omega)\Delta_{MT} + \omega'\mu_{Q_TM} \right\} \\ &= \left\{ \frac{gnL_{Q_TM}}{2(1+gn)} < \omega'(\mu_{Q_TM} - 6\Delta_{MT}) \right\}. \end{aligned}$$

By the union bound we have that

$$\mathbb{E}(p(M \mid \mathbf{y}, \boldsymbol{\theta})) \leq \int_0^1 P\left(\frac{gnL_{Q_T T}}{2(1+gn)} \geq \ln\left(\frac{e^{A_T}}{\frac{1}{u}-1}\right)\right) du + P\left(\frac{gnL_{Q_T M}}{2(1+gn)} < \omega'(\mu_{Q_T M} - 6\Delta_{MT})\right). \quad (17\text{pp.170})$$

The third and final step of the proof is to upper bound each of the terms in the right-hand side of (17pp.170). The intuition is that both T and M are nested within Q_T , and therefore $L_{Q_T T}$ and $L_{Q_T M}$ follow chi-squared distributions. We first bound the first term. If $M \subset T$ then $Q_T = T$, $L_{Q_T T} = 0$, and this term is zero. Suppose now that $M \not\subset T$. Then, by Lemma 10.1, $L_{Q_T T} \sim \chi_{|Q_T \setminus T|}^2(\mu_{Q_T T})$ with $|Q_T \setminus T| = |M \setminus T|$. By assumption, $A_T > 0$, $|M \setminus T| = o(\ln(e^{A_T}))$ and $\mu_{Q_T T} = o(\ln(e^{A_T}))$, then by Lemma 13.6, for $\alpha \in (\psi, 1)$, and every n large enough,

$$\int_0^1 P\left(L_{Q_T T} > 2\left(1 + \frac{1}{gn}\right) \ln\left(\frac{e^{A_T}}{1/u - 1}\right)\right) du < e^{-\alpha A_T}. \quad (17\text{pp.171})$$

We now bound the second term in (17pp.170). If $M \supset T$, then $Q_T = M$, $L_{Q_T M} = 0$, $\mu_{Q_T M} = 0$, and this term is zero. Alternatively, if $M \not\supset T$ then, by Lemma 10.1, $L_{Q_T M} \sim \chi_{|Q_T \setminus M|}^2(\mu_{Q_T M})$ with $|Q_T \setminus M| = |T \setminus M|$. Clearly, when $\mu_{Q_T M} \leq 6\Delta_{MT}$ this term is also zero, so suppose that $\mu_{Q_T M} > 6\Delta_{MT}$. We have

$$P\left(L_{Q_T M} < 2\left(1 + \frac{1}{gn}\right)(\mu_{Q_T M} - 6\Delta_{MT})\right) \leq P\left(L_{Q_T M} < \frac{2(1+gn)\omega'}{gn}\mu_{Q_T M}\right)$$

and, by Lemma 13.4 and using that $\omega' \in (0, \frac{1}{12})$, we obtain that

$$P\left(L_{Q_T M} < 2\left(1 + \frac{1}{gn}\right)\mu_{Q_T M}\right) \leq \left(2\omega'(1 + \frac{1}{gn})\right)^{\frac{|T \setminus M|}{4}} e^{-\frac{1}{2}\left(1 - \sqrt{2\omega'(1 + \frac{1}{gn})}\right)^2 \mu_{Q_T M}}$$

Since $\omega' < \frac{1}{12}$ and $\frac{1}{gn} \rightarrow 0$, for n large enough $2\omega'(1 + \frac{1}{gn}) < \frac{1}{6}$ and $\frac{1}{2}\left(1 - \sqrt{2\omega'(1 + \frac{1}{gn})}\right)^2 > \frac{1}{2}\left(1 - \sqrt{1/6}\right)^2 > 1/6$, we have

$$P\left(L_{Q_T M} < 2\left(1 + \frac{1}{gn}\right)\mu_{Q_T M}\right) \leq \left(\frac{1}{6}\right)^{\frac{|T \setminus M|}{4}} e^{-\frac{1}{6}\mu_{Q_T M}}.$$

Since $\mu_{Q_T M} > 6\Delta_{MT}$, we get

$$A_T = \omega'\mu_{Q_T M} + \omega\Delta_{MT} < \frac{1-\omega}{6}\mu_{Q_T M} + \frac{\omega}{6}\mu_{Q_T M} = \frac{1}{6}\mu_{Q_T M}.$$

We then get

$$P\left(L_{Q_{TM}} < 2\left(1 + \frac{1}{gn}\right)\omega'(\mu_{Q_{TM}} - 6\Delta_{MT})\right) \leq \left(\frac{1}{6}\right)^{\frac{|T \setminus M|}{4}} e^{-A_T} \leq e^{-A_T} \leq e^{-\alpha A_T} \quad (17\text{pp.172})$$

Summing the bounds in (17pp.171) and (17pp.172) gives that for every n large enough $\mathbb{E}(p(M \mid \mathbf{y}, \boldsymbol{\theta})) < 2e^{-\alpha A_T}$. Since $\psi < \alpha$, for every n large enough, we have that $\mathbb{E}(p(M \mid \mathbf{y}, \boldsymbol{\theta})) < e^{-\psi A_T}$ as claimed. $\square$

17.1.2 Concentration of posterior model probabilities on a reduced set of models

Lemma 17.2 is analogous to Theorem 3.7 and shows, under no assumption on signal strength other than minimal technical requirements, that posterior model probabilities concentrate on the reduced set of models

$$\mathcal{T}(\bar{\kappa}) := \left\{ M \in \mathcal{M} \mid M = S^L(\bar{\kappa}) \cup R, R \in \mathcal{P}(S^I(\bar{\kappa})) \right\}. \quad (17\text{pp.173})$$

The set $\mathcal{T}(\bar{\kappa})$ contains all the models that are the union of $S^L(\bar{\kappa})$ (large signals) and some subset of $S^I(\bar{\kappa})$ (intermediate signals).

Lemma 17.2. *Assume A1 and A4ter, $|S^I(\bar{\kappa})| = O(1)$ and $|S_j^S(\bar{\kappa})| = O(p_j - s_j)$ (cf (??)) for every $j = 1, \dots, b$, then,*

$$\lim_{n \rightarrow \infty} \sum_{M \notin \mathcal{T}(\bar{\kappa})} E(p(M \mid \mathbf{y}, \boldsymbol{\theta})) = 0$$

Lemma 17.2 is a generalization of Theorem ?? . I use it to prove both Theorem ?? and Theorem ??. The Lemma is not included in the main text of Section 3.4 for the sake of brevity and because it is analogous to Theorem 3.7.

The proof strategy for Lemma 17.2 is the same as that of Theorem 3.7, with suitable adjustments. The first step is to use Lemma 17.1 to bound, for every $M \notin \mathcal{T}(\bar{\kappa})$, $\mathbb{E}(p(M \mid \mathbf{y}, \boldsymbol{\theta})) \leq e^{-\psi A_{TM}}$ for every large enough n and any $\psi \in (0, 1)$, where $A_{TM} = \omega \Delta_{MT^M} + \frac{1-\omega}{6} \mu_{Q_{TM}M}$ (cf (17pp.165) and (40)), $\omega \in (1/2, 1)$ is defined in (??) and $Q_{TM} = M \cup T^M$.

The second step is to obtain a lower bound for A_{T^M} , which gives an upper bound for $\mathbb{E}(p(M \mid \mathbf{y}, \boldsymbol{\theta}))$. The final step is to get an upper-bound on $\sum_{M \in \mathcal{M} \setminus \mathcal{T}(\bar{\kappa})} \mathbb{E}(p(M \mid \mathbf{y}, \boldsymbol{\theta}))$ that vanishes (as n grows) under the assumptions of Lemma 17.2.

Proof. For any $M \notin \mathcal{T}(\bar{\kappa})$, take the unique $T^M \in \mathcal{T}(\bar{\kappa})$ such that $T^M = [M \cap S^I(\bar{\kappa})] \cup S^L(\bar{\kappa})$, which implies $M \cap S^I(\bar{\kappa}) = T^M \cap S^I(\bar{\kappa})$. That is, T^M contains all the large signals plus the intermediate signals in M . To show that $\mathbb{E}(p(M \mid \mathbf{y}, \boldsymbol{\theta})) \leq e^{-\psi A_{T^M}}$ we show that A_{T^M} satisfies the conditions of Lemma 17.1, taking $T = T^M$. That is, we wish to show that three conditions hold: $A_{T^M} > 0$, $|M \setminus T^M| = o(A_{T^M})$, and $\mu_{Q_{T^M} T^M} = o(A_{T^M})$.

Observe that Δ_{MT^M} , defined in (17pp.165), can be rewritten as $\Delta_{MT^M} = \sum_{j=1}^b (|M_j \setminus T_j^M| - |T_j^M \setminus M_j|) \bar{\kappa}_j$ and then:

$$A_{T^M} = \sum_{j=1}^b |M_j \setminus T_j^M| \omega \bar{\kappa}_j + \frac{1-\omega}{6} \mu_{Q_{T^M} M} - \sum_{j=1}^b |T_j^M \setminus M_j| \omega \bar{\kappa}_j$$

Since $\omega < 1$, $(1 - \omega)/6 > 0$, by Lemma 13.2, for every $n \in \mathbb{N}$ we have

$$A_{T^M} \geq \sum_{j=1}^b |M_j \setminus T_j^M| \omega \bar{\kappa}_j + \sum_{j=1}^b |T_j^M \setminus M_j| \left(\frac{1-\omega}{6} n \rho(\mathbf{X}) \min_{i \in T_j^M \setminus M_j} \beta_i^{*2} - \omega \bar{\kappa}_j \right). \quad (17\text{pp.174})$$

We have that $T^M \subseteq (S^I(\bar{\kappa}) \cup S^L(\bar{\kappa}))$ since $T^M \in \mathcal{T}(\bar{\kappa})$ and that $T^M \cap S^I(\bar{\kappa}) = M \cap S^I(\bar{\kappa})$. It follows that $T^M \setminus M \subseteq S^L(\bar{\kappa})$. By definition of $S^L(\bar{\kappa})$, the rightmost component in (17pp.174) is nonnegative and, if $|T^M \setminus M| \neq 0$, it is positive, for every n large enough. By Assumption A4ter, component $\omega \sum_{j=1}^b |M_j \setminus T_j^M| \bar{\kappa}_j$ is nonnegative, and, if $|M \setminus T^M| \neq 0$, positive. Now, $M \neq T^M$ implies that necessarily $|M \setminus T^M| \neq 0$ or $|T^M \setminus M| \neq 0$ and then, for every n large enough, $A_{T^M} > 0$, establishing the first condition required by Lemma 17.1. Regarding its second condition, if $|M \setminus T^M| = 0$, we immediately have $|M \setminus T^M| = o(A_{T^M})$. If $|M \setminus T^M| \neq 0$, since the rightmost component in (17pp.174) is nonnegative, we have

$$\frac{|M \setminus T^M|}{A_{T^M}} \leq \left[\omega \sum_{j=1}^b \frac{|M_j \setminus T_j^M|}{|M \setminus T^M|} \bar{\kappa}_j \right]^{-1} \leq \left[\omega \min_{j=1, \dots, b} \bar{\kappa}_j \right]^{-1}$$

where the last inequality follows from $\sum_{j=1}^b \frac{|M_j \setminus T_j^M|}{|M \setminus T^M|} = 1$. By Assumption A4ter, $\min_{j=1, \dots, b} \bar{\kappa}_j \rightarrow \infty$ as $n \rightarrow \infty$ and hence $|M \setminus T^M| = o(A_{T^M})$ when $|M \setminus T^M| \neq 0$ too.

Finally, consider the third condition in Lemma 17.1. Note that, since $T^M \in \mathcal{T}(\bar{\kappa})$, we have that $S^L(\bar{\kappa}) \subseteq T^M$. Moreover we have $M \cap S^I(\bar{\kappa}) = T^M \cap S^I(\bar{\kappa})$. It follows that $M \setminus T^M \subseteq (S^I(\bar{\kappa}) \cup S^L(\bar{\kappa}))^C$, that is the elements of $M \setminus T^M$ are either inactive or belong to $S^S(\bar{\kappa})$. If $M \cap S^S(\bar{\kappa}) = \emptyset$ then $M \setminus T^M \subseteq S^C$ and we immediately get $\mu_{Q_{TM}T^M} = o(A_{TM})$ because $\beta_{M \setminus T^M}^* = \mu_{Q_{TM}T^M} = 0$. Assume now that $M \cap S^S(\bar{\kappa}) \neq \emptyset$. Using that the rightmost component in (17pp.174) is nonnegative and Lemma 13.9, we have

$$\frac{\mu_{Q_{TM}T^M}}{A_{TM}} \leq \frac{\mu_{Q_{TM}T^M}}{\omega \sum_{j=1}^b |M_j \setminus T_j^M| \bar{\kappa}_j} \leq \frac{n \bar{\lambda} \sum_{j=1}^b |(S_j \cap M_j) \setminus T_j^M| \max_{i \in (S_j \cap M_j) \setminus T_j^M} \beta_i^{*2}}{\omega \sum_{j=1}^b |M_j \setminus T_j^M| \bar{\kappa}_j}.$$

Observe that for all $j = 1, \dots, b$, $M_j \setminus T_j^M \supseteq (S_j \cap M_j) \setminus T_j^M$ and then $|M_j \setminus T_j^M| \geq |(S_j \cap M_j) \setminus T_j^M|$. We then get

$$\frac{\mu_{Q_{TM}T^M}}{A_{TM}} \leq \frac{n \bar{\lambda} \sum_{j=1}^b |(S_j \cap M_j) \setminus T_j^M| \max_{i \in (S_j \cap M_j) \setminus T_j^M} \beta_i^{*2}}{\omega \sum_{j=1}^b |(S_j \cap M_j) \setminus T_j^M| \bar{\kappa}_j}.$$

Moreover, since $M \setminus T^M \subseteq (S^I(\bar{\kappa}) \cup S^L(\bar{\kappa}))^C$ as discussed earlier, we have, for all $j = 1, \dots, b$, $(S_j \cap M_j) \setminus T_j^M \subseteq S_j^S(\bar{\kappa})$. It follows $\max_{i \in (S_j \cap M_j) \setminus T_j^M} \beta_i^{*2} \leq \max_{i \in S_j^S(\bar{\kappa})} \beta_i^{*2}$ and

$$\frac{\mu_{Q_{TM}T^M}}{A_{TM}} \leq \frac{n \bar{\lambda} \sum_{j=1}^b |(S_j \cap M_j) \setminus T_j^M| \max_{i \in S_j^S(\bar{\kappa})} \beta_i^{*2}}{\omega \sum_{j=1}^b |(S_j \cap M_j) \setminus T_j^M| \bar{\kappa}_j}. \quad (17pp.175)$$

Let $\bar{r} := \sum_{j=1}^b n \bar{\lambda} \max_{i \in S_j^S(\bar{\kappa})} \beta_i^{*2} / (\omega \bar{\kappa}_j)$. We show next that $\bar{r}$ is an upper bound on $\mu_{Q_{TM}T^M} / A_{TM}$. By restricting the sum in $\bar{r}$ to the j such that $|(S_j \cap M_j) \setminus T_j^M| \neq 0$ and multiplying the numerator and denominator of the summand by $|(S_j \cap M_j) \setminus T_j^M|$, we get

$$\bar{r} \geq \sum_{j=1, |(S_j \cap M_j) \setminus T_j^M| \neq 0}^b \frac{|(S_j \cap M_j) \setminus T_j^M| n \bar{\lambda} \max_{i \in S_j^S(\bar{\kappa})} \beta_i^{*2}}{|(S_j \cap M_j) \setminus T_j^M| \omega \bar{\kappa}_j} \quad (17pp.176)$$

Using the property of (15pp.93) in the right-hand side of (17pp.176), we get

$$\begin{aligned} \bar{r} &\geq \frac{\sum_{j=1, |(S_j \cap M_j) \setminus T_j^M| \neq 0}^b |(S_j \cap M_j) \setminus T_j^M| n \bar{\lambda} \max_{i \in S_j^S(\bar{\kappa})} \beta_i^{*2} \left(1 + \sum_{l \neq j} \frac{|(S_l \cap M_l) \setminus T_l^M| \omega \bar{\kappa}_l}{|(S_j \cap M_j) \setminus T_j^M| \omega \bar{\kappa}_j} \right)}{\sum_{j=1, |(S_j \cap M_j) \setminus T_j^M| \neq 0}^b |(S_j \cap M_j) \setminus T_j^M| \omega \bar{\kappa}_j} \\ &\geq \frac{\sum_{j=1}^b |(S_j \cap M_j) \setminus T_j^M| n \bar{\lambda} \max_{i \in S_j^S(\bar{\kappa})} \beta_i^{*2}}{\sum_{j=1}^b |(S_j \cap M_j) \setminus T_j^M| \omega \bar{\kappa}_j} \end{aligned}$$

where last inequality follows from $\left(1 + \sum_{l \neq j} \frac{|(S_l \cap M_l) \setminus T_l^M| \omega \bar{\kappa}_l}{|(S_j \cap M_j) \setminus T_j^M| \omega \bar{\kappa}_j}\right) \geq 1$ for all j and from the identity $\sum_{j=1}^b |(S_j \cap M_j) \setminus T_j^M| \omega \bar{\kappa}_j = \sum_{j=1}^b |(S \cap M_j) \setminus T_j^M| \omega \bar{\kappa}_j$. Then, by (17pp.175),

$$\frac{\mu_{Q_{TM}TM}}{A_{TM}} \leq \bar{r} = \sum_{j=1}^b \frac{n\bar{\lambda} \max_{i \in S_j^S(\bar{\kappa})} \beta_i^{*2}}{\omega \bar{\kappa}_j}.$$

By definition of $S_j^S(\bar{\kappa})$, $n\bar{\lambda} \max_{i \in S_j^S(\bar{\kappa})} \beta_i^{*2} = o(\bar{\kappa}_j)$ for all j and $\mu_{Q_{TM}TM} = o(A_{TM})$ when $M \cap S^S(\bar{\kappa}) \neq \emptyset$ too. We can now apply Lemma 17.1 and get that for every $M \in \mathcal{M} \setminus \mathcal{T}(\bar{\kappa})$, for any $\psi \in (0, 1)$ and every n large enough, $\mathbb{E}(p(M | \mathbf{y}, \boldsymbol{\theta})) \leq e^{-\psi A_{TM}}$.

The second step of the proof is to lower-bound $A_{TM} = \omega \Delta_{TM} + \frac{1-\omega}{6} \mu_{Q_{TM}TM}$. Let

$$A_{TM}^* := \omega \sum_{j=1}^b |M_j \setminus T_j^M| \bar{\kappa}_j + \sum_{j=1}^b |T_j^M \setminus M_j| \left(\frac{1-\omega}{6} n \rho(\mathbf{X}) \min_{i \in S_j^L(\bar{\kappa})} \beta_i^{*2} - \omega \bar{\kappa}_j \right)$$

Recall that, for every j , $T_j^M \setminus M_j \subseteq S_j^L(\bar{\kappa})$. We have then $\min_{i \in T_j^M \setminus M_j} \beta_i^{*2} \geq \min_{i \in S_j^L(\bar{\kappa})} \beta_i^{*2}$, and by (17pp.174), $A_{TM} \geq A_{TM}^*$. It follows that for any $\psi \in (0, 1)$ and for all n large enough,

$$\mathbb{E}(p(M | \mathbf{y}, \boldsymbol{\theta})) \leq e^{-\psi A_{TM}^*}. \quad (17pp.177)$$

To conclude the second part of the proof we lower-bound $\psi A_{TM}^* = \sum_{j=1}^b |M_j \setminus T_j^M| \psi \omega \bar{\kappa}_j + \sum_{j=1}^b |T_j^M \setminus M_j| \psi \left(\frac{1-\omega}{6} n \rho(\mathbf{X}) \min_{i \in S_j^L(\bar{\kappa})} \beta_i^{*2} - \omega \bar{\kappa}_j \right)$. To do this, we obtain a lower bound for $\psi \omega \bar{\kappa}_j$ and for $\psi \left(\frac{1-\omega}{6} n \rho(\mathbf{X}) \min_{i \in S_j^L(\bar{\kappa})} \beta_i^{*2} - \omega \bar{\kappa}_j \right)$.

The definition of $S_j^L(\bar{\kappa})$ implies there exists some $e'_j \rightarrow \infty$ such that

$$\frac{(1-\omega)n\rho(\mathbf{X})}{6} \min_{i \in S_j^L(\bar{\kappa})} \beta_i^{*2} - \bar{\kappa}_j = \ln(s_j) + g'_j. \quad (17pp.178)$$

Let $\delta \in (0, 1)$ and denote $\bar{m}_j = \max \left\{ \frac{2\ln(p_j - s_j)}{f_j}, \frac{2\ln(s_j)}{g'_j} \right\}$, where f_j is given in Assumption A4ter. Take $\psi = \max_{j=1, \dots, b} \frac{\xi + \delta + \bar{m}_j}{1 + \bar{m}_j}$ for some $\xi \in (0, 1 - \delta)$ then $\psi \in (0, 1)$ and we have, for every $j = 1, \dots, b$,

$$\psi > \frac{\delta + \frac{2\ln(p_j - s_j)}{f_j}}{1 + \frac{2\ln(p_j - s_j)}{f_j}} = \frac{\delta f_j / 2 + \ln(p_j - s_j)}{f_j / 2 + \ln(p_j - s_j)} \quad (17pp.179)$$

$$\psi > \frac{\delta + \frac{2\ln(s_j)}{e'_j}}{1 + \frac{2\ln(s_j)}{e'_j}} = \frac{\delta e'_j / 2 + \ln(s_j)}{e'_j / 2 + \ln(s_j)} \geq \frac{\delta e'_j / 2 + \ln(s_j)}{e'_j + \ln(s_j)}. \quad (17pp.180)$$

Recall that Assumption A4ter defines $f_j = \bar{\kappa}_j - \ln(p_j - s_j)$ and $\omega = \frac{1}{2}(1 + \max_j \frac{\ln(p_j - s_j)}{\bar{\kappa}_j})$ respectively. Hence,

$$\omega \bar{\kappa}_j \geq \frac{1}{2} \left(1 + \frac{\ln(p_j - s_j)}{\bar{\kappa}_j} \right) \bar{\kappa}_j = \ln(p_j - s_j) + \frac{1}{2} (\bar{\kappa}_j - \ln(p_j - s_j)) = \ln(p_j - s_j) + \frac{1}{2} f_j.$$

Hence, by (17pp.179), we have

$$\psi \omega \bar{\kappa}_j \geq \psi \left(\ln(p_j - s_j) + \frac{1}{2} f_j \right) \geq \ln(p_j - s_j) + \delta \frac{1}{2} f_j. \quad (17pp.181)$$

Further,

$$\psi \left(\frac{1-\omega}{6} n \rho(\mathbf{X}) \min_{i \in S_j^L(\bar{\kappa})} \beta_i^{*2} - \omega \bar{\kappa}_j \right) \geq \psi \left(\ln(s_j) + e'_j \right) \geq \ln(s_j) + \delta \frac{1}{2} e'_j. \quad (17pp.182)$$

where the first inequality follows from (17pp.178) and the second inequality from (17pp.180).

In (17pp.177),

$$\psi A_{T^M}^* = \sum_{j=1}^b |M_j \setminus T_j^M| \psi \omega \bar{\kappa}_j + \sum_{j=1}^b |T_j^M \setminus M_j| \psi \left(\frac{1-\omega}{6} n \rho(\mathbf{X}) \min_{i \in S_j^L(\bar{\kappa})} \beta_i^{*2} - \omega \bar{\kappa}_j \right)$$

. Then by (17pp.181) and (17pp.182), we get

$$\mathbb{E}(p(M | \mathbf{y}, \boldsymbol{\theta})) \leq \exp \left\{ - \sum_{j=1}^b |M_j \setminus T_j^M| (\ln(p_j - s_j) + \delta \frac{f_j}{2}) - \sum_{j=1}^b |T_j^M \setminus M_j| (\ln(s_j) + \delta \frac{e'_j}{2}) \right\}. \quad (17pp.183)$$

For the final step of the proof, denote $\mathcal{S} = \sum_{M \in \mathcal{M} \setminus \mathcal{T}(\bar{\kappa})} \mathbb{E}(p(M | \mathbf{y}, \boldsymbol{\theta}))$ for convenience. By (17pp.183) we have

$$\mathcal{S} \leq \sum_{M \in \mathcal{M} \setminus \mathcal{T}(\bar{\kappa})} e^{-\sum_{j=1}^b |M_j \setminus T_j^M| \left(\ln(p_j - s_j) + \delta \frac{f_j}{2} \right) - \sum_{j=1}^b |T_j^M \setminus M_j| \left(\ln(s_j) + \delta \frac{e'_j}{2} \right)}.$$

We split the sum in the right-hand side above into sums over models M such that $T^M = T$ for some common $T \in \mathcal{T}(\bar{\kappa})$. Denote for any $T \in \mathcal{T}$, $\mathcal{M}(T) := \{M \in \mathcal{M} \setminus \mathcal{T}(\bar{\kappa}) | T^M = T\}$, then

$$\mathcal{S} \leq \sum_{T \in \mathcal{T}(\bar{\kappa})} \sum_{M \in \mathcal{M}(T)} e^{-\sum_{j=1}^b |M_j \setminus T_j| \left(\ln(p_j - s_j) + \delta \frac{f_j}{2} \right) - \sum_{j=1}^b |T_j \setminus M_j| \left(\ln(s_j) + \delta \frac{e'_j}{2} \right)}. \quad (17pp.184)$$

The right hand-side of (17pp.184) is composed of a double sum over $T \in \mathcal{T}(\bar{\kappa})$ and over $M \in \mathcal{M}(T)$. Consider the sum over $M \in \mathcal{M}(T)$, add T to it, and denote it

$$\mathcal{S}(T) = \sum_{M \in \mathcal{M}(T) \cup T} e^{-\sum_{j=1}^b |M_j \setminus T_j| \left(\ln(p_j - s_j) + \delta \frac{f_j}{2} \right) - \sum_{j=1}^b |T_j \setminus M_j| \left(\ln(s_j) + \delta \frac{e'_j}{2} \right)}. \quad (17pp.185)$$

In the summand in the right-hand side of (17pp.185), the case $M = T$ correspond to $|M_j \setminus T_j^M| = |T_j^M \setminus M_j| = 0$ for all j and the summand is then 1. By (17pp.184), we then get that

$$\mathcal{S} \leq \sum_{T \in \mathcal{T}(\bar{\kappa})} (\mathcal{S}(T) - 1). \quad (17pp.186)$$

For each $T = T^M$, we further split $\mathcal{S}(T)$ into sums over subsets of models M that have u_j more parameters than T^M in block j , and are missing w_j parameters from T^M in block j . Specifically, consider models M such that, for all j , $|M_j \setminus T_j^M| = u_j$ and $|T_j^M \setminus M_j| = w_j$ with $u_j \in \{0, \dots, p_j - s_j, \dots, p_j - s_j + |S_j^S(\bar{\kappa})|\}$ and $w_j \in \{0, \dots, |S_j^L(\bar{\kappa})|\}$. Denote by

$$S_{\mathbf{w}}^{\mathbf{u}}(T) = \sum_{M \in \mathcal{M}(T) \cup T: \forall j |M_j \setminus S_j| = u_j, |S_j \setminus M_j| = w_j} e^{-\sum_{j=1}^b u_j \left(\ln(p_j - s_j) + \delta \frac{f_j}{2} \right) - \sum_{j=1}^b w_j \left(\ln(s_j) + \delta \frac{e'_j}{2} \right)}.$$

We get

$$\mathcal{S}(T) = \sum_{w_1=0}^{|S_1^L(\bar{\kappa})|} \cdots \sum_{w_b=0}^{|S_b^L(\bar{\kappa})|} \sum_{u_1=0}^{p_1 - s_1 + |S_1^S(\bar{\kappa})|} \cdots \sum_{u_b=0}^{p_b - s_b + |S_b^S(\bar{\kappa})|} S_{\mathbf{w}}^{\mathbf{u}}(T). \quad (17pp.187)$$

The number of models missing, for all j , w_j out of the $|S_j^L(\bar{\kappa})|$ large active parameters and having u_j inactive or small active parameters from B_j is $\prod_{j=1}^b \binom{p_j - s_j + |S_j^S(\bar{\kappa})|}{u_j} \binom{|S_j^L(\bar{\kappa})|}{w_j}$. We thus have

$$\begin{aligned} S_{\mathbf{w}}^{\mathbf{u}}(T) &= \left(\prod_{j=1}^b \binom{p_j - s_j + |S_j^S(\bar{\kappa})|}{u_j} \binom{|S_j^L(\bar{\kappa})|}{w_j} \right) e^{-\sum_{j=1}^b u_j \left(\ln(p_j - s_j) + \delta \frac{f_j}{2} \right) - \sum_{j=1}^b w_j \left(\ln(s_j) + \delta \frac{e'_j}{2} \right)} \\ &= \prod_{j=1}^b \binom{p_j - s_j + |S_j^S(\bar{\kappa})|}{u_j} e^{-u_j \left(\ln(p_j - s_j) + \delta \frac{f_j}{2} \right)} \binom{|S_j^L(\bar{\kappa})|}{w_j} e^{-w_j \left(\ln(s_j) + \delta \frac{e'_j}{2} \right)}. \end{aligned}$$

Inputting the expression above in (17pp.187) and factorizing over terms in u_j and w_j gives

$$\begin{aligned} \mathcal{S}(T) &\leq \prod_{j=1}^b \left(\sum_{u_j=0}^{p_j - s_j + |S_j^S(\bar{\kappa})|} \binom{p_j - s_j + |S_j^S(\bar{\kappa})|}{u_j} e^{-u_j \left(\ln(p_j - s_j) + \delta \frac{f_j}{2} \right)} \right) \\ &\quad \cdot \left(\sum_{w_j=0}^{|S_j^L(\bar{\kappa})|} \binom{|S_j^L(\bar{\kappa})|}{w_j} e^{-w_j \left(\ln(s_j) + \delta \frac{e'_j}{2} \right)} \right). \end{aligned}$$

By the bound in (15pp.78) and taking the terms in $u_j = 0$ and $w_j = 0$ out of the sums

above, we have

$$\mathcal{S}(T) \leq \prod_{j=1}^b \left(1 + \sum_{u_j=1}^{p_j-s_j+|S_j^S(\bar{\kappa})|} e^{-u_j \left(\delta \frac{f_j}{2} - \ln \left(1 + \frac{|S_j^S(\bar{\kappa})|}{p_j-s_j} \right) - 1 \right)} \right) \quad (17\text{pp.188})$$

$$\cdot \left(1 + \sum_{w_j=1}^{|S_j^L(\bar{\kappa})|} e^{-w_j \left(\delta \frac{e'_j}{2} + \ln \left(\frac{s_j}{|S_j^L(\bar{\kappa})|} \right) - 1 \right)} \right).$$

Denote

$$d_j = e^{1 + \ln \left(1 + \frac{|S_j^S(\bar{\kappa})|}{p_j-s_j} \right) - \delta \frac{f_j}{2}}, \quad h_j = e^{1 - \ln \left(\frac{s_j}{|S_j^L(\bar{\kappa})|} \right) - \delta \frac{e'_j}{2}}.$$

By assumption $|S_j^S(\bar{\kappa})| = O(p_j - s_j)$, and by definition of $|S_j^L(\bar{\kappa})|$ we have $s_j \geq |S_j^L(\bar{\kappa})|$. By Assumptions A4ter and the definition of $|S_j^L(\bar{\kappa})|$, we also have $f_j \rightarrow \infty$ and $e'_j \rightarrow \infty$, and then $\lim_{n \rightarrow \infty} d_j = \lim_{n \rightarrow \infty} h_j = 0$. Using the properties of geometric series, for every j we have

$$1 + \sum_{u_j=1}^{p_j-s_j+|S_j^S(\bar{\kappa})|} e^{-u_j \left(\delta \frac{f_j}{2} - \ln \left(1 + \frac{|S_j^S(\bar{\kappa})|}{p_j-s_j} \right) - 1 \right)} = \frac{1 - d_j^{p_j-s_j+|S_j^S(\bar{\kappa})|+1}}{1 - d_j}$$

$$1 + \sum_{w_j=1}^{|S_j^L(\bar{\kappa})|} e^{-w_j \left(\delta \frac{e'_j}{2} + \ln \left(\frac{s_j}{|S_j^L(\bar{\kappa})|} \right) - 1 \right)} = \frac{1 - h_j^{|S_j^L(\bar{\kappa})|+1}}{1 - h_j},$$

where both expressions converge to 1 as n grows. By (17pp.186) and (17pp.188):

$$\mathcal{S} \leq \sum_{T \in \mathcal{T}} \left(\prod_{j=1}^b \left(\frac{1 - d_j^{p_j-s_j+|S_j^S(\bar{\kappa})|+1}}{1 - d_j} \right) \left(\frac{1 - h_j^{|S_j^L(\bar{\kappa})|+1}}{1 - h_j} \right) - 1 \right).$$

Each of the summand vanishes as $n \rightarrow \infty$. Moreover, by assumption $|S^I(\bar{\kappa})| = O(1)$ and then $|\mathcal{T}(\bar{\kappa})| = 2^{|S^I(\bar{\kappa})|} = O(1)$. We thus have $\lim_{n \rightarrow \infty} \mathcal{S} = \lim_{n \rightarrow \infty} \sum_{M \in \mathcal{M} \setminus \mathcal{T}(\bar{\kappa})} \mathbb{E}(p(M | \mathbf{y}, \boldsymbol{\theta})) = 0$. $\square$

17.1.3 Convergence of the expected sum of posterior inclusion probabilities

Lemma 17.3 is analogous to Proposition 3.9 and shows asymptotic bounds on the expected sum of posterior inclusion probabilities, under no assumption on signal strength other minimal technical requirements. It is used in the proof of Theorem ??.

Lemma 17.3. Assume A1 and A4ter, $|S^I(\bar{\kappa})| = O(1)$ and $|S_j^S(\bar{\kappa})| = O(p_j - s_j)$ for every $j = 1, \dots, b$. Then

$$\frac{|S_j^L(\bar{\kappa})|}{p_j} \leq \lim_{n \rightarrow \infty} E \left(\frac{1}{p_j} \sum_{i: z_i=j} p(m_i = 1 | \mathbf{y}, \boldsymbol{\theta}) \right) \leq \frac{s_j - |S_j^S(\bar{\kappa})|}{p_j} \quad \text{for all } j = 1, \dots, b.$$

Proof. Denote

$$A_j := \frac{1}{p_j} \sum_{i:z_i=j} \sum_{M \in \mathcal{T}(\bar{\kappa}) | i \in M} p(M | \mathbf{y}, \boldsymbol{\theta}) \quad \text{and} \quad C_j := \frac{1}{p_j} \sum_{i:z_i=j} \sum_{M \notin \mathcal{T}(\bar{\kappa}) | i \in M} p(M | \mathbf{y}, \boldsymbol{\theta})$$

where $\mathcal{T}(\bar{\kappa})$ is defined in (17pp.173). For every $j = 1, \dots, b$, we have the decomposition

$$\frac{1}{p_j} \sum_{i:z_i=j} p(m_i = 1 | \mathbf{y}, \boldsymbol{\theta}) = \frac{\sum_{i:z_i=j} \sum_{M | i \in M} p(M | \mathbf{y}, \boldsymbol{\theta})}{p_j} = A_j + C_j. \quad (17\text{pp.189})$$

To show the lower bound, we decompose A_j

$$\begin{aligned} A_j &= \sum_{M \in \mathcal{T}(\bar{\kappa})} p(M | \mathbf{y}, \boldsymbol{\theta}) \sum_{i:z_i=j} \frac{I(i \in M)}{p_j} \\ &= \sum_{M \in \mathcal{T}(\bar{\kappa})} p(M | \mathbf{y}, \boldsymbol{\theta}) \sum_{i:z_i=j, i \in S_j^L(\bar{\kappa})} \frac{I(i \in M)}{p_j} + \sum_{M \in \mathcal{T}(\bar{\kappa})} p(M | \mathbf{y}, \boldsymbol{\theta}) \sum_{i:z_i=j, i \notin S_j^L(\bar{\kappa})} \frac{I(i \in M)}{p_j} \\ &= \frac{|S_j^L(\bar{\kappa})|}{p_j} \sum_{M \in \mathcal{T}(\bar{\kappa})} p(M | \mathbf{y}, \boldsymbol{\theta}) + \sum_{M \in \mathcal{T}(\bar{\kappa})} p(M | \mathbf{y}, \boldsymbol{\theta}) \sum_{i:z_i=j, i \notin S_j^L(\bar{\kappa})} \frac{I(i \in M)}{p_j}. \end{aligned} \quad (17\text{pp.190})$$

where the last equality follows from $I(i \in M) = 1$ for all $i \in S_j^L(\bar{\kappa})$ when $M \in \mathcal{T}(\bar{\kappa})$. The rightmost term above and C_j are nonnegative, then by the linearity of the expectation

$$E\left(\frac{1}{p_j} \sum_{i:z_i=j} p(m_i = 1 | \mathbf{y}, \boldsymbol{\theta})\right) \geq \frac{|S_j^L(\bar{\kappa})|}{p_j} \sum_{M \in \mathcal{T}(\bar{\kappa})} E(p(M | \mathbf{y}, \boldsymbol{\theta})).$$

By Lemma 17.2, $\lim_{n \rightarrow \infty} \sum_{M \in \mathcal{T}(\bar{\kappa})} \mathbb{E}(p(M | \mathbf{y}, \boldsymbol{\theta})) = 1$. It follows that

$$\lim_{n \rightarrow \infty} E\left(\frac{1}{p_j} \sum_{i:z_i=j} p(m_i = 1 | \mathbf{y}, \boldsymbol{\theta})\right) \geq \frac{|S_j^L(\bar{\kappa})|}{p_j} \quad \text{for every } j = 1, \dots, b.$$

We now prove the upper bound. Recall that $\mathcal{T}(\bar{\kappa})$ by definition includes models that have no small signals, i.e. all parameters are in $S^L(\bar{\kappa}) \cup S^I(\bar{\kappa})$. That is, for all $M \in \mathcal{T}(\bar{\kappa})$, we have that $I(i \in M) = 0$ for all $i \in \{i : z_i = j, i \notin (S_j^L(\bar{\kappa}) \cup S_j^I(\bar{\kappa}))\}$. Hence, A_j in (17pp.190) satisfies

$$\begin{aligned} A_j &= \frac{|S_j^L(\bar{\kappa})|}{p_j} \sum_{M \in \mathcal{T}(\bar{\kappa})} p(M | \mathbf{y}, \boldsymbol{\theta}) + \sum_{M \in \mathcal{T}(\bar{\kappa})} p(M | \mathbf{y}, \boldsymbol{\theta}) \sum_{i \in S_j^I(\bar{\kappa})} \frac{I(i \in M)}{p_j} \\ &\leq \frac{|S_j^L(\bar{\kappa})|}{p_j} \sum_{M \in \mathcal{T}(\bar{\kappa})} p(M | \mathbf{y}, \boldsymbol{\theta}) + \frac{|S_j^I(\bar{\kappa})|}{p_j} \sum_{M \in \mathcal{T}(\bar{\kappa})} p(M | \mathbf{y}, \boldsymbol{\theta}) \end{aligned}$$

where the inequality follows from $\sum_{i \in S_j^I(\bar{\kappa})} I(i \in M) \leq |S_j^I(\bar{\kappa})|$ for all M . By (17pp.189), we then have

$$\frac{1}{p_j} \sum_{i: z_i=j} p(m_i = 1 \mid \mathbf{y}, \boldsymbol{\theta}) \leq \frac{|S_j^L(\bar{\kappa})| + |S_j^I(\bar{\kappa})|}{p_j} \sum_{M \in \mathcal{T}(\bar{\kappa})} p(M \mid \mathbf{y}, \boldsymbol{\theta}) + C_j. \quad (17\text{pp.191})$$

Moreover, for every $j = 1, \dots, b$, C_j satisfies

$$C_j = \sum_{M \notin \mathcal{T}(\bar{\kappa})} p(M \mid \mathbf{y}, \boldsymbol{\theta}) \sum_{i: z_i=j} \frac{I(i \in M)}{p_j} \leq \sum_{M \notin \mathcal{T}(\bar{\kappa})} p(M \mid \mathbf{y}, \boldsymbol{\theta}) \quad (17\text{pp.192})$$

where the inequality follows from $\sum_{i: z_i=j} \frac{I(i \in M)}{p_j} \leq 1$ for all M . Taking expectations in (17pp.191) and (17pp.192) gives

$$E\left(\frac{1}{p_j} \sum_{i: z_i=j} p(m_i = 1 \mid \mathbf{y}, \boldsymbol{\theta})\right) \leq \frac{|S_j^L(\bar{\kappa})| + |S_j^I(\bar{\kappa})|}{p_j} \sum_{M \in \mathcal{T}(\bar{\kappa})} \mathbb{E}(p(M \mid \mathbf{y}, \boldsymbol{\theta})) + \sum_{M \notin \mathcal{T}(\bar{\kappa})} \mathbb{E}(p(M \mid \mathbf{y}, \boldsymbol{\theta})).$$

By Lemma 17.2, we have on one hand $\lim_{n \rightarrow \infty} \sum_{M \in \mathcal{T}(\bar{\kappa})} \mathbb{E}(p(M \mid \mathbf{y}, \boldsymbol{\theta})) = 1$ and, on the other hand, $\lim_{n \rightarrow \infty} \sum_{M \notin \mathcal{T}(\bar{\kappa})} \mathbb{E}(p(M \mid \mathbf{y}, \boldsymbol{\theta})) = 0$. It follows that

$$\lim_{n \rightarrow \infty} E\left(\frac{1}{p_j} \sum_{i: z_i=j} p(m_i = 1 \mid \mathbf{y}, \boldsymbol{\theta})\right) \leq \frac{|S_j^L(\bar{\kappa})| + |S_j^I(\bar{\kappa})|}{p_j} = \frac{s_j - |S_j^S(\bar{\kappa})|}{p_j}$$

for every $j = 1, \dots, b$, which proves the upper bound. $\square$

17.2 Proof of Theorem ??

Theorem ?? is a particular case of Lemma 17.2. Observe that under assumption A5ter, $S^L(\bar{\kappa}) = S$ and $S^S(\bar{\kappa}) = S^I(\bar{\kappa}) = \emptyset$. It follows that $\mathcal{T}(\bar{\kappa}) = S$. Assumptions A1 and A4ter are also assumed to hold and, trivially, $|S^I(\bar{\kappa})| = O(1)$ and $|S_j^S(\bar{\kappa})| = O(p_j - s_j)$ for every $j = 1, \dots, b$, then by Lemma 17.2, $\lim_{n \rightarrow \infty} \sum_{M \neq S} \mathbb{E}(p(M \mid \mathbf{y}, \boldsymbol{\theta})) = 0$.

17.3 Proof of Theorem ??

The proof strategy is to apply Theorem ?? that requires Assumptions A1, A3, A4ter and A5ter. Recall that Theorem ?? makes Assumption A1, A3, hence it suffices to show that A4ter and A5ter hold. We first derive a convenient decomposition of the empirical Bayes penalties $\bar{\kappa}_j^{(1)}$. The second step of the proof is to show that, with probability going to 1, these

$\bar{\kappa}_j^{(1)}$ satisfy Assumption A4ter. The third step consists in showing that Assumption A8bis implies Assumption A5ter for $\bar{\kappa}_j^{(1)}$. The convergence of $E \left(p(S \mid \mathbf{y}, \boldsymbol{\theta}^{(1)}) \right)$ then follows from Theorem ??.

In Step 1, from the choice $\boldsymbol{\theta}^{(0)}$ and the ensuing penalty $\bar{\kappa}^{(0)}$, we have

$$\begin{aligned} \frac{1}{p_j} \sum_{i:z_i=j} p(m_i = 1 \mid \mathbf{y}, \boldsymbol{\theta}^{(0)}) &= \frac{1}{p_j} \sum_{i:z_i=j} \sum_M p(M \mid \mathbf{y}, \boldsymbol{\theta}^{(0)}) I(i \in M) \\ &= \frac{1}{p_j} \sum_M p(M \mid \mathbf{y}, \boldsymbol{\theta}^{(0)}) \sum_{i:z_i=j} I(i \in M) \\ &= \frac{s_j}{p_j} p(S \mid \mathbf{y}, \boldsymbol{\theta}^{(0)}) + \sum_{M \mid M \neq S} \frac{|M_j|}{p_j} p(M \mid \mathbf{y}, \boldsymbol{\theta}^{(0)}). \end{aligned}$$

Using that $p(S \mid \mathbf{y}, \boldsymbol{\theta}^{(0)}) = 1 - \sum_{M \mid M \neq S} p(M \mid \mathbf{y}, \boldsymbol{\theta}^{(0)})$, we get

$$\frac{1}{p_j} \sum_{i:z_i=j} p(m_i = 1 \mid \mathbf{y}, \boldsymbol{\theta}^{(0)}) = \frac{s_j}{p_j} + \sum_{M \mid M \neq S} \frac{|M_j| - s_j}{p_j} p(M \mid \mathbf{y}, \boldsymbol{\theta}^{(0)}).$$

Consider the decomposition of the sum in the right-hand side above between the sum over models M that contain more parameters than S in block b and the sum over those that contain less parameters than S in block b . Denote

$$\mathcal{O}_j^{(0)} := \sum_{M \mid |M_j| > s_j} \frac{|M_j| - s_j}{p_j} p(M \mid \mathbf{y}, \boldsymbol{\theta}^{(0)}) \quad \text{and} \quad U_j^{(0)} := \sum_{M \mid |M_j| < s_j} \frac{s_j - |M_j|}{p_j} p(M \mid \mathbf{y}, \boldsymbol{\theta}^{(0)}).$$

We have

$$\frac{1}{p_j} \sum_{i:z_i=j} p(m_i = 1 \mid \mathbf{y}, \boldsymbol{\theta}^{(0)}) = \frac{s_j}{p_j} + \mathcal{O}_j^{(0)} - U_j^{(0)}. \quad (17\text{pp.193})$$

Observe that we have the following decomposition of Step 2 penalties

$$\begin{aligned} \bar{\kappa}_j^{(1)} &= \log(p_j - s_j) + \log \left(\frac{\sqrt{1 + gn}}{s_j} \right) + \log \left(\frac{p_j - \sum_{i:z_i=j} p(m_i = 1 \mid \mathbf{y}, \boldsymbol{\theta}^{(0)})}{p_j - s_j} \right) \\ &\quad + \log \left(\frac{s_j}{\sum_{i:z_i=j} p(m_i = 1 \mid \mathbf{y}, \boldsymbol{\theta}^{(0)})} \right). \end{aligned}$$

By (17pp.193), it follows that

$$\begin{aligned} \bar{\kappa}_j^{(1)} &= \log(p_j - s_j) + \log \left(\frac{\sqrt{1 + gn}}{s_j} \right) + \log \left(1 - \frac{p_j(\mathcal{O}_j^{(0)} - U_j^{(0)})}{p_j - s_j} \right) \\ &\quad + \log \left(\frac{s_j}{\sum_{i:z_i=j} p(m_i = 1 \mid \mathbf{y}, \boldsymbol{\theta}^{(0)})} \right), \end{aligned} \quad (17\text{pp.194})$$

completing the first step of the proof.

We continue with the second step of the proof: showing that the $\bar{\kappa}_j^{(1)}$'s satisfy Assumption A4ter with probability going to 1. Recall that Assumption A4ter states that there exists $f_j \rightarrow \infty$ (as $n \rightarrow \infty$) such that for every sufficiently large n ,

$$\bar{\kappa}_j = \log(p_j - s_j) + f_j.$$

Since $U_j^{(0)}$ is nonnegative, a lower bound on $\bar{\kappa}_j^{(1)}$ is

$$\bar{\kappa}_j^{(1)} \geq \log(p_j - s_j) + \log\left(\frac{\sqrt{1+gn}}{s_j}\right) + \log\left(1 - \frac{p_j O_j^{(0)}}{p_j - s_j}\right) + \log\left(\frac{s_j}{\sum_{i:z_i=j} p(m_i = 1 \mid \mathbf{y}, \boldsymbol{\theta}^{(0)})}\right).$$

Plugging in the definition of $O_j^{(0)}$, we have that

$$\frac{p_j O_j^{(0)}}{p_j - s_j} = \sum_{M \mid |M_j| > s_j} \frac{|M_j| - s_j}{p_j - s_j} p(M \mid \mathbf{y}, \boldsymbol{\theta}^{(0)}) \leq \sum_{M \mid |M_j| > s_j} p(M \mid \mathbf{y}, \boldsymbol{\theta}^{(0)})$$

where the inequality follows from $(|M_j| - s_j)/(p_j - s_j) \leq 1$ for all M . Note that if M is such that $|M_j| > s_j$, then $M \notin \mathcal{T}(\bar{\kappa}^{(0)})$ (this follows immediately from the definition of $\mathcal{T}(\bar{\kappa})$ in (17pp.173)) and therefore $\sum_{M \mid |M_j| > s_j} p(M \mid \mathbf{y}, \boldsymbol{\theta}^{(0)}) \leq \sum_{M \notin \mathcal{T}(\bar{\kappa}^{(0)})} p(M \mid \mathbf{y}, \boldsymbol{\theta}^{(0)})$. Moreover, $\bar{\kappa}^{(0)}$ satisfies Assumption A4ter and the assumptions of Lemma 17.2 are met for $\bar{\kappa}^{(0)}$. Then, by Lemma 17.2, $\lim_{n \rightarrow \infty} \sum_{M \notin \mathcal{T}(\bar{\kappa}^{(0)})} E(p(M \mid \mathbf{y}, \boldsymbol{\theta}^{(0)})) = \lim_{n \rightarrow \infty} \sum_{M \mid |M_j| > s_j} E(p(M \mid \mathbf{y}, \boldsymbol{\theta}^{(0)})) = 0$. It follows that $\frac{p_j O_j^{(0)}}{p_j - s_j}$ vanishes in probability and so does $\log\left(1 - \frac{p_j O_j^{(0)}}{p_j - s_j}\right)$. By Assumption A4, we also have that $\log(\sqrt{1+gn}s_j^{-1}) \rightarrow \infty$. Then, to show that Assumption A4ter holds with probability going to 1, it is enough to show that with probability going to 1, $\log(s_j / \sum_{i:z_i=j} p(m_i = 1 \mid \mathbf{y}, \boldsymbol{\theta}^{(0)}))$ is nonnegative. Observe that all assumptions in Lemma 17.3 are also met for $\bar{\kappa}^{(0)}$. By (17pp.191) and (17pp.192) in the proof of Lemma 17.3, we have

$$\begin{aligned} \frac{\sum_{i:z_i=j} p(m_i = 1 \mid \mathbf{y}, \boldsymbol{\theta}^{(0)})}{p_j} &\leq \frac{|S_j^L(\bar{\kappa}^{(0)})| + |S_j^I(\bar{\kappa}^{(0)})|}{p_j} \sum_{M \in \mathcal{T}(\bar{\kappa}^{(0)})} p(M \mid \mathbf{y}, \boldsymbol{\theta}^{(0)}) \\ &\quad + \frac{1}{p_j} \sum_{M \notin \mathcal{T}(\bar{\kappa}^{(0)})} p(M \mid \mathbf{y}, \boldsymbol{\theta}^{(0)}). \end{aligned}$$

By Lemma 17.2, $\sum_{M \in \mathcal{T}(\bar{\kappa}^{(0)})} p(M \mid \mathbf{y}, \boldsymbol{\theta}^{(0)})$ and $\sum_{M \notin \mathcal{T}(\bar{\kappa}^{(0)})} p(M \mid \mathbf{y}, \boldsymbol{\theta}^{(0)})$ converge in probability to 1 and 0 respectively. We then have that, with probability going to 1,

$$\frac{\sum_{i:z_i=j} p(m_i = 1 \mid \mathbf{y}, \boldsymbol{\theta}^{(0)})}{p_j} \leq \frac{|S_j^L(\bar{\kappa}^{(0)})| + |S_j^I(\bar{\kappa}^{(0)})|}{p_j},$$

which implies, with probability going to 1,

$$\log \left(\frac{s_j}{\sum_{i:z_i=j} p(m_i = 1 \mid \mathbf{y}, \boldsymbol{\theta}^{(0)})} \right) \geq \log \left(\frac{s_j}{|S_j^L(\bar{\kappa}^{(0)})| + |S_j^I(\bar{\kappa}^{(0)})|} \right) \geq 0.$$

We then obtain that, with probability going to 1,

$$\bar{\kappa}_j^{(1)} \geq \log(p_j - s_j) + \log \left(\frac{\sqrt{1 + gn}}{s_j} \right) \quad (17\text{pp.195})$$

and that the $\bar{\kappa}_j^{(1)}$ satisfy Assumption A4ter, completing the second part of the proof.

For the third and final part of the proof, we now show that Assumption A8bis implies Assumption A5ter for the $\bar{\kappa}_j^{(1)}$ with probability going to 1. Recall that Assumption A5ter for the $\bar{\kappa}_j^{(1)}$ states that for each block j there exists $e_j \rightarrow \infty$ such that for large enough n ,

$$\sqrt{\frac{(1 - \omega)n\rho(\mathbf{X})}{6}} \beta_{\min,j}^* - \sqrt{\bar{\kappa}_j^{(1)}} = \sqrt{\log(s_j)} + e_j.$$

where ω takes value

$$\omega = \frac{1}{2} \left(1 + \max_j \frac{\log(p_j - s_j)}{\bar{\kappa}_j^{(1)}} \right). \quad (17\text{pp.196})$$

Observe that Assumption A8bis and Assumption A5ter take the same form. To show that Assumption A8bis implies Assumption A5ter for the $\bar{\kappa}_j^{(1)}$ with probability going to 1, it suffices to show that the following two inequalities

$$\sqrt{\frac{(1 - \omega)n\rho(\mathbf{X})}{6}} \beta_{\min,j}^* \geq \sqrt{\frac{(1 - \omega')n\rho(\mathbf{X})}{6}} \beta_{\min,j}^*, \quad (17\text{pp.197})$$

$$- \sqrt{\bar{\kappa}_j^{(1)}} \geq - \sqrt{\log \left(\frac{p}{|S_j^L(\bar{\kappa}^{(0)})|} - 1 \right) + \frac{1}{2} \log(1 + gn)} \quad (17\text{pp.198})$$

hold with probability going to 1 for ω as in (17pp.196) and $\omega' = \frac{1}{2} \left(1 + \max_j \frac{\log(p_j - s_j)}{\log(p_j/s_j - 1) + 0.5 \log(1 + gn)} \right)$ (defined in Assumption A8bis). We first show (17pp.197) holds with probability going to 1 and then that (17pp.198) does too.

By (17pp.195), with probability going to 1,

$$\frac{\log(p_j - s_j)}{\bar{\kappa}_j^{(1)}} \leq \frac{\log(p_j - s_j)}{\log(p_j - s_j) + \log\left(\frac{\sqrt{1+gn}}{s_j}\right)} = \frac{\log(p_j - s_j)}{\log(p_j/s_j - 1) + 0.5 \log(1 + gn)}.$$

It follows that:

$$\omega = \frac{1}{2} \left(1 + \max_{j=1, \dots, b} \frac{\log(p_j - s_j)}{\bar{\kappa}_j^{(1)}} \right) \leq \frac{1}{2} \left(1 + \max_j \frac{\log(p_j - s_j)}{\log(p_j/s_j - 1) + 0.5 \log(1 + gn)} \right) = \omega'$$

and (17pp.197) holds with probability going to 1.

We now upper bound $\bar{\kappa}_j^{(1)}$ to show (17pp.198) holds with probability going to 1. Observe that

$$\begin{aligned} \log\left(\frac{\sum_{i: z_i=j} p(m_i = 1 \mid \mathbf{y}, \boldsymbol{\theta}^{(0)})}{s_j}\right) &= \log\left(1 + \frac{\sum_{i: z_i=j} p(m_i = 1 \mid \mathbf{y}, \boldsymbol{\theta}^{(0)}) - s_j}{s_j}\right) \\ &= \log\left(1 + \frac{p_j(O_j^{(0)} - U_j^{(0)})}{s_j}\right) \end{aligned}$$

where the second equality follows from (17pp.193). Plugging this expression into (17pp.194), and using that $\mathcal{O}_j^{(0)} \geq 0$, we have that

$$\bar{\kappa}_j^{(1)} \leq \log(p_j - s_j) + \log\left(\frac{\sqrt{1+gn}}{s_j}\right) + \log\left(\frac{1 + \frac{p_j}{p_j - s_j} U_j^{(0)}}{1 - \frac{p_j}{s_j} U_j^{(0)}}\right). \quad (17pp.199)$$

We split the sum in $U_j^{(0)}$ between models in $\mathcal{T}(\bar{\kappa}^{(0)})$ and those not in $\mathcal{T}(\bar{\kappa}^{(0)})$.

$$U_j^{(0)} = \sum_{M \in \mathcal{T}(\bar{\kappa}^{(0)}) \mid |M_j| < s_j} \frac{s_j - |M_j|}{p_j} p(M \mid \mathbf{y}, \boldsymbol{\theta}^{(0)}) + \sum_{M \notin \mathcal{T}(\bar{\kappa}^{(0)}) \mid |M_j| < s_j} \frac{s_j - |M_j|}{p_j} p(M \mid \mathbf{y}, \boldsymbol{\theta}^{(0)}).$$

If $M \in \mathcal{T}(\bar{\kappa}^{(0)})$, then by definition $|M_j| \geq |S_j^L(\bar{\kappa}^{(0)})|$ and thus $s_j - |M_j| \leq s_j - |S_j^L(\bar{\kappa}^{(0)})|$.

A bound on $s_j - |M_j|$ for $M \notin \mathcal{T}(\bar{\kappa}^{(0)})$ is simply $s_j - |M_j| \leq s_j$. It follows that

$$U_j^{(0)} \leq \frac{s_j - |S_j^L(\bar{\kappa}^{(0)})|}{p_j} \sum_{M \in \mathcal{T}(\bar{\kappa}^{(0)}) \mid |M_j| < s_j} p(M \mid \mathbf{y}, \boldsymbol{\theta}^{(0)}) + \frac{s_j}{p_j} \sum_{M \notin \mathcal{T}(\bar{\kappa}^{(0)}) \mid |M_j| < s_j} p(M \mid \mathbf{y}, \boldsymbol{\theta}^{(0)})$$

By Lemma 17.2, $\sum_{M \in \mathcal{T}(\bar{\kappa}^{(0)}) \mid |M_j| < s_j} p(M \mid \mathbf{y}, \boldsymbol{\theta}^{(0)})$ and $\sum_{M \notin \mathcal{T}(\bar{\kappa}^{(0)}) \mid |M_j| < s_j} p(M \mid \mathbf{y}, \boldsymbol{\theta}^{(0)})$ converge in probability to 1 and 0 respectively. We then get that, with probability going to 1,

$$\begin{aligned} \frac{p_j}{p_j - s_j} U_j^{(0)} &\leq \frac{s_j - |S_j^L(\bar{\kappa}^{(0)})|}{p_j - s_j} \\ \frac{p_j}{s_j} U_j^{(0)} &\leq \frac{s_j - |S_j^L(\bar{\kappa}^{(0)})|}{s_j}. \end{aligned} \quad (17pp.200)$$

By the bounds above and (17pp.199), we have that with probability going to 1,

$$\begin{aligned}\bar{\kappa}_j^{(1)} &\leq \log(p_j - s_j) + \log\left(\frac{\sqrt{1+gn}}{s_j}\right) + \log\left(\frac{1 + \frac{s_j - |S_j^L(\bar{\kappa}^{(0)})|}{p_j - s_j}}{1 - \frac{s_j - |S_j^L(\bar{\kappa}^{(0)})|}{s_j}}\right) \\ &= \log(p_j - s_j) + \log\left(\frac{\sqrt{1+gn}}{s_j}\right) + \log\left(\frac{\frac{p_j - |S_j^L(\bar{\kappa}^{(0)})|}{p_j - s_j}}{\frac{|S_j^L(\bar{\kappa}^{(0)})|}{s_j}}\right) \quad (17pp.201)\end{aligned}$$

$$= \log(p_j / |S_j^L(\bar{\kappa}^{(0)})| - 1) + \frac{1}{2} \log(1 + gn) \quad (17pp.202)$$

which shows (17pp.198) holds with probability going to 1 and that Assumption A8bis implies Assumption A5ter holds for the $\bar{\kappa}_j^{(1)}$ with probability going to 1.

Since Assumptions A4ter and A5ter hold with probability going to 1, by Theorem ??, $\lim_{n \rightarrow \infty} E[p(S | \mathbf{y}, \hat{\theta}^{(1)})] = 1$, as claimed.

17.4 Proof of Theorem ??

Since $p(\mathbf{y} | \boldsymbol{\theta}) = \sum_M p(\mathbf{y} | M)p(M | \boldsymbol{\theta})$ and $\nabla_{\boldsymbol{\theta}} \log p(M | \boldsymbol{\theta}) = [\nabla_{\boldsymbol{\theta}} p(M | \boldsymbol{\theta})] / p(M | \boldsymbol{\theta})$, we have that

$$\begin{aligned}\nabla_{\boldsymbol{\theta}} \log p(\mathbf{y} | \boldsymbol{\theta}) &= \frac{\nabla_{\boldsymbol{\theta}} p(\mathbf{y} | \boldsymbol{\theta})}{p(\mathbf{y} | \boldsymbol{\theta})} = \frac{\sum_M p(\mathbf{y} | M) \nabla_{\boldsymbol{\theta}} p(M | \boldsymbol{\theta})}{p(\mathbf{y} | \boldsymbol{\theta})} = \\ &= \frac{\sum_M p(\mathbf{y} | M) p(M | \boldsymbol{\theta}) \nabla_{\boldsymbol{\theta}} \log p(M | \boldsymbol{\theta})}{p(\mathbf{y} | \boldsymbol{\theta})} = \sum_M p(M | \mathbf{y}, \boldsymbol{\theta}) \nabla_{\boldsymbol{\theta}} \log p(M | \boldsymbol{\theta}),\end{aligned}$$

as claimed.

For $p(M | \boldsymbol{\theta}) = \prod_{i=1}^p \text{Bern}(m_i; \pi_i(\boldsymbol{\theta}))$ for arbitrary differentiable $\pi_i(\cdot)$, we get

$$\nabla_{\boldsymbol{\theta}} \log p(M | \boldsymbol{\theta}) = \sum_{i=1}^p \nabla_{\boldsymbol{\theta}} \log p(m_i | \boldsymbol{\theta}) = \sum_{i=1}^p \frac{\nabla_{\boldsymbol{\theta}} p(m_i | \boldsymbol{\theta})}{p(m_i | \boldsymbol{\theta})}.$$

Since $p(m_i | \boldsymbol{\theta}) = \pi_i(\boldsymbol{\theta})^{m_i} [1 - \pi_i(\boldsymbol{\theta})]^{1-m_i}$

$$\begin{aligned}\nabla_{\boldsymbol{\theta}} p(m_i | \boldsymbol{\theta}) &= \begin{cases} \nabla_{\boldsymbol{\theta}} \pi_i(\boldsymbol{\theta}) = \frac{\nabla_{\boldsymbol{\theta}} \pi_i(\boldsymbol{\theta})}{\pi_i(\boldsymbol{\theta})} p(m_i | \boldsymbol{\theta}), & \text{if } m_i = 1 \\ -\nabla_{\boldsymbol{\theta}} \pi_i(\boldsymbol{\theta}) = -\frac{\nabla_{\boldsymbol{\theta}} \pi_i(\boldsymbol{\theta})}{1 - \pi_i(\boldsymbol{\theta})} p(m_i | \boldsymbol{\theta}), & \text{if } m_i = 0 \end{cases} \\ &= p(m_i | \boldsymbol{\theta}) \nabla_{\boldsymbol{\theta}} \pi_i(\boldsymbol{\theta}) \left[\frac{m_i}{\pi_i(\boldsymbol{\theta})} - \frac{1 - m_i}{1 - \pi_i(\boldsymbol{\theta})} \right] = p(m_i | \boldsymbol{\theta}) \frac{\nabla_{\boldsymbol{\theta}} \pi_i(\boldsymbol{\theta})}{\pi_i(\boldsymbol{\theta}) [1 - \pi_i(\boldsymbol{\theta})]} [m_i - \pi_i(\boldsymbol{\theta})]. \quad (17pp.203)\end{aligned}$$

This gives that

$$\begin{aligned}\nabla_{\boldsymbol{\theta}} \log p(\mathbf{y} \mid \boldsymbol{\theta}) &= \mathbb{E} \left[\sum_{i=1}^p \frac{\nabla_{\boldsymbol{\theta}} \pi_i(\boldsymbol{\theta})}{\pi_i(\boldsymbol{\theta})[1 - \pi_i(\boldsymbol{\theta})]} [m_i - \pi_i(\boldsymbol{\theta})] \mid \mathbf{y}, \boldsymbol{\theta} \right] \\ &= \sum_{i=1}^p \frac{\nabla_{\boldsymbol{\theta}} \pi_i(\boldsymbol{\theta})}{\pi_i(\boldsymbol{\theta})[1 - \pi_i(\boldsymbol{\theta})]} [p(m_i = 1 \mid \mathbf{y}, \boldsymbol{\theta}) - p(m_i = 1 \mid \boldsymbol{\theta})],\end{aligned}$$

as claimed. In the particular case where $\pi_i(\boldsymbol{\theta}) = (1 + e^{\mathbf{z}_i^\top \boldsymbol{\theta}})^{-1}$, we have $\nabla_{\boldsymbol{\theta}} \pi_i(\boldsymbol{\theta}) = \mathbf{z}_i e^{\mathbf{z}_i^\top \boldsymbol{\theta}} / (1 + e^{\mathbf{z}_i^\top \boldsymbol{\theta}})^2 = \mathbf{z}_i \pi_i(\boldsymbol{\theta}) [1 - \pi_i(\boldsymbol{\theta})]$, and then

$$\nabla_{\boldsymbol{\theta}} \log p(\mathbf{y} \mid \boldsymbol{\theta}) = \sum_{i=1}^p \mathbf{z}_i [p(m_i = 1 \mid \mathbf{y}, \boldsymbol{\theta}) - p(m_i = 1 \mid \boldsymbol{\theta})].$$

17.5 Proof of Corollary ??

The gradient of $f(\boldsymbol{\theta}) = \sum_M p(M \mid \mathbf{y}, \hat{\boldsymbol{\theta}}^{(k)}) \log p(M \mid \boldsymbol{\theta})$ is $\nabla_{\boldsymbol{\theta}} f(\boldsymbol{\theta}) = \sum_M p(M \mid \mathbf{y}, \hat{\boldsymbol{\theta}}^{(k)}) \nabla_{\boldsymbol{\theta}} \log p(M \mid \boldsymbol{\theta})$. Using (17pp.203), we obtain that

$$\begin{aligned}\nabla_{\boldsymbol{\theta}} f(\boldsymbol{\theta}) &= \sum_M p(M \mid \mathbf{y}, \hat{\boldsymbol{\theta}}^{(k)}) \sum_{i=1}^p \frac{\nabla_{\boldsymbol{\theta}} \pi_i(\boldsymbol{\theta})}{\pi_i(\boldsymbol{\theta})[1 - \pi_i(\boldsymbol{\theta})]} [m_i - p(m_i = 1 \mid \boldsymbol{\theta})] \\ &= \sum_{i=1}^p \frac{\nabla_{\boldsymbol{\theta}} \pi_i(\boldsymbol{\theta})}{\pi_i(\boldsymbol{\theta})[1 - \pi_i(\boldsymbol{\theta})]} [p(m_i = 1 \mid \mathbf{y}, \hat{\boldsymbol{\theta}}) - p(m_i = 1 \mid \boldsymbol{\theta})].\end{aligned}$$

In the particular case where $\pi_i(\boldsymbol{\theta}) = (1 + e^{\mathbf{z}_i^\top \boldsymbol{\theta}})^{-1}$, from Theorem ?? we have that

$[\nabla_{\boldsymbol{\theta}} \pi_i(\boldsymbol{\theta})] / (\pi_i(\boldsymbol{\theta})[1 - \pi_i(\boldsymbol{\theta})]) = \mathbf{z}_i$ and then $\nabla_{\boldsymbol{\theta}} f(\boldsymbol{\theta}) = \sum_{i=1}^p \mathbf{z}_i [p(m_i = 1 \mid \mathbf{y}, \boldsymbol{\theta}) - p(m_i = 1 \mid \boldsymbol{\theta})] = \mathbf{Z}^\top [\hat{\mathbf{r}}(\boldsymbol{\theta}) - \boldsymbol{\pi}(\boldsymbol{\theta})]$. The expression for the hessian $\nabla_{\boldsymbol{\theta}}^2 f(\boldsymbol{\theta})$ follows from straightforward calculus.

17.6 Proof of Proposition ??

The goal is to find $\boldsymbol{\theta}$ such that $\nabla_{\boldsymbol{\theta}} f(\boldsymbol{\theta}) + \nabla_{\boldsymbol{\theta}} \log p(\boldsymbol{\theta}) = \mathbf{0}$. Plugging in the expression for $\nabla_{\boldsymbol{\theta}} f(\boldsymbol{\theta})$ and $\nabla_{\boldsymbol{\theta}} \log p(\boldsymbol{\theta}) = -\frac{1}{g} \mathcal{V}^{-1} \boldsymbol{\theta}$, where $\mathcal{V}^{-1} = (\mathbf{Z}^\top \mathbf{Z})/p$, we seek $\boldsymbol{\theta}$ such that

$$\mathbf{Z}^\top [\hat{\mathbf{r}} - \boldsymbol{\pi}(\boldsymbol{\theta})] - \frac{1}{gp} \mathbf{Z}^\top \mathbf{Z} \boldsymbol{\theta} = 0 \Leftrightarrow \mathbf{Z}^\top \left[\boldsymbol{\pi}(\boldsymbol{\theta}) + \frac{1}{gp} \mathbf{Z} \boldsymbol{\theta} \right] = \mathbf{Z}^\top \hat{\mathbf{r}},$$

where $\pi_i(\boldsymbol{\theta}) = 1/(1 + e^{-\mathbf{z}_i^\top \boldsymbol{\theta}})$. The $q \times q$ matrix $\mathbf{U}$ contains the unique rows in $\mathbf{Z}$ and is full-rank, hence $\tilde{\mathbf{Z}} = \mathbf{Z} \mathbf{U}^{-1}$ is a matrix whose unique rows equal the identity matrix. Define $\tilde{\boldsymbol{\theta}} = \mathbf{U} \boldsymbol{\theta}$ and consider the problem of finding $\tilde{\boldsymbol{\theta}}$ such that $\tilde{\mathbf{Z}}^\top [\tilde{\boldsymbol{\pi}}(\tilde{\boldsymbol{\theta}}) - (gp)^{-1} \tilde{\mathbf{Z}} \tilde{\boldsymbol{\theta}}] = \tilde{\mathbf{Z}}^\top \hat{\mathbf{r}}$,

where $\tilde{m}_j(\tilde{\boldsymbol{\theta}}) = 1/(1 + e^{-\tilde{\mathbf{z}}_j^\top \tilde{\boldsymbol{\theta}}})$. we first show how to obtain $\tilde{\boldsymbol{\theta}}$, and subsequently that one can find $\boldsymbol{\theta}$ from $\tilde{\boldsymbol{\theta}}$.

To obtain $\tilde{\boldsymbol{\theta}}$, intuitively this is simple because the subset of unique rows in $\tilde{\mathbf{Z}}$ is the identity matrix. Specifically, the j^{th} equation in $\tilde{\mathbf{Z}}^\top [\tilde{\pi}(\tilde{\boldsymbol{\theta}}) - (gp)^{-1} \tilde{\mathbf{Z}} \tilde{\boldsymbol{\theta}}] = \tilde{\mathbf{Z}}^\top \hat{\mathbf{r}}$ is

$$\begin{aligned} \sum_{i=1}^p \mathbb{I}(\tilde{z}_{ij} = 1) \left[\frac{1}{1 + e^{-\tilde{\mathbf{z}}_i^\top \tilde{\boldsymbol{\theta}}}} + \frac{\tilde{\mathbf{z}}_i^\top \tilde{\boldsymbol{\theta}}}{gp} \right] &= \tilde{\mathbf{z}}_{.j}^\top \hat{\mathbf{r}} \Rightarrow \\ \frac{1}{1 + e^{-\theta_j}} + \frac{\tilde{w}_j}{gp} &= \frac{\tilde{\mathbf{z}}_{.j}^\top \hat{\mathbf{r}}}{\sum_{i=1}^p \mathbb{I}(\tilde{z}_{ij} = 1)} = \frac{\tilde{\mathbf{z}}_{.j}^\top \hat{\mathbf{r}}}{\tilde{\mathbf{z}}_{.j}^\top \mathbf{1}}, \end{aligned}$$

where $\tilde{\mathbf{z}}_{.j}$ is the j^{th} column in $\tilde{\mathbf{Z}}$, and we used that $\tilde{\mathbf{z}}_i^\top \tilde{\boldsymbol{\theta}} = \theta_j$, since $\tilde{z}_{il} = 1$ for $l = j$ and $= 0$ otherwise. We hence have that $\theta_j = h(\tilde{\mathbf{z}}_{.j}^\top \hat{\mathbf{r}} / \tilde{\mathbf{z}}_{.j}^\top \mathbf{1}, gp)$, where $h(a, c)$ denotes the solution (in terms of w) to $1/(1 + e^{-w}) + w/c = a$. For the case $c = \infty$, $h(a, \infty) = \log(a/(1 - a))$. More generally, the solution to this univariate equation can be easily found via Newton-Raphson.

To complete the proof, define $\hat{\boldsymbol{\theta}} = \mathbf{U}^{-1} \tilde{\boldsymbol{\theta}}$, so that $\tilde{\mathbf{Z}} \tilde{\boldsymbol{\theta}} = \mathbf{Z} \mathbf{U}^{-1} \mathbf{U} \hat{\boldsymbol{\theta}} = \mathbf{Z} \hat{\boldsymbol{\theta}}$, and therefore $\pi(\hat{\boldsymbol{\theta}}) = \tilde{\pi}(\tilde{\boldsymbol{\theta}})$. By definition of $\tilde{\boldsymbol{\theta}}$ we have that

$$\tilde{\mathbf{Z}}^\top [\tilde{\pi}(\tilde{\boldsymbol{\theta}}) - \frac{\tilde{\mathbf{Z}} \tilde{\boldsymbol{\theta}}}{gp}] = \tilde{\mathbf{Z}}^\top \hat{\mathbf{r}} \Rightarrow (\mathbf{U}^{-1})^\top [\mathbf{Z}^\top \pi(\hat{\boldsymbol{\theta}}) - \frac{\mathbf{Z} \hat{\boldsymbol{\theta}}}{gp}] = (\mathbf{U}^{-1})^\top \mathbf{Z}^\top \hat{\mathbf{r}}$$

which implies that $\mathbf{Z}^\top [\pi(\hat{\boldsymbol{\theta}}) - (gp)^{-1} \mathbf{Z} \hat{\boldsymbol{\theta}}] = \mathbf{Z}^\top \hat{\mathbf{r}}$, as claimed.